KTH ROYAL INSTITUTE
OF TECHNOLOGY

Degree Project in Human-Computer Interaction Design

Second cycle,30 credits

Explainable AI for supporting operators in manufacturing machines maintenance

Evaluating different techniques of explainable AI for a machine learning model that can be used in a manufacturing environment

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Stockholm, Sweden, 2022

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Master’s Programme, Interactive Media Technology, 120 credits
Date:June 23, 2022

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Host company:Nytt
Swedish title:Förklarlig AI för att stödja operatörer vid underhåll av tillverkningsmaskiner

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Abstract

Monitoringmachineswithinamanufacturingenvironmentisofvital importanceforcompaniesthatwanttoreachhighlevelsofprofitand productivity. Moreover,itisfundamentaltomaintainmachinesina reliablestateandsolvetheirfailuresquicklytolimitfinanciallosses. Therefore,informationtechnologyissteadilyincreasinginsidefactoriesto monitor machines' behavior and assist operators and managers in performing predictive maintenance.However, Artificial Intelligence AIs is still not widely adopted within the manufacturing field.Hence, the purpose of this study is to integrateMachineLearningMLstechnologies,whichcansupportoperators inmonitoringmanufacturingmachines'behavior,withinamanufacturing setup.Inparticular,aninvestigationonhowtorealizeaclassifierableto predict machines' state has been conducted.In addition, different techniques ofExplainableArtificialIntelligenceXAIshavebeentestedtodiscoverthe methodology allowing people without a MLs background to trust predictions. Four XAIs facilities have been tested: three identified through literature review and an experimental one based on a MLs model for outliers identification in productionparameters.Topredictfuturemachines'states,classifiersbased on different algorithms were built, and after evaluating them, it emerged that theRandomForest-basedmodelwastheonewiththebestperformance.In addition, a user study was carried out in order to discern end-users cognitive perspectivesaboutthefourXAIsinterfaces.Finalresultsshowedhowan XAIs facility based on situational data, such as the one representing anomalous production values, gained high approval among users with none or basic MLs knowledge.

Keywords

ExplainableArtificialintelligence,MachineLearning,RandomForest, Anomalies Identification, User Understandability, User Trustability, Interface, Evaluation, User study, Isolation Forest, Thematic analysis, Surveys

Acknowledgments

Stockholm, June2022
Francesco Di Flumeri

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List of acronyms and abbreviations

AIArtificial Intelligence

FNFalse Negative

FPFalse Positive

MLMachine Learning

PCAPrincipal Component Analysis

RFRandom Forest

TNTrue Negative

TPTrue Positive

TSTime Series

UXUser Experience

XAIExplainable Artificial Intelligence

Chapter 1

Introduction

In the last years machine learning has been incorporated in various fields, such as medicine, finance, law and defense as an artificially intelligent partner for expert workers [1, 2].Numerous studies have also been carried out about using machine learning within a manufacturing setup, in order to support managers andoperatorsinplantmaintenance[3,4].However,inthemajorityofthe cases,resultsprovidedbyaMLsmodelneedstobeexplainedsothatcan userscanactuallytrusttheresultandunderstandtheircause[5]. Dueto this necessity,the field of explainable AI (XAIs) has been explored with the purpose of providing users with intelligible behaviour for MLs models [6].

1.1 MLs background

Machinelearninginamanufacturingsettingcanbeusedbypredicting upcomingproblemswiththemachinery.MLhasnotbeenwidelyexplored in a manufacturing setting.However, a few previous studies have investigated the topic.

In a study by Je-Kang Park et al.[3], an automated method for performing productqualitychecksontheproductlinewasproposed.Theirexperiment showed that having anartificial system detecting goods' defects during their production brings advantages in terms of time, cost-saving, and performance comparedtotraditionalhumaninspection[3].AnotherexampleofAIsin manufacturing setup is in the study of Bernard Espiau et al.[7].He proposed a new method for controlling robot pose using a vision-based automated system. Asaresult,theymanagedtoconstructasystemthatwasabletomonitor robotmovesbasedongeometricrules,sincethesystemcouldautomatically recognize breakdowns on a robot line and assist operators [7].Finally, there

istheworkofYuanCheng[8]whereasystemtoautomaticallyinspectthe componentsofafinalproductwasdeveloped.Fromtheevaluationofthe system, it emerged that it was able to detect broken parts with high accuracy, so that it could substitute traditional supervisor [8].Not so much research has been carried out in predicting a manufacturing machine's behavior by a MLs system.Yet, predicting if a manufacturing system is going to fail or slow down its productivity could be valuable knowledge for managers and operators.

1.1.1 Anomalies

Anomaliesaredefinedasabnormalvalueswhicharenotalignedwiththe dataaverage.Inamachine,anomaliesarealwayscausedbytheproduction parameters by the different tools adopted during the manufacturing process.

Previousworkshavebeenidentifiedforanomaliesidentificationin manufacturing such as the one of Zhang et al.[9].In this research, they built ananomalydetectionmodelabletoidentifyabnormalvaluesintimeseries representing production parameters of CNC machines.Another relevant work istheoneconductedbyPugginietal.[10],wheretheyshowedtheefficacy of the Isolation Forest algorithm for identifying anomalies in semiconductor manufacturing.Finally, there is the study of Xu et al.[11].Here, they showed howitispossibletoidentifyclustersofanomalieswithhighaccuracyand precision by using the isolation forest algorithm.Although these researchers suggested possible approaches to implement anomaly detection, none of them attempted to use it as an XAIs technique.

1.2 XAIs background

Unfortunately, having a model that produces reliable predictions is not enough. In order to use the model's result, operators and managers need to trust it, and thismayrequirethesystemtoprovideexplanationsforthepredictedresult. Onewayofgeneratingusertrustisto"explain"howthemodelarrivedat apredictionwithanXAIsinterface.ExplainableAIs(XAIs)isdefinedas,"machin elearningalgorithmsandotherArtificialIntelligencesystemsthat produceoutcomesthathumanscanreadilyunderstandandtrackbacktothe origins" [12].

NumerousstudieshavebeencarriedoutindesigningeffectiveXAIs interfaces.Wangetal.[13]proposedanewhuman-centeredandtheory-drivenframeworkforbuildingexplainableAIs.Theytestedtheframework bydevelopingarealMLsmodelforassistingphysiciansandsuggested

generalrecommendationsonhowtoimprovetheintelligibilityofanXAIs system[13].Theyincluded:*supporthypothesisgeneration*,whichmeans thatusers'suppositionsaboutMLsoutputsneedtobeborefromtheXAIs techniques;*supportforward(data-driven)reasoning*,whichrepresentsthe needofshowingdataonwhichthepredictionisbased;*Supportcoherent factors* which indicates the necessity of underlining correlations among data used in the MLs model; *Support access to source and situational data* which standsfortheneedoflettingusersconsultrawdataonwhichMLsinputis based; *Support Bayesian reasoning* which means that it is important to show predictionsprobabilities;*Integratingmultipleexplanations*whichsignifies thatmultipleXAIsfacilitiesshouldbecombinedtogether.Anotherstudy wascarriedoutbyLundebergetal.[14]in2019.Inthisresearch,they proposeddifferentmethodsforimprovingtheinterpretabilityoftree-based MLs models.In detail, they found out that, by focusing on local understanding, users can learn about every single aspect of the prediction process and build global knowledge in an effective way.Petkovic et al.[15] conducted a study in2018,wheretheypresentedan*Explainabilitysummaries*asanefficient XAIs technique for interpreting MLs in the medical domain.This method is based on showing a summary of the MLs model performances in a one-page layout.After performing user evaluation with 13 expert and non-expert users, theyconcludedthatbyusingtheexplainabilitysummary,usersexperienced increasedconfidenceinthemodelpredictionsandtheybuildanaccurate understanding of how the system produced the final outcomes.Finally, there is the study of Hrnjica et al.[16] in 2020, where they implemented an AIs system usingdatafromasimulatedmanufacturingenvironment.Theyconcluded thatbyusingvisualizationsasanexplainabilitytool,suchasverticalbar chartsshowingmachinedata,operatorsmightbeabletoperformpredictive maintenance. However,itisimportanttomentionthatthisresearchwas not performed in real-world manufacturing settings, which may influence the validity of the results.

1.3 Problem

Assistingmanagersandoperatorsinpredictingmanufacturingsystems'failures can bring huge advantages to a company in terms of time and money. However, even with an extremely accurate MLs model people tend to not trust itsoutputbecausetheydonotpossessknowledgeaboutwhatcausesthose results.Hence, three research questions devised for this study are:

What is the reliability of a Machine Learning model for predicting future behaviour of manufacturing machines?
Is it possible to identify anomalies in the time series representing analytical data collected from the machines during the production cycles?
Which techniques of XAIs can lead end users to trust and have confidence in a MLs model that assists them in predicting manufacturing machines'behaviour?

1.3.1 Original problem and definition

Inordertostaycompetitiveonthemarket,manufacturingcompaniesneeds to maintain high level of efficacy and convenience [17] during the production process.Anunforeseenmachinebreakdowncanbefinanciallycostlyfora manufacturing company.However,predicting the behaviour of a production setup is not an easy task for humans.Most of the time, this due to two reasons: 1)alackofannotationsaboutpreviousfailures,and2)theheterogeneityof manufacturing data.
ThisproblemhasbeentackledbythecompanyNytt. Theyhelp customers take a step by step approach to improve the communication of their manufacturingsetup.Theyaimtofacilitatebettercommunicationbetween theoperators,themanagersandthemachinesandhelpthembecomemore information-driven in their decision making.The idea is to have a bottom-up approach to decision making and make the manufacturing companies operator-driven.They strive to provide simple and easy to use solutions developed from
readilyavailabletechnologytohelpcustomerstakethefirststeptogetdata from their machines, empower operators to make informed decisions, impart knowledge between the operators and managers and become more profitable. Theirgoalistohelpoutsmallandmedium-sizedmanufacturingenterprises improve their information flow, productivity and utilization of resources, thus making them more profitable [18].The system developed by Nytt consists of a dashboard that provides actionable insights to the managers and the operators to make informed decisions regarding the machines in a manufacturing setup. The insights are based on time series data that is being continuously collected using our data collection platform.The system has paved the way to companies
tomakereactivedecisionsbasedonthedatasothattheycanimprovetheir strategiesandhenceimproveproductivityoftheirmanufacturingresources. However,Nyttwantstoextendthisproductsothatitcanprovidinginsights about the future behaviour of manufacturing setup in order to assist managers and operators and to act in advance before a breakdown occurs.

1.3.2 Scientific and engineering issues

ThesystemproposedbyNyttperformproductionmachines'monitoringby visual recognition of stack lights' colors.This indicators can be classified in different categories based on the number of colors:two, three or four colors. The most diffused are the stack light producing three tones:if the green light isonthenthemachineisoperatingnormally,whileifanyissuesoccurthe lightturnsimmediatelyred.Theyellowlightinstead,representsawaiting state [19].Figure 1.1 gives a description overview.Stack lights are necessary forkeepinghighefficiencylevelsontheproductionline,sothatabnormal operations or missing inputs can be detected and solved [20].However, stack lightsdonotprovideworkerswithanyinsightsaboutfuturebreakdownsor issues on the product line.Yet, storing their status on long periods can provide usefuldatafordevelopingTSsmachinelearningmodelthateachtimecan foreseen the machines' behaviour.

Colors	Status of production line of each color
Green	A production line operates normally.
Red	Any trouble happens in a production line.
Yellow	Manager and/or operator have resolved the issue.

Figure 1.1:Stack light indicator

AnotherwaythatNyttfollowstomonitorproductionplantsisthe involvementoftheMTconnectprotocol. MTconnectisastandardized vocabulary for gathering analytical data from manufacturing machine via a 3rd party micro-controller [21].With this approach,data have a uniform format and they do not need to be translated.Hence, as the method is scalable, data canbecollectedandanalysedfromdifferenttypesofmachines.Production parametersmeasuredthroughMTconnectwillbeinvolvedinthestudyas well, in order to build another model for predicting machines' behaviour and compare it with the one based on stack lights data.

1.4 Purpose

The main purpose of the thesis is to investigate whether it is possible to forecast future stoppages of manufacturing machines and to identify the most suitable XAIs tool so that the model can assist managers and operators in performing

predictive maintenance.The main challenge of the project consists in building a model that, by using techniques of XAIs, can inspire trust and confidence in theendusers so they can take into consideration its results. It is necessary to mention that the MLs model does not have the purpose of substituting employees, but just of helping them in solving issues on the production line. From an academic perspective the main purpose of the project is to understand which techniques of XAIs are most suitable for a MLs model used in the manufacturing field.

1.5 Goals

The goal of this project is to provide the Nytt company with a starting point for integrating an explainable MLs model within their system.This has been divided into the following three sub-goals:

1. Developing a MLs model that can predict manufacturing machines behaviour.

2. Implementing different techniques of XAIs so that the MLs technology can inspire end-users' confidence and reduce biases.

3. Developing a MLs model that can identify anomalies on the production data and using its results as a XAIs technique based on concrete manufacturing values.

4. Testing the different XAIs techniques and selecting the most suitable for manufacturing setup.

1.6 Research Methodology

This study utilized a combination of analytical [22] and empirical [22] methodologies. Analytical methodologies for model evaluations and empirical ones for user responses examinations.

First, a machine learning model was built and evaluated by using the metrics **accuracy**, **precision**, **recall**, **F1-score** [23] but also **Confusion Matrix**, **ROC AUC**, **PR AUC** and **features importance** [24].

Secondly, four different XAIs techniques were applied in order to explain the MLs model: 1) **explainability summaries** displaying feature importance [15,25], 2) showing **features values** [26] 3) describing the extraction rules

by showing the **model decision** process [27], and 4) showing **past anomalies** preceding the breakdown [13].

Lastly, a usability evaluation [28] was conducted in order to:

- comparethedifferentXAIsmethodsandselectthemostsuitablefor end-users with a strong features' knowledge but not a MLs one;
- obtain user feedback about the utility of model's explanation.

Theevaluationswascarriedoutwithagroupof10usersdividedinthree groups:MLsexperts,featuresexpertswithnoMLsbackground(operators) and features experts with basic MLs knowledge.It is hypothesized that one of themostrelevantresultfromthisstudywillberepresentedbytheincreased capabilityofmanufacturinglineoperatorsindetectingafaultandwhathad caused it [29].

Moreover, the XAI-interface will bridge the gap among the ML predictions and the experts deductions, so that to increase trust and confidence in the model output [30].Furthermore, it is supposed that users will select more than one XAI facilities for obtaining predictions insights [31, 32] and that, for building up their trust, they might want to see raw data and make their own assumptions before applying the model results [13].

1.7 Delimitations

This project will only constitute a starting point for integrating an explainable MLsmodelwithintheNyttmobileapplication,butitwillnotaprovidea completeinteractiveinterface. Indeed,evaluationtestswillbeperformed onlyforobtainingfeedbackaboutthedifferenttechniquesofXAIsbutnot for assessing the whole UXs or designing the users' interface.

1.8 Structure of the thesis

The thesis will have the following structure:

- Chapter2presentsrelevantbackgroundinformationaboutMLsfield, the modern topic of XAIs and theory behind users' tests;
- Chapter3presentstheMLstechniquesusedforbuildingthemodel, theXAIsadoptedfortheresultsexplanationandtheevaluationtests structure;

- Chapter 4 presents the results about model's performances and the analysis about the feedback gathered from the end users about the different XAIs techniques used;
- Chapter 5 presents the discussion about the results obtained, whether the hypothesis formulated at the beginning were confirmed and if an answer to the research question was provided;
- Chapter 6 presents what was concluded from the results and some inspirations for future work.

Chapter 2

Theory behind the methods

This chapter provides an overview of the basic theoretical concepts needed to understand the methodology illustrated in Chapter 3.

2.1 Data

Data is a crucial element in Data Science that needs to be shaped and adapted tothepurposeoftheanalysis.Beforeperforminganyactions,cleaningand formatting are necessary in order to obtain best results from their elaboration.

2.1.1 Time Series

Datainvolvedinthisstudyisrepresentedbytimeseries. Tobegin,itis importanttounderlinethedifferenceamongtimeseriesandsignals.The formerisinformationrelatedtotimeandeachattribute(y)inthecollection is related to time (y = y(t)).Hence, any kind of facts that is measured in time can be represented by this function, such as stock market trends, physiological impulsesormachines'productionpatterns.Thelatterisamoregeneralized version of the previous one, where data does not depend from time but it still canberepresentedwithothertypesofcontinuousfunction,suchasspatial coordinates (y = y(x, y)) or distance from the source (y = y(r)) [33].Moreover, it is relevant to mention that information stored in a TSs dataset are discrete intimeastheyareonlydefinedatspecifictime-steps.Forexample,inthe collection involved in this study, each measure is realized in an interval of one second.

2.1.2 Time series cleaning and preparation

Before proceeding with data processing, information needs to be cleaned and preparedinordertoincreasethethequalityandtoavoidissuesduringthe elaboration.Different measures are available for understanding if the data is of good quality.In the case of TSs, the data can be considered acceptable when it is possible to identify cycles that repeat themselves on time.This property can be defined as data seasonality [34].Another important component to consider inTSsevaluationisthetrend,whichrepresentsthefunctioncourseduring time.Whetherthesetwocomponentsareidentifiable,thentheTSscanbe modeled using the Classical Decomposition Model [34] as follows:

$$X_t = m_t + s_t + Y_t \qquad t = 1, \ldots, n$$

$$\text{where} \quad E(Y_t) = 0, \quad s_t + d = s_t \quad \text{and} \quad \sum_{j=1} s_j = 0 \tag{2.1}$$

In the formula above (from [35]) $X_t$represents the entire time series, m_t is the trend component, s_t is the seasonality and Y_t is the random noise.

2.1.3 Coordinates aggregation formula

In the case features contains 2D or 3D coordinates, they need to be aggregate before feeding MLs algorithms, as they accept only numerical values as input. Hence an aggregation formula was applied in order to gather the coordinates in one number.As it is both additive and multiplicative it allows to keep the coordinates' uniqueness and shape.The formula is shown in the equation 2.2, where "Max_Dig" stands for the maximum number of digits of one coordinate.

$$k = x + (y * 10^{MaxDigX}) + (z * 10^{(MaxDigX+MaxDigY)}) \tag{2.2}$$

2.1.4 Mean and Standard

In order to identify outliers, sample mean $\bar{x}$ and standard deviation σ have been involved.The formula of the standard deviation is represented by equation 2.3.

$$\sigma = \frac{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2}}{n-1} \tag{2.3}$$

where the formula of $\bar{x}$ is expressed by equation 2.4.

$$\bar{x} = \frac{\sum_{i=1}^{n} x_j}{n} \tag{2.4}$$

2.1.5 Principal Components Analysis

Principal Component Analysis PCAs is one of the most common techniques formultivariatestatisticalinspection[36].Itallowstoapproximateadata matrix, X, by considering the product of two small matrices T and P'. In this way the most significant patterns in X are captured and they can be described by plotting the columns of T and the rows of P' [37].PCAs captures the most important data by taking into consideration the attributes with highest values of variance.This metric measures how far each attribute is from the mean value: the higher the distance the more significant a feature will be [38].For instance, if two dimensions of a dataset allow to represent the 95% of the variance, then eliminating the other will cause a loss only of the 5%.Hence, a combination of the most representative data is kept while the rest is discarded.

2.2 Timeseriesanomaliesidentification: Isolation Forest

For the time series anomalies identification Isolation Forest algorithm has been applied.Isolationforest(IF),aswellasRandomforest,isbasedondecision trees.It is an unsupervised model,since there are no predefined label in the training process.It relies on two important features of anomalies:exceptions occurinsmallproportioncomparedtotherestofdataandabnormaldata strongly differs from regular one.
IntheIsolationForestrandomfeaturesareselectedandusedforprocessing datainatree-basedstructure.Ifasamplereachesaleafoftheforestinthe first layers, then it will be considered an anomaly.On the other hand, points that go deep into the forest are less likely to be considered anomalies [11].

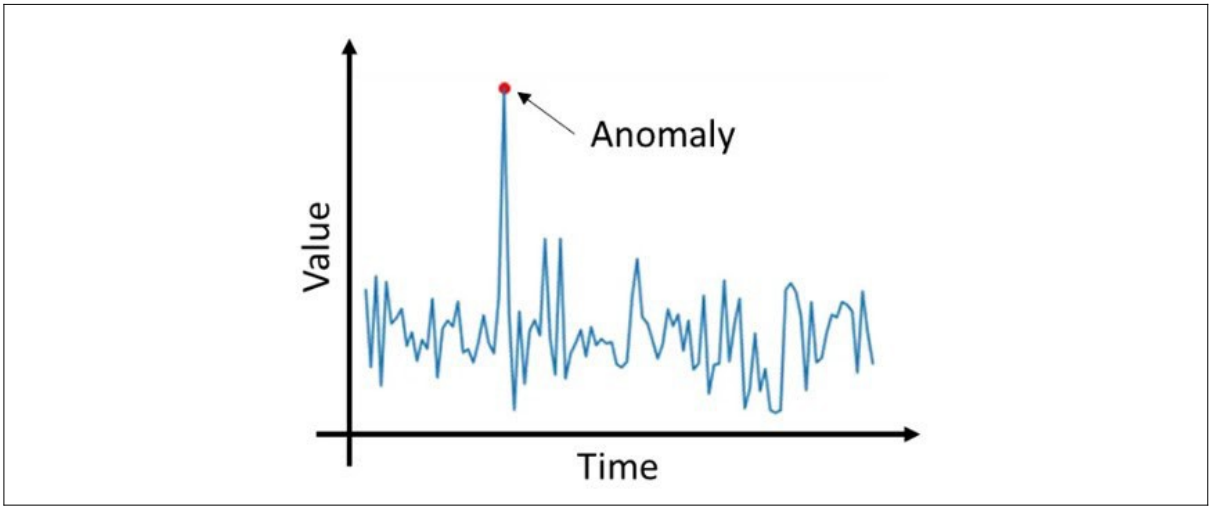

Figure 2.1:Abnormal values in one machine parameter

2.3 Future state classification

Forthefuturemachines'stateclassificationdifferentalgorithmcanbe involved.Ashortdescriptionofeachofthemisprovidedinthefollowing paragraphs.

2.3.1 Logistic Regression

Logistic regression separates the set of points by using a linear function, where the final label is expected to be a value included in an interval of 0 and 1.In this technique a Sigmoid function it is applied in order to avoid values outside of the [0,1] interval [39].

2.3.2 KNN Classifier

In the KNN classifier a record's label is predicted by taking into consideration the known results of its neighbours.In detail, we can consider the following query:

$$(x', y') \in \{(x_i, y_i)\}_{N\ i=1} \tag{2.5}$$

Theclasslabely'isobtainedbyfirstcomputingtheEuclideandistance between x' and each (x_i,y_i):

$$d(x', x_i) = ||x' - x_i|| \tag{2.6}$$

Subsequently all the distances are arranged in ascending order and weights are assigned to the nearest neighbour in the following way:

$$w_i = \frac{1}{d(x', x_i)^2} \quad (2$$

in case two records x_i and x_j have the same KNN rule then the weight is set to 1.In the end the predicted label by finding the most common among the all nearest neighbours:

$$y' = \arg\max \sum w_i d(x = y'_i) \quad (2.8)$$

where X is a class label, $y'_i$is the label of the nearest neighbour among the k nearest neighbours and *d* is the Diradic Delta function which returns 1 if the argument is true or 0 in the opposite case [40].

2.3.3 Decision Tree Classifier

Adecisiontreeclassifierarrangesthedatasetinahierarchicalmodelthat isstructuredasatree:itincludesrootnode,branchesandleafnodes.In order to perform the prediction it compares the independent variable with each internal node, which contains one feature.The comparison's result represents the branch that should be taken in order to reach the leaf node, which contains the label to be predicted [41].

2.3.4 Random Forest classifier

TheRandomForestclassifierworksbuildingseveraldecisiontrees,usually a few hundreds or thousands depending on the problem size, and comparing the results of all the decision tree to pick the most common outcome [43], as briefly shown in the figure 2.3.

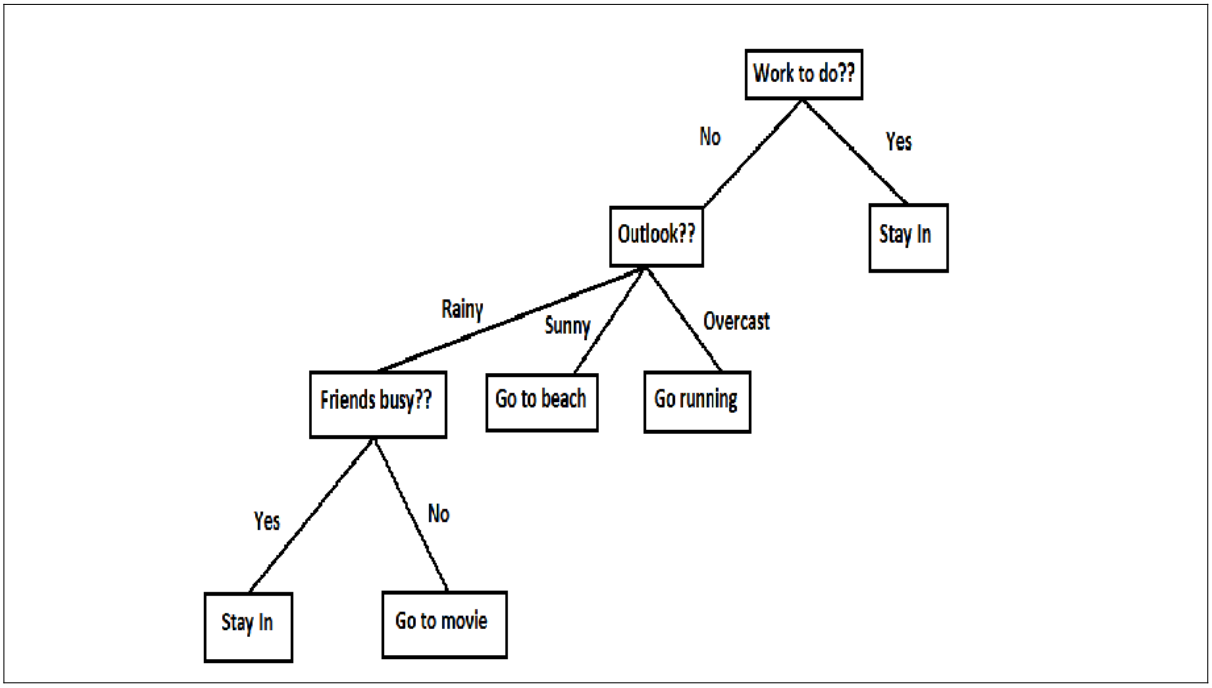

Figure 2.2:Decision Tree example [42]

2.3.5 Neural Network

In this type of classifier, a network based on human brain is built and trained withtheinformationcontainedwithinadata-set. Thebasicprocessing elementsaretheneuronswhichareorganizedinsetscalledlayers.There arealwaysNlayersandLneurons:theinterconnectionsamongbasicunits representthetransitionfunctions.Theentirestructureofinterconnections allows to reach the terminal neurons ($n_{m,j}$) from the source ones ($s_{l,i}$) and hence obtaininganoutputfromapreciseinput.Theinformationalwaysmovesin one direction among all the hidden layers,which are the ones not connected to the external world, without creating any loops [44].Figure 2.4 illustrate an example of Neural Network structure.

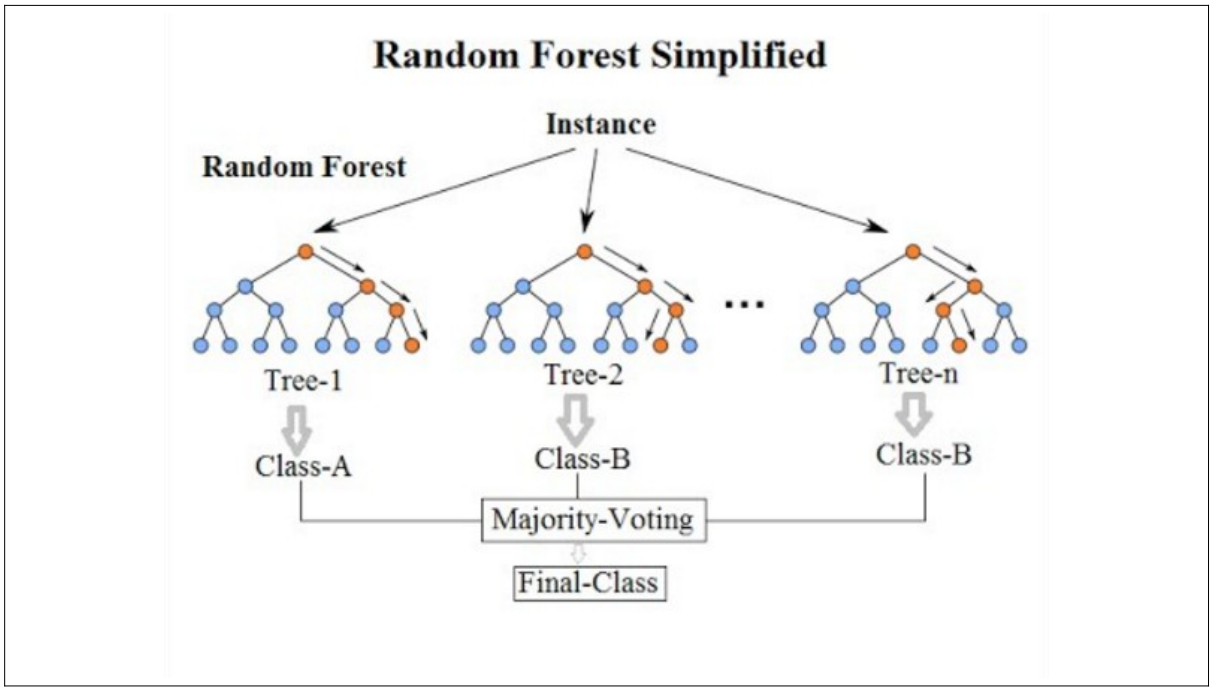

Figure 2.3:Intuitive explanation of the Random Forest model

2.4 Evaluation metrics

Different evaluation metrics have been involved for both the classifier and the anomaly detection validation.A general overview is provided below [23].

2.4.1 Accuracy

Accuracy is defined as the percentage of correct classifications.In particular it is the ratio of TP (true positive) and TN (true negative) cases out of all the predictions (see Equation 2.9).

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{2.9}$$

2.4.2 Precision

Precisionisdefinedasthepercentageofcorrectpositivelabelsclassified. InparticularitistheratioofTP(truepositive)casesoutofallthepositive predictions (see Equation 2.10).

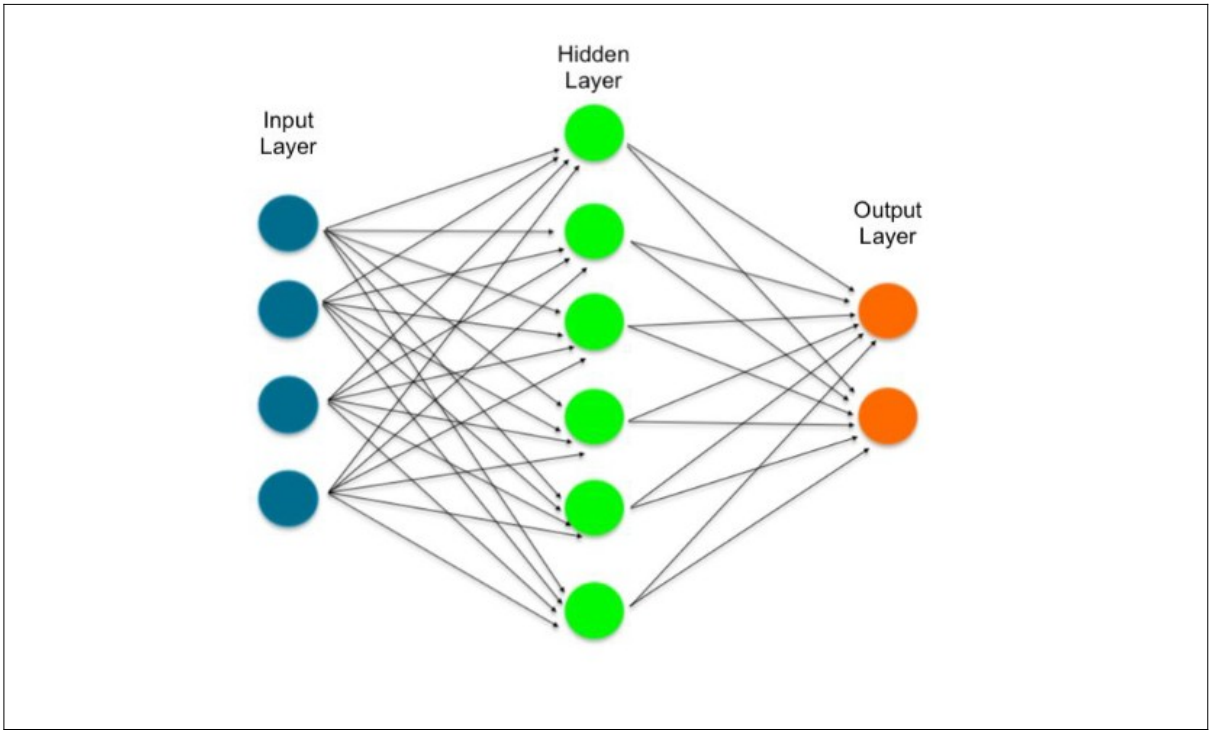

Figure 2.4:Neural Network illustration

$$Precision = \frac{TP}{TP + FP} \tag{2.10}$$

2.4.3 Recall or True Positive Rate

Precision is defined as the sensitivity of the model or True Positive Rate (TPR). In particular it is the ratio of TP (true positives) cases out of all the real, actual positive labels (see Equation 2.11).

$$Recall = \frac{TP}{TP + FN} \tag{2.11}$$

2.4.4 F1 score

F1 score is defined as the combination of Precision and Recall exclude the TN (true negative) cases (see Equation 2.12).It is specially used as an evaluation metric with unbalanced classes.

$$F1Score = 2 * Precision * Recall \tag{2.12}$$

2.4.5 Confusion matrix

Theconfusionmatrixisaatwo-by-twocontingencytablewhererealvalues are displayed as rows and predicted labels are presented as columns (see Figure 2.5).

		Prediction outcome		
		p	n	total
actual value	p′	True Positive	False Negative	P′
	n′	False Positive	True Negative	N′
	total	P	N	

Figure 2.5:Confusion matrix representation [45]

2.4.6 False Positive Rate

Thefalsepositiverate(FPR)isthepercentageofincorrectnegativeclasses classified.In particular it is calculated as the ratio of FP (false positive) cases out of all the actual negative values (see Equation 2.13).

$$FPR = \frac{FP}{TN + FP} \tag{2.13}$$

2.4.7 ROC AUC

TheROCAUCgraph,whichliterallymeansAreaUndertheCurveof theReceiverOperatingCharacteristics,isausefultwodimensionaltool whichrepresentsthetradeoffbetweenthetruepositiverate(TPR)andthe falsepositiverate(FPR)(seeequation2.14tounderstandhowtheareais calculated).

$$\int_0^1 (TPR)d(FPR) \tag{2.14}$$

2.4.8 PR AUC

ThePRAUCgraph,whichliterallymeansAreaUndertheCurveofthe Positive Rate, is an additive two dimensional tool which represents the trade off between the true positive rate (TPR) and the whole set of labels predicted as positive (P) (see equation 2.15 to understand how the area is calculated).It is used to clarify what is displayed in the ROC AUC graph.

$$\int_0^1 (TPR)d(P) \tag{2.15}$$

2.4.9 Features importance

Feature importance study allows to understand which feature/attributes had the highest impact on the predicted label.The importance is calculated by using theShapleymethod,wherethemodelistrainedmultipletimeswithholding onefeature.Inparticular,foreachiterationafeature*i*isisolatedfromthe setofallfeaturesF,inordertoobtainthesubsetS.Tocomputetheimpact of *i* on the prediction, a model $f_{S\cup\{i\}}$including the feature is trained together withanotheronefsexcluding*i*.Subsequently,acomparisonisperformed between the first model's results $f_{S\cup\{i\}}(x_{S\cup\{i\}})$ and the second one $f_S(x_S)$, where x represents the input record of the model.This operation is repeated for all thepossiblefeaturesindataset.Attheendoftheprocess,theimportance value ϕ_i of feature *i* is calculate using the equation 2.16).

$$\phi_i = \frac{\sum_{S\subset F/\{i\}} |S|!(|F|-|S| \quad f_{S\cup\{i\}}(x_{S\cup\{i\}}) - f_S(x_S)}{|F|!} \tag{2.16}$$

2.4.9.1 Human reasoning techniques

These XAIs techniques are intended to be based on the way humans think.In other words, the explanations for the MLs predictions should be similar to the ones involved by the human brain for making a prediction.An XAIs interface based on this technique should thereby explain the prediction as the indented user of the interface would arrive at the prediction.For example, in a study of Wang et al [13], physicians were comparing the predictions with clinical data in order to justify the diagnosis produced by the MLs model.This technique requires the end user to have domain specific knowledge (e.g.operators and managers in a manufacturing setting).

Below,thethreemainmethodologiesinvolvedinthehumanreasoning technique for explaining predictions to MLs non expert users, are described.

- •Bayesian probability:This methodology is based on the Bayes theorem which states that the probability of an event is related to the likelihood ofpreviousobservedconditionsthatdeterminetheexaminedevent [46].Thismethodsupportend-usersbyemphasizingwhicharethe

predominantevents,andwhichfeaturesinfluencedthefinaloutcome (See figure 2.6).

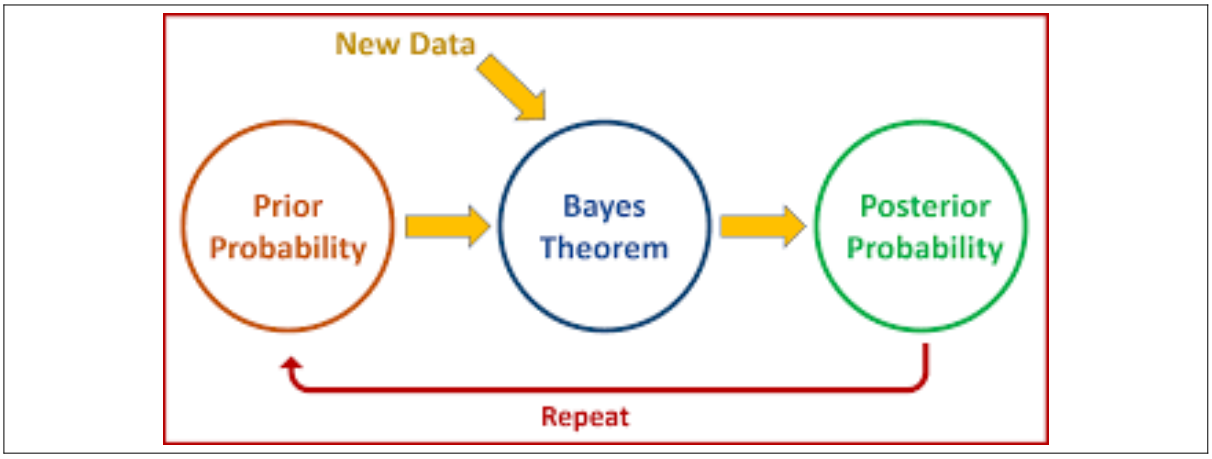

Figure 2.6:Bayesian Probability

•SimilarityModeling: Thismethodologyisbasedonthefactthat peoplebasetheirknowledgeongeneralconcepts,andforsolving newproblems,theyseeksimilaritiesintheirpastexperiences.Inthe same way,supervised MLsmodels,such as classifiers,distance-based methods, or dimensionality reduction, are built on previously seen data (training set) which provides the model with the structure of a general concept. Analogicalandinductivereasoningareinvolvedinthese mechanisms to clarify the outcomes of certain predictions.For instance, when operators discover new failures in manufacturing machines, they attempt to explain them by seeking similarities in the breakdowns that occurred in the past (See figure 2.7).

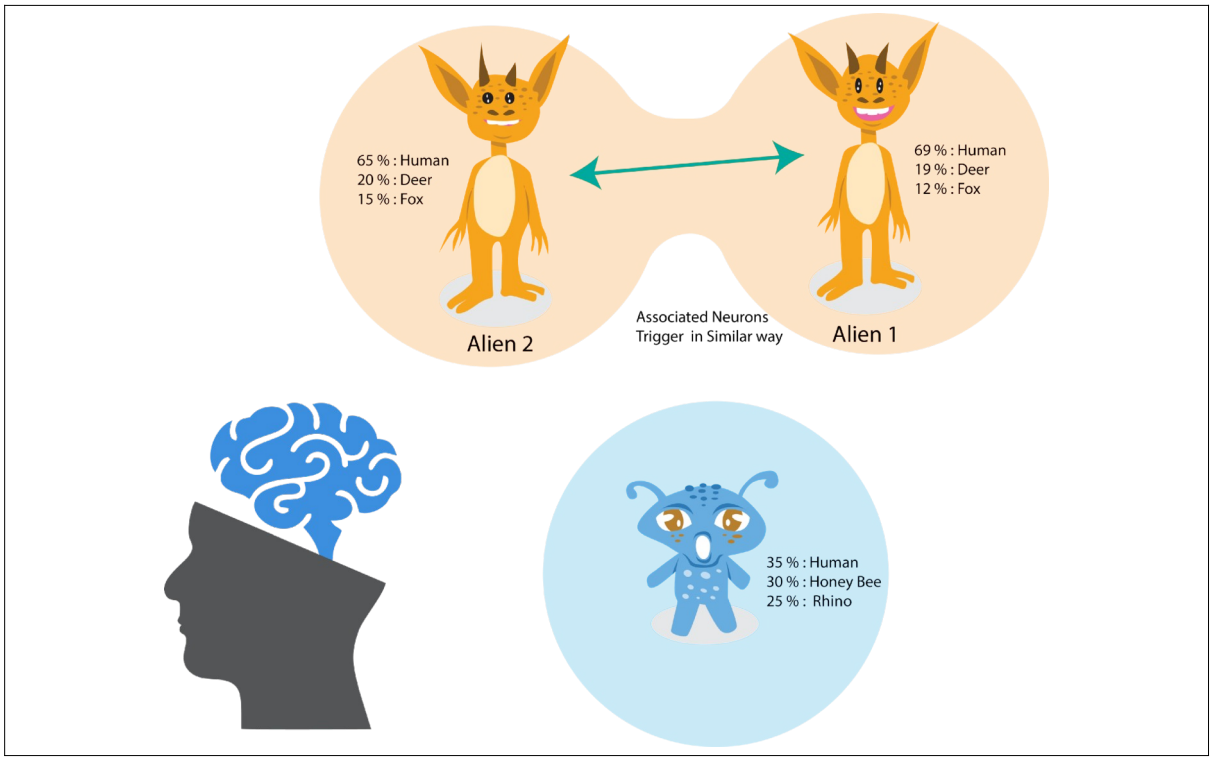

Figure 2.7:Similarity models

•IntelligibilityQueries:Thismethodologyisbasedonusingdifferent queries that end-users may want to ask a MLs model for building trust andconfidencetowardsthesmartsystem.AstudyconductedbyLim and Dey [47] identified different questions that people might pose to the system,regardingthestate(Inputs,WhatOutput,WhatElseOutputs, Certainty)andtheunderlyingprocessforthepredictions'production (Why,Why Not,What If,How To).They reached the conclusion that non-expertsubjectsuseawidevarietyofintelligibilityquerieswhich depends also on the context where MLs is involved (See figure 2.8).

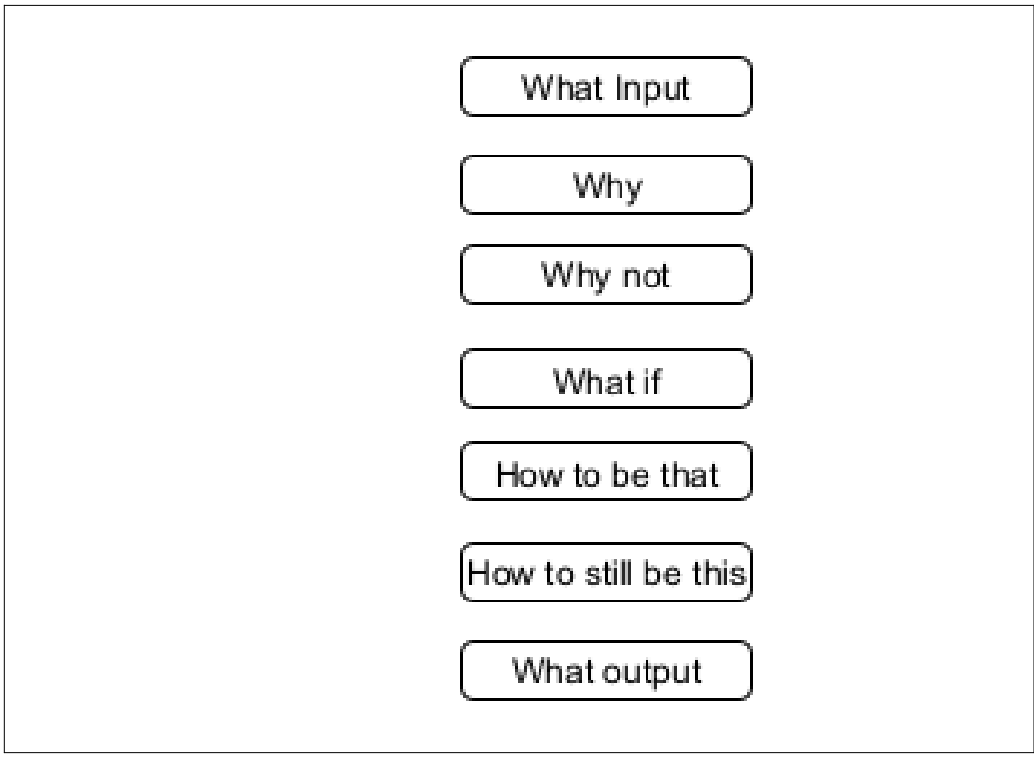

Figure 2.8:Intelligibility queries

2.4.9.2 Representing explanations

This technique utilizes different visual blocks provided to end-users in order to provide them a higher level of interpretability and transparency for the MLs predictions.Studies by Caruana and Krause [48, 49] identified that the most adopted tools for XAIs are features importance or influence, which emphasize the impact of the different factors on the final predictions.Other approaches includerepresentinginstancesfromthetrainingdata,prototypes,criticism, and counter-examples [50, 51].In addition,XAI elements may comprehend theillustrationoffeaturesthresholdorthedescriptionofinputandoutput values. Thefollowinglistwilldefinethedifferentmethodologiesforpresenting explanations:

•Explainabilitysummaries(Seefigure2.9): Thismethodologyfirst provides an high-level overview of the model structure and the process behind the output production.Then lists and visualization are used for answering a set of questions that non-expert users usually ask for gaining

trust and confidence within the model.The study conducted by Petkovic etal.[15]showedthatthesequeriesregardthefeaturesranking,the **trade-off in accuarcy when dropping some of the input factors and the** direction of the features.

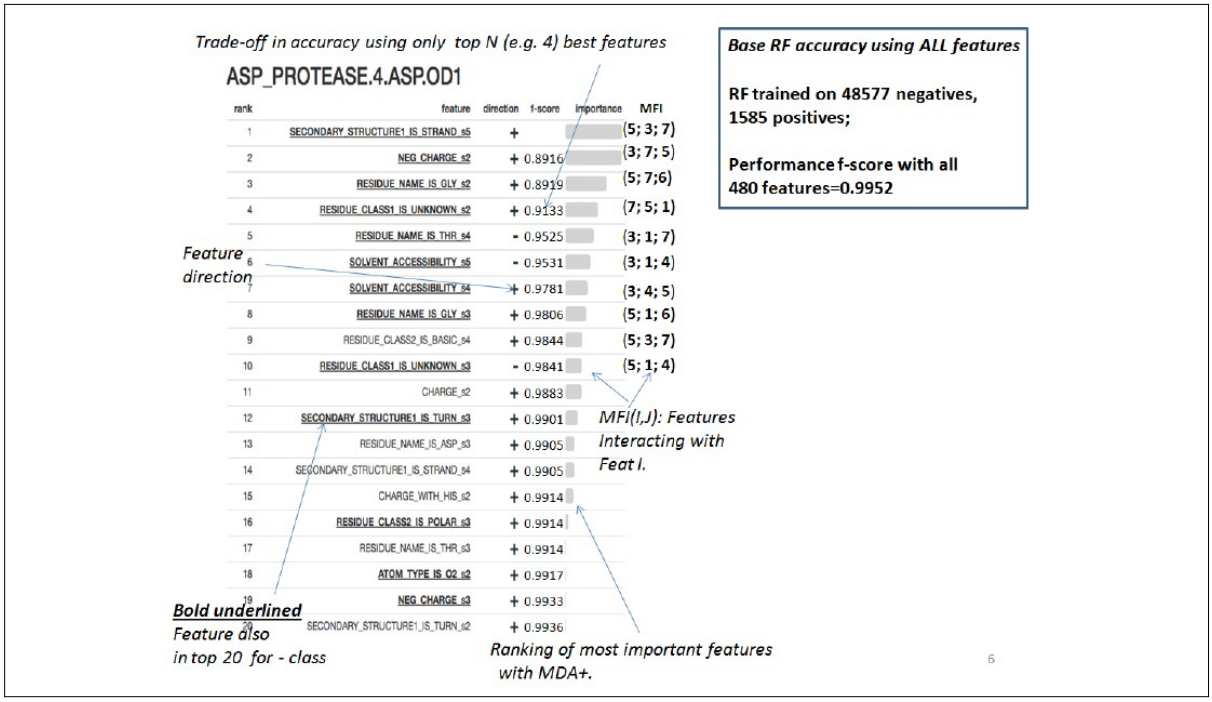

Figure 2.9:Explainability summaries example

•Visualizations(seefigure2.10): asanalternativeoftextandlist visualizationcanbeinvolvedforexplaininginaeasierwaycomplex concepts.Representationsmightregardrawdata,themodelstructure orfeaturesrelationship.Forexample,innaturallanguageprocessing highlights of the words are provided or in image recognition colors are used for identifying pixels that influenced the classification.Charts and graphs, on the other hand, are used in predictions models for illustrating similaritiesamongthedifferentattributes,partialdependenciesofthe final outcome from the input features [48, 49] or how the sensitivity of the model changes under some specific conditions.

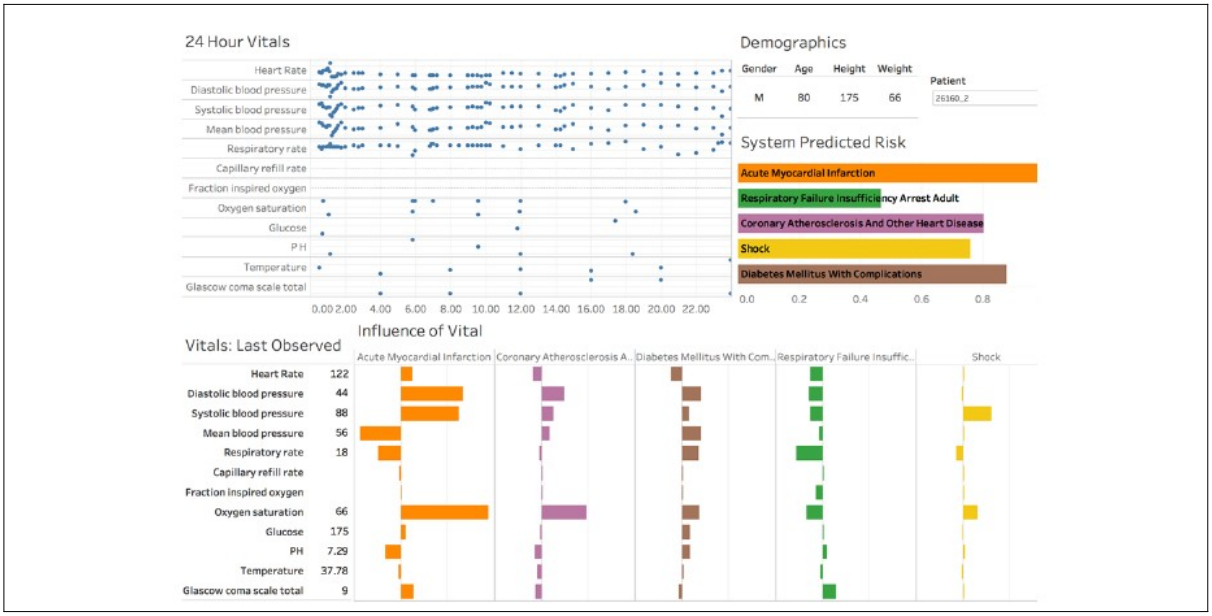

Figure 2.10:Visualizations examples

•DataStructures(seefigure2.11):thismethodologyimpliestheusage ofdifferentdatastructuresthatareusuallytaughtinundergraduate computer science courses, in order to clarify the MLs model's operating process even to non-expert users.The simplest structure for providing explanations is with lists.For example, classification classes are always insertedintolist[49,48].Ontheotherhand,decisiontreesareused forrepresentingpredictingrulesandgraphsforcomplexrelationships or conceptual models among training data.

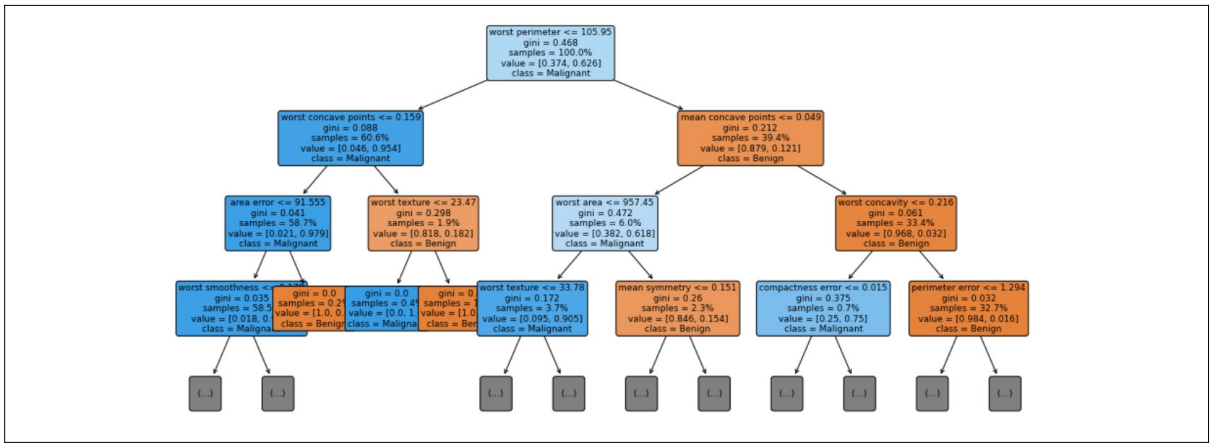

Figure 2.11:Data structure example

•Anomalousvaluesprecedingaprediction(seefigure2.12): this techniqueisanexperimentaloneandisbasedontheresultsofthe study of Wang from 2019 [13], which demonstrated how humans tend to examine previous anomalous values in order to justify a prediction. InTimeSeriesforexample,whichisthedatatypeinvolvedinthis study (see Section 2.1.1), values repeat themselves cyclically, therefore anomalies are represented by figures not aligned with the average.

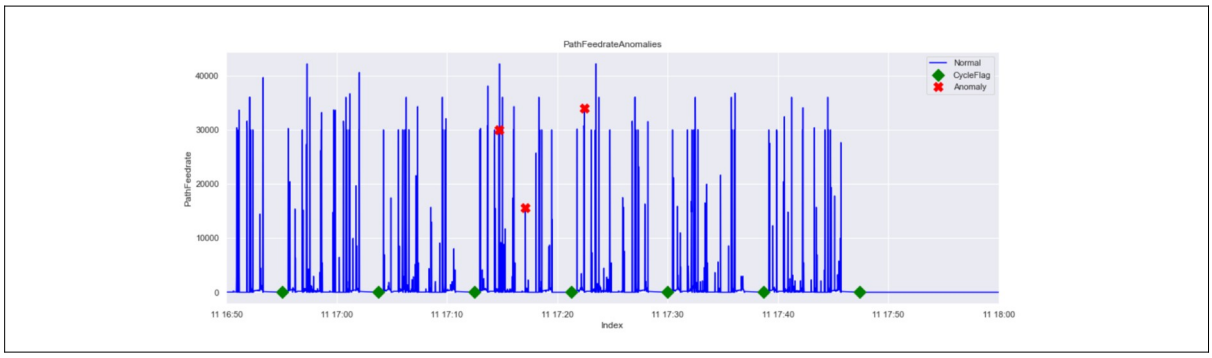

Figure 2.12:Anomalies example

2.4.10 Cognitive metrics for explainable AI

AstudybyHsiaoetal.identifieddifferentcognitivemetricsforevaluating XAIstechniques[52].Theyareconsideredusefulforobtainingiftheuser hasreachedapragmaticunderstandingoftheAIsystem.Belowageneral description of each of them is provided.

- Explanation Goodness:this metric refers to the explanation's precision and clarity in providing a description of how the MLs model produced thefinalresults.Itisimportanttomentionthatgoodnessisstrongly context-dependent which means that it varies on user's perception.For an explanation to be considered sufficient the intersection of amount of detailandlevelofcomprehensibilityisrequired.Differentqualitative methodologiescanbeadoptedformeasuringthegoodnessofan explanation, such as interviews and questionnaires with a Likert scale.

- User satisfaction:this metric refers to the level of fulfillment users are experiencingwhenseeinganexplanation.Indeed,wheneverpeople interacts with an explanation, they go through different mental processes that determine the satisfaction degree.In detail,the necessary criteria forobtainingsatisfactionisthatsubjectsunderstandreasonsbehinda specific phenomena.However, there are also cognitive biases that may influence the perception and the preference about a certain explanation. Theassessmentofusersatisfactioncanbedonebyusingsubjective measures, such as the Explanation Satisfaction Scale.Satisfaction and goodness are particularly related with each other.Indeed, understanding theclarityofanexplanationnotonlyincreasesgoodnessbutuser's gratificationaswell.Yet,thefirstcanbeassociatedtoperceptionof external stimuli, while the second one concerns the user's internal state.

- User Curiosity/Attention Engagement:this metric helps the practitioner inassessingthelevelofinformation-seekingandactivityoftheuser. Curiosityariseswheneversubjectsexperienceatheorygapandthey trytofindanexplanationinstinctively.Anhigherlevelofcuriosity corresponds to a major interest from the user towards the explanation. AcuriositychekclistisprovidedbyHoffmanetal.(2018)[53]asa subjectivemeasureandsuggestedthatcuriosityshouldbeevaluated every time users seek for clarifications.

- User Trust/Reliance:people usually tend to consider a system reliable whenevertheytrustit.Hence,thismetricprovideanoverviewabout howmuchtheXAIstoolhelptheminbuildingconfidencetowards theMLsmodeloutcomes.Forexample,ifanAIssystemproduces impractical and not intuitive predictions, people are not going to trust the results even though they are correct.In order to measure users' reliance, subjectivescalescanbeadopted,suchtheLikertscaleproposed byHoffmanetal. (2018)[53]. Basicallythisinstrumentgather

informationaboutfouraspectsoftheXAIsystem: predictability, reliability, efficiency, and believability.

•Userunderstanding: thismetricreferstothecompletenessofthe mental model users have of the AIs system.It can be divided in three typologies:global,localandfunctionalunderstanding.Thefirstone isrelatedtothegeneralperceptionapersonhasofasystem. On theotherhand,thesecondonerefersjusttoaspecificelementofa system.Finally,thethirdoneisconnectedwiththeunderstandingof asystem'sfunctionsanduses.Apossiblemethodforevaluatinguser understanding might be represented by a qualitative questionnaire which asksparticipantsthedescribetheanalyzedsystem[53].Ontheother hand,theunderstandingcanbeassessedgradually,throughagrasp-ability test [12].

•UserPerformance/Productivity:thismetricreferstosubjectseffec-tivenessandefficiencyinusingtheAIssystemafterhavingseen explanations.Inthisway,itpossibletodefinealearningcurvefor each user.Potential methodologies for assessing user performance are usabilitytest,wherepractitionerscancomparethenumberoftrials with the number of successful cases where users understand the reasons behind the prediction [53].

•System Controllability/Interactivity: this metric evaluates the simplicity ofusersininteractingwiththeXAIssystem. Inordertoinspire willingness in using an explanation, a goal might be building it flexible, context-awareandtailored.SystemControllabilitycanbemeasured through questionnaires or other qualitative methods.These may include probes regarding explanations' likeability and competence.

Chapter 3

Methods

Thischapterisdividedinthreemainsections:thefirstonepresentsan overviewoftheresearchprocessappliedinthisproject,thesecondone providesadescriptionofthemethodologyusedintheAIsandXAIs development phase, and the third one describes the user study.

3.1 Research Process

As stated before the research has been divided in three main phases, iwth each phase featuring multiple steps (see Figure 3.1).Each step can be categorized either as simple or complex, in case it includes additional blocks.

3.1.1 Classifier realization

The first phase is the ”classifier realization”, as seen in 3.2.This phase consists of three steps.

3.1.1.1 Data cleaning and preparation

First,thedatawascleanedandpreparedforthetrainingoffivedifferent modelsbasedonfivealgorithms(seeSection2.3).Thisphaseincludesan additional complex box, which represents the set of steps needed to create the”Previoustool’sanomaliesexplanation”.Indeed,unliketheotherinterface, this one involved a further MLs model development for identifying anomalies (see Section 2.2) in the machines’ analytical data.Finally,the third phase is represented by a complex block including all the steps for carrying out the user evaluation.All the code was developed in Python since it offers a wide variety of libraries for artificial intelligence.

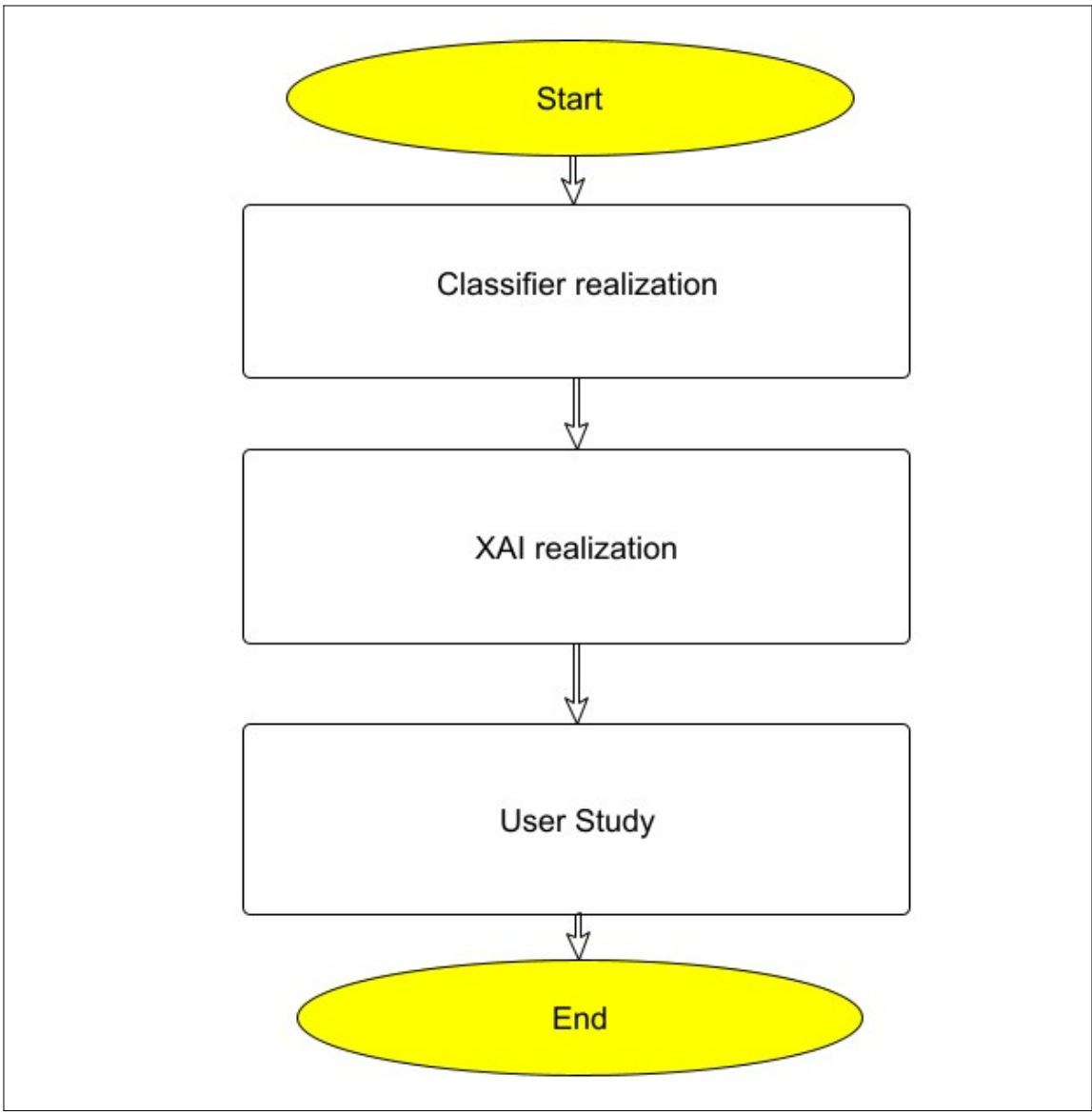

Figure 3.1: Research Process: figure 3.2, 3.3, 3.4 elaborates the complex boxes

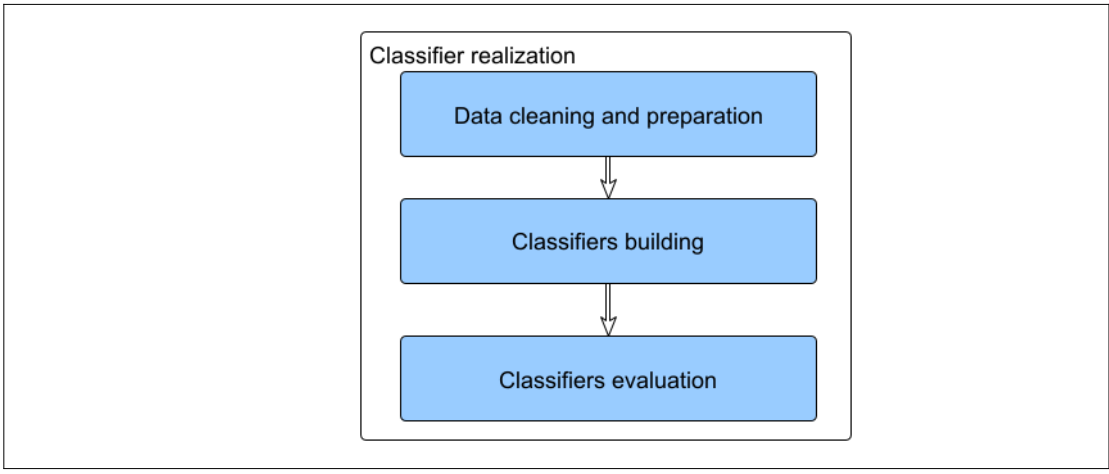

Figure 3.2:Classifier realization

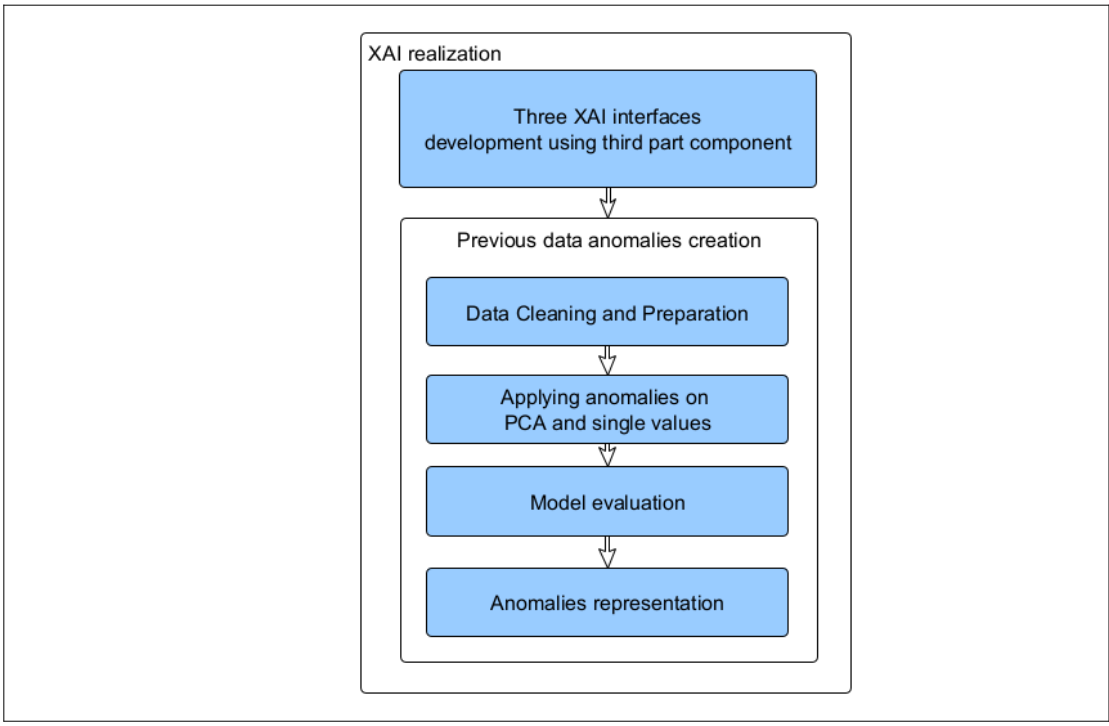

Figure 3.3:XAIs realization

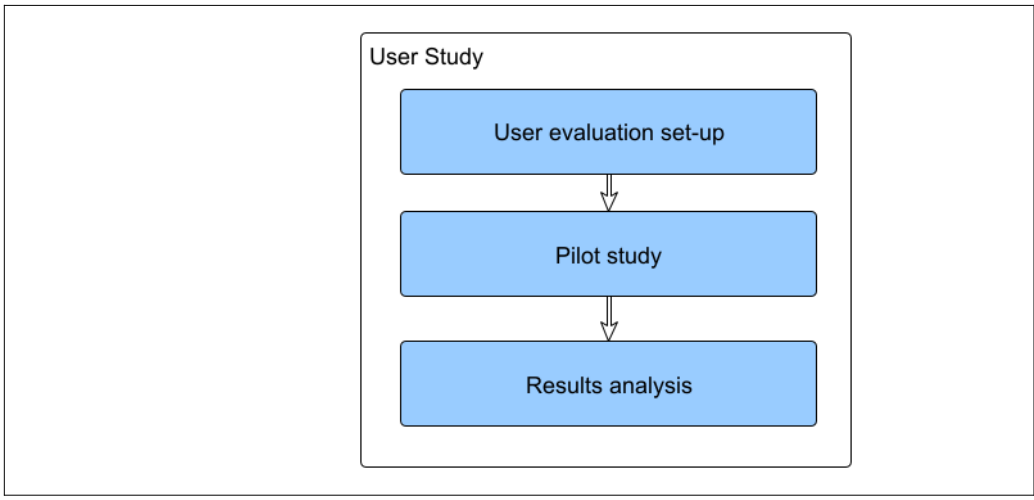

Figure 3.4:User evaluation

3.2 Classifier realization

Thissectionsdescribesthesetsofmethodologiesappliedfortheclassifier implementation.It is important to remind that the role of the classifier is the oneofpredictingthestateofthemachineinthefutureproductioncycle.In Appendix C it is possible to find the code implemented for accomplishing this operation.

3.2.1 Data collections

Twodatacollectionshavebeeninvolvedintheclassifierdevelopment:the former contains data regarding the stack light indicators, and the latter stores analytical data gathered through MTconnect (see Section 1.3.2).It is important to mention that all the data sets involved, collect real manufacturing machines'information.At the end, the models built on the different data collections were compared in order to find the best performing one.

3.2.1.1 Stack lights data

Thisdatasetstoredinformationaboutthemachinestaterepresentedbythe stacklightandithadasizeof125kilobytes.Duetothelowperformances of the model (see Section 4) only machine 50's data were considered for the stack lights.Each record contained 10 features:table 3.1 shows the shape of the data collection.

Name	**Description**
LogID	Record identifier
Duration	Duration of the
DurationLogId	Identifier for the
Color	Stack lights color
Accuracy	Accuracy of the light
OriginalTime	Timestampofthedetection(itkeepstraceofyea
InsertTime	Timestamp of the
MachineID	Machine's identifier
UserID	Identifieroftheoperatorwhoisworkingonthat

Table 3.1:Dataset stack ligth data

3.2.1.2 MTconnect data

Thisdatasetstoresmachines'analyticalinformationintheshapeoftime series(seeSection2.1.1).Twodatacollectionswereinvolved:onewitha sizeof51.5megabytesstoringmachine50dataandtheotherwithasizeof 31megabytesstoringmachine54data.Indetail,eachrecordscontained17 features regarding machines' production parameters measured per each second (see table 3.2).

Name	Description
FactoryID	Factory's identifier
MachineID	Production machine's
State	Current state of the
ControllerMode	It indicates the mode
EmergencyStop	Description of the last
ToolId	Machine's tool that is
Line	Production line
RotatoryLoad	Pressure applied to the
PathFeedRate	Thespeedatwhichthec utterengagesthepart,
PathPosition	Cutter location on the
RotatoryActiveVelo	Rotational speed of
RotatoryOverrideVel o	Binaryflagwhichindic atesiftheoperatormani
Xload	Feeding work into the
Yload	Feeding work into the
Zload	Feeding work into the
Partcount	Binaryflagtocommuni catewhenaparthasbee
OriginalTime	Timestampofthedetect ion(itkeepstraceofyear

Table 3.2:Dataset MTconnect

3.2.2 Data cleaning and preparation

Before starting with the MLs model implementation, it was necessary to clean andpreparedata.Having*dirty*datausuallyresultsinunreliablepredictions which might create serious consequences.Issues such as duplicates, missing values,integrityconstraintsviolations,andoutliers[54]havebeensolved

applyingdifferentmethodologies. *Human-guided*techniques[55]were employed for duplicated records removal,empty attributes deletion and data formatting[56].Ontheotherhand,one*automated*techniquewasusedfor missing values retrieval [57]:the k-Nearest Neighbor (KNN) algorithm which replacesthemissingdatabylookingtothekclosestneighborswithanot nullvalue(seeSection2.3.2).Thechoiceofthistechniquewasledbythe study conducted by Zhang in 2012, where it is shown that the KNN algorithm isgenerallyefficientwithnumericalvalues[58]. Thelaststepinvolved coordinates mapping.Some of the features are stored as tuple of coordinates, suchasthePathpositionandtheLoad. Thesevalueswereaggregating throughaggregationformula(seeSection2.1.3).Inbothdatasets,afterthe cleaningprocedure,dataweregroupedperiodically.Inordertoachievethis goal, attributes were grouped using different aggregation formula, in particular mean values.In the stack lights dataset, where values were stored each time themachines'statewaschanging,therewastheneedofgatheringdataper hours in order to make them homogeneous and cyclic.On the other hand in theMTconnectdataset,therewastheneedofgroupingthemforproduction cycles.Indeed,thedatasetshapewasmanipulatedinsuchawaythateach recordcorrespondedtopartproducedbythemachine.Inaddition,columns storingthetotalrunning,stopping,readyandinterruptiontimewereadded tothedataset.Subsequently,acolumncalled"Prediction"containingthe next production cycle state was created:it stored the label which needed to be predictedbytheclassifier(1ifthemachineisstoppedand0ifthemachine isactive).Tables3.4and3.3representrespectivelytheshapeofthestack lights and of the MTconnect data sets.These operations were all implemented throughacombinationof"Pandas"and"Scikit-learn"librariesavailablein Python [59, 60].

Name	Description
CumulativePartCount	Part’s identifier
Running	Running time of the machine during the
Ready	Readytimeofthemachineduringtheproductioncy
Stopped	Stopping time of the machine during the
Interrupted	Interruption time of the machine during the
RotatoryLoad	Pressure applied to the
PathFeedRate	The speed at which the
RotatoryActiveVelo	Rotational speed of the
Load	Feeding work into the
PathPosition	Cutter location on the
EmergencyStop	Description of the last
Partcount	Binary flag to
CombRotatory	Combination of Rotatory
State	Current state of the
Prediction	Label to be predicted
OriginalTime	Timestampoftheproductioncyclestartingpoint(i

Table 3.3:Preprocessed MTconnect dataset

Name	Description
OriginalTime	Timestamprepresentingthehour(itkeepstrace
Running	Running time of the machine during the
Ready	Readytimeofthemachineduringtheproductio
Stopped	Stopping time of the machine during the
Interrupted	Interruption time of the machine during
ProdCycles	Number of production
PartsProduced	Number of parts
CorrectPartsUntilThen	Cumalativecounterwhichkeepstraceofthetot
State	Current state of the
Prediction	Label to be predicted

Table 3.4:Preprocessed stack lights dataset

3.2.3 EDA exploratory data analysis

The purpose of the EDA (Exploratory Data Analysis) was to explore the data structure in order to find the prediction labels distribution and to understand the correlation among features.Indeed, strongly correlated features can lead to misleading results especially in regression models.Hence, as we compared differentmodelsthiscouldhavebiasedtheperformancesevaluation.The correlationwasdisplayedusingacorrelationmatrixandtheattributeswith either high positive or negative correlation were discarded (see Appendix A). Subsequentlyclassbalancingwaspreformed.Itisimportanttoremindthat bothdatasetsincludetwoclasses(1ifthemachineisstoppedand0ifthe machineisactive).Classbalancingwasnecessarytohaveamodelwhich generalized well among all possible classes.At the end of the process, positive (1)andnegative(0)labelswereequallydistributed.Theseoperationswere implemented using the SMOTE method from the ”imbalanced-learn” library available in Python [61].

3.2.4 Model building

Duringthephaseofmodelbuilding,thedatacollectionsweresplitintwo sub sets:the training set corresponding to the 75% of the whole size and the testingsetcorrespondingtothe25%.Thechoiceoflargetestingsetswas madeinordertoallowthemodeltoperformcorrectpredictionsevenwith complexrecordsasinput.Aftertheoperationofdata-splitting,parameters Cross Validation was performed.In this techniques each model is tested with differentparametersinordertofindtheoptimalonesforthespecificcase.
However, as the validation is not performed on the real test set (see Figure 3.5, duringtheevaluationitwillbenecessarytoconsideralsothemodeltrained with standard parameters.

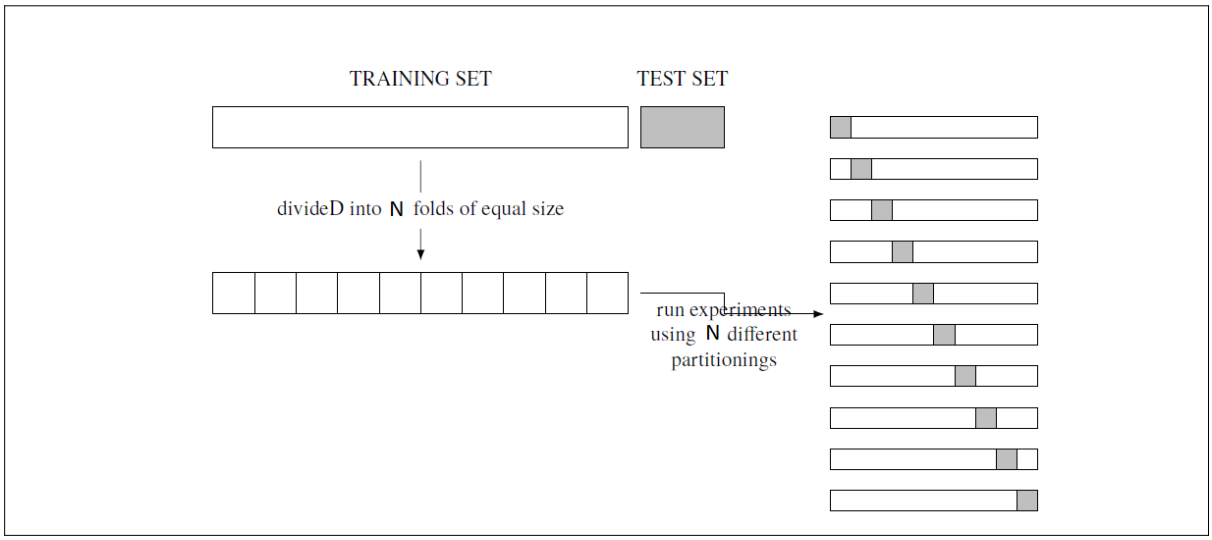

Figure 3.5:Train-Test set vs.parameters Cross Validation [45]

Five different typologies of models based on five algorithms respectively weretrainedandtestedusingbothstandardandoptimalparameters.The various algorithms involved in the construction were:

- **Logistic Regression**:this algorithm separates the set of points by using a linear function (see Section 2.3.1).

- **KNNClassifier**:inthisalgorithmtherecord'slabelispredictedby takingintoconsiderationtheknownresultsofitsneighbours(see Section 2.3.1).

- **DecisionTreeClassifier**:thisalgorithmarrangesthedatasetina hierarchical model that is structured as a tree (see Section 2.3.3).

- **NeuralNetworkClassifier**:inthisalgorithmanetworkbasedon human brain is built and trained with the information contained within a dataset (see Section 2.3.5).
- **RandomForestClassifier**: thisalgorithmworksbuildingseveral decision trees (see Section 2.3.4).

The model building was realized using methods available in the ”Scikit-learn”library [59].

3.2.5 Model evaluation

Themodelevaluationwasdividedintwosteps:firstglobalmetricswere applied on the five models, then on the best performing model more detailed measures were applied.Four standard metrics were involved in the first phase: **Accuracy**, **Precision**, **Recall**, **F-score**.On the other hand, during the second phase,**confusionmatrix**,**roc-auc**,**pr-auc**and**featureimportance**were used in order to perform a complete model analysis (see Section 2.4).These operationswereimplementedusingfunctionfromthe”Scikit-learn”library [59].

3.3 XAI realization

ThefollowingsectiondescribeshowtheXAIstoolwasdesignedand implemented.Thetoolconsistsoffourinterfaces,eachonebuiltoneofthe methods described in section **??**.
Alltheexplanationswerereferredtoonepredictionaboutthemachine behaviourinthemonthofMarch.Theinputfeatureswererepresentedby theinformationoftheproductioncycleprecedingtheonepredicted. For therealizationofthefourXAIsinterfaces,differentmethodologieswere involved:forfeatures’impact,features’valuesandtreegraphvisualization, the necessary data were obtained during the classifier building and evaluation. On the other hand, for the anomalies visualization it was necessary to perform an anomalies detection of the MTconnect time series (see Section 3.2.1.2).

3.3.1 Features’impact, features’valuesandtree graph visualization

ThreeexplainableAIsinterfacewerebuiltusingdataproducedduringthe classifierrealizationandwiththesupportofaPythonthirdpartcomponent

explaierdashboard [62].

Thefirstinterfacerepresentedthefeatureimpactonthefinalprediction. Thishasbeenimplementedusingabarchartgraphwhichshowedthe importance values, calculated with the SHAPLEY method (see Section 2.4.9), inanascendingorder. Inaddition,themodelaccuracyisalsodisplayed (seeSection2.4.1).Thesecondinterface,instead,presentedinputfeatures'valuesc omparedtotheiroverallaverageinnormalworkingcondition.In particular, only the attributes highly distant from the average were visualized using an horizontal bar chart.The third interface,on the other hand showed theasimplifiedversionofmodel´sdecisionprocess. Indetail,sincethe RandomForestincludesNdecisiontreewithinitsstructure(seeSection 2.3.4),averticalbargraphshowedthenumberofdecisiontreespredicting astoppage(positivelabel)againsttheonespredictinganormalbehaviour (negative label).Moreover, web page rendering has been performed in order topresenttheexplanationsinauserfriendlyway,byusinganotherPython third part component [63].

3.3.2 Anomalies in time series

The fourth explainable AIs interface was based on time series analysis of the MTconnectdata(seeSection3.2.1.2).Itisimportanttomentionthatthis modelwasalsotestedwithotherdatacollectionswithsizeof262and247 megabytes,respectivelyformachine50andmachine54.Inordertofind outliersinthedataset,ananomaliesidentificationmodelwasbuilt.The modelwasappliedoneachoneoftheworkingmachineparameters(Load, RotatoryLoad,RotatoryActVelo,PathFeedrate,PathPosition)andontheir combinationcalculatedthroughPCAtechnique(seeSection2.1.5).Allthe operationsdescribedbelowwererealizedwiththeaidofthe"Scikit-learn"library[59].InAppendixCitispossibletofindthecodeimplementedfor accomplishing this operation.

3.3.2.1 Data cleaning and preparation

Thecleaningprocedurefortheanomaliesidentificationintimeserieswas brief compared to the one performed during the classifier realization.Indeed, pre-definedfunctionscreatedinSection3.2.2wereusedforpre-processing theMTconnectdataset.Asunderlinedinsection2.1.2,therandomnoise needs to be zero so a cleaning procedure needs to be applied.*Human-guided* and*automated*techniques[55]canbeappliedfordatascrubbing.Thefirst ones required deep human observation while the second ones involved usage

ofautomatedalgorithms.Oneexamplewasremovingnotactivesequences, whichcanberealizedwitha*human-guided*approach:detecttimeperiods where there is no data collected and then discard them.However, during this task,valueswerenotgroupedperproductioncyclesinceitwasnecessary tomaintainthetimeseriesshape,butaflagrepresentingthebeginningand theendofthecyclewaskept(seeAppendixB).Furthermore,anattribute containing the combination of the machine production parameters was created using the PCA method and selecting the most significant principal component (see Section 2.1.5).At the end of the process values were split in N data set, where N is the production cycle's length in seconds.In particular, production cycles were separated in N chunks, each one with a length of 1 second.Then, values belonging to different production cycles but occupying the same time position, were saved in the same data set.This technique allowed to have data aligned in a such a way that it was possible to identify anomalies only among values with similar temporal characteristics.

3.3.2.2 Applying anomalies on PCA and single production parameters

Before starting with the model building, it was necessary to generate anomalies labelstocomparepredictionsresults.

Indeed,MTconnectdatasetwas notstoringflagsindicatingabnormalvaluestoassessthefinaloutcome. AsimilarmethodtotheoneofHochenbaumetal.[64]wasadoptedfor labelsgeneration:allvaluesfourtimesgreaterthenthemeanplusfour timesthestandarddeviation(seeSection2.1.4)orlowerthemeanminus fourtimesthestandarddeviationwerelabeledasanomalous(seeEquation 3.1). Itisimportanttomentionthattheflag1representedananomalies while0representednormalvalues.Thistechniqueswasappliedtoallthe production parameters (Load, RotatoryLoad, RotatoryActVelo, PathFeedrate, PathPosition) and to the PCA values.In addition, labels generation allowed to calculate the outliers fraction, that was passed as parameter during the isolation foresttraining.Indetail,theoutliersfractionrepresentsthepercentageof anomalies expected to be within the data set.

$$\begin{aligned} lowThreshold &= \bar{x} - 4 * \sigma \\ upThreshold &= \bar{x} + 4 * \sigma \\ if(x > upThreshold | x < lowThreshold) &\Rightarrow label = 1 \end{aligned} \tag{3.1}$$

The model used for anomalies identification was the Isolation Forest (see Section 2.2).It was trained with 75% of the data and tested on the 25%.At

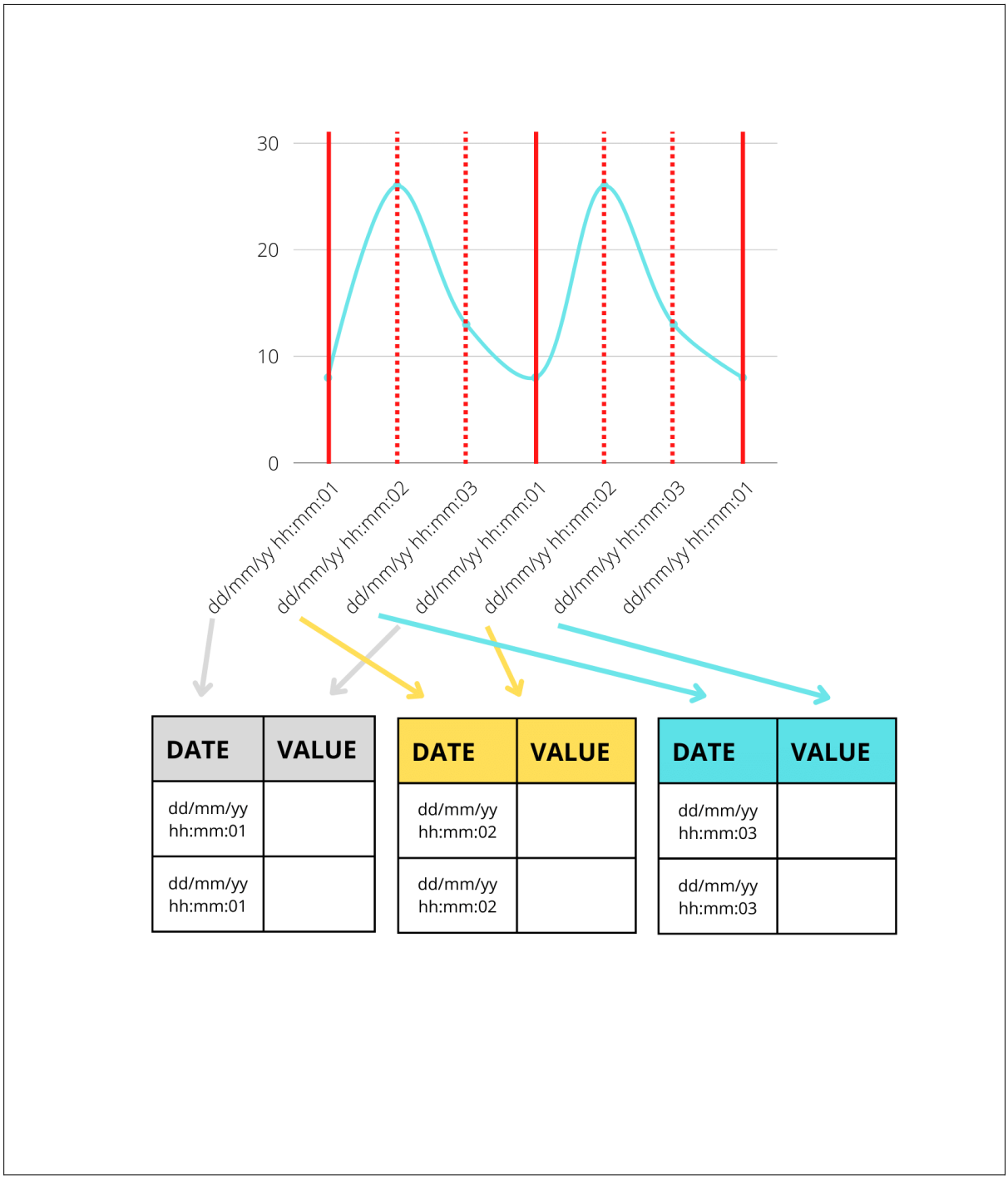

Figure 3.6:Time series division

the 6*N models were generated:six is the number of production parameters

plus the PCA values and N is the number of data collections generated during data preparation.

3.3.2.3 Model evaluation

Inordertoevaluatethemodel,anomaliesidentifiedwerecomparedtothe labelsmanuallygenerated. Afterthecomparison,themetricsAccuracy, Precision,RecallandF1-scorewereappliedinordertoquantifythemodel performances.The other methodologies, adopted in the classifier evaluation, were not applicable for the model structure.In addition, a visual assessment wasperformed.Inparticular,numbersofanomalieswereregroupedbyhours anditwascheckediftherewerehighpercentagesofanomaliesbeforea machine stoppage.

3.3.2.4 Anomalies representation

TwotechniqueswereusedintheanomaliesrepresentationwithintheXAIs interface.In the first one abnormal values were represented as red dots within atemporalgraphcontainingthewholetimeseries.Ontheotherhand,the anomalies detected before the production cycle's prediction were summed up per each parameter and visualized in different tables.

3.4 User study

The user study where conducted in order to understand which one of the XAIs interfaceswasthemostunderstandableandleadusersintrustingtheMLs modelresults.ThetechniquefortheuserstudywastheNearestNeighbour (see Section **??**).As no standard structure has been identified for this approach, participants were asked to perform a combination of Lab Studies and Cognitive Walkthroughs.

3.4.1 User evaluation set-up

Withintheuserevaluationset-upthestudystructure,thenumberof participants and the tools to be used during the user research were defined.

3.4.1.1 Sampling

Inordertoensurethereliabilityofthedataacombinationoftwosampling techniqueswasadopted.Astheresearchwasplacedintheindustryfield,

conveniencesampling[28]wasadopted. Inthiswayparticipantswere recruitedfromaconvenientsubsetofthepopulation. However,aswith convenience sampling there might have been the risk of not truly representing thetargetpopulation,**purposivesampling**[28]wasnecessarytointegrate othersubjectshavingadditionalcharacteristicsofinterestforthestudy. Indeed,inordertoincreasereliabilityandvalidityofthestudy,itwas necessary to include individuals with a machine learning background instead of an industrial one.

3.4.1.2 Sample Size

Attheendofsamplingactivities,itwaspossibletodefinethefinalsample sizeandtodifferentiateparticipantsinvariouscategories:manufacturing features experts with no MLs knowledge, manufacturing features experts with basicbackgroundinMLsandfinallyMLsexpertswithnobackgroundin manufacturing.In the end a total of ten participants was involved in the study, with three operators, four managers and three MLs experts respectively to the categories mentioned before.The managers were in the highest number since they were able to provide a wide variety of significant insights because of their broad background.

3.4.1.3 Target Population

The target population was represented by the three participants categories:the first group represented production line operators, the second one corresponded to manufacturing managers and the third one symbolized MLs developers who might modify with the XAIs interfaces in the future.

3.4.1.4 Nearest neighbour

Asmentionedbefore,notemplateswereidentifiedforapplyingtheNearest neighbour approach where users could indicate which explainability technique inspired them the highest confidence levels.Hence, it was decided to proceed withacombinationof**labstudy**and**cognitivewalkthroughs**[28]. In particularparticipantswerebroughtinadedicatedtestingspace,usually acompanyoffice,wheretheywereaskedtoconsulttheXAIsinterfaces displayedonscreen.

Beforetheinteraction,anexplanationofthedata displayedintheinterfacewasprovidedbythepractitioner,sinceforallthe participants it was the first time using XAIs tools.It is important to mention that the purpose of the study was not to assess the participants' effectiveness

orefficiencyincompletingatask,butwhichinterfacewasexplainingthe results in the most trustful way.After seeing each interface, participants were requiredtofillaquestionnaireonGoogleformtoevaluatetheexplanation andthenproceedingwiththenextXAIstechnique.Thesurveycontained binaryquestions,eachonereferredtoonecognitivemetricmentionedin section2.4.10. Withthisapproachparticipantswereabletoprovidea complete cognitive overview about each one of the XAIs techniques.As one questionnaire was used for all the four interfaces, this allowed to compare the results and derive the best XAIs method from humans perspective.However, sincewithalabstudytheremighthavebeentheriskofecologicalvalidity lack, cognitive walkthrough with an open ended interview were conducted as lastactivityofthestudy.Theinterviewwasstructuredwithfourquestions whereparticipantshadthepossibilitytoclarifythesurvey'sanswersandto provide additional insights whether the information provided by the interfaces were meeting their expectations.

Topic	**Choice**
Participants	10
Group	3 participants
Format	Lab studies and
Activities	Surveys and open
Duration	20 minutes
Medium	In person with PC
Recording	Audio recording and

Table 3.5:User study procedure

3.4.2 Pilot study

Thepilotstudywasconductedbeforetheuserevaluationwithonemanager from the Nytt company.During the session the subject was asked to answer questions as a normal participant and then to provide feedback regarding the whole evaluation structure.It allowed to reduce the study duration from 40 to 20 minutes, to improve the four XAIs interfaces and to redefine some questions in the survey which might have been unclear or required an high mental effort. Prior,numericalandmathematicalinformationwerecompletelyremoved fromtheexplanationsastheywerenotunderstandablefrompeoplewithno MLs knowledge.After,questions providing multiple options about reliance, completeness and satisfaction were removed as considered misleading for the

users.Inaddition,toallthebinaryquestions,an”*Idon’tknow*”optionwas added in order to reduce participant acquiescence and dissent biases [28].

3.4.3 Result analysis

The results obtained from the user evaluation were stored in a Google drive: the answers to the surveys were saved using the Google Form storage while the interviews were audio recorded and stored in the folder.This implied to ask the participants signing a consent form before starting the user evaluation.Data gatheredthroughquestionnaireswereofthequantitativetypeandtheywere analyzedusinggraphsprovidedbyGoogleForm,suchasbarorpiecharts displaying percentages.On the other hand responses from the interviews were qualitative data.In this case the results were analyzed using thematic analysis [65].Answersfromparticipantwerescannedandlabelswereassignedto wordsorphrasesrepresentingimportantorrecurringtopics. Onlysmall expressionwereselectedinordertomaintaintheanalysiseffective.After identifyingtopics,theseweregroupedtogetherinthemesbasedontheir similarities(seeFigure3.7).Thismethodallowstoorganizeandcategorize the results so that they can be presented in a direct and concise.

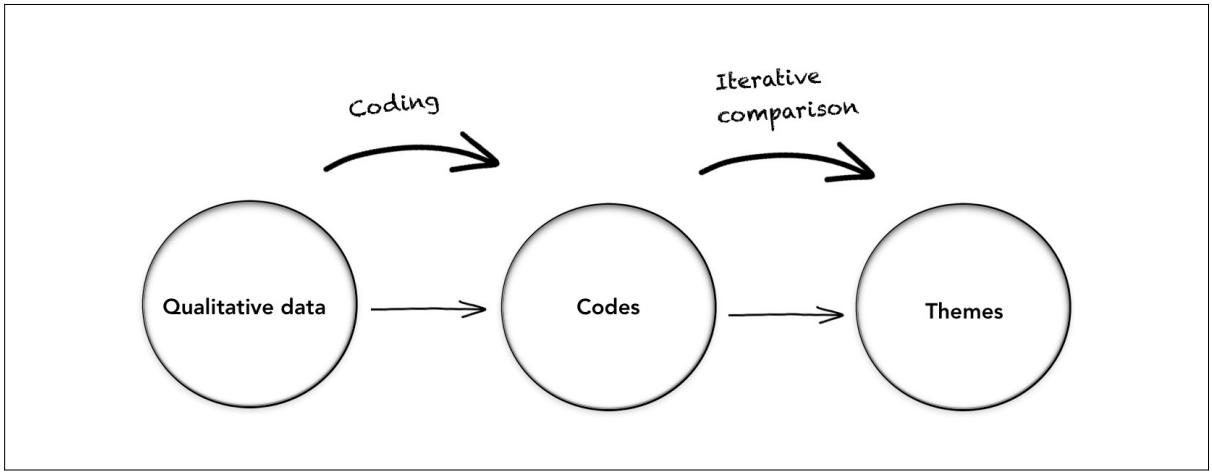

Figure 3.7:Thematic Analysis structure

Chapter 4

Results and Analysis

This chapter is divided in two parts:the first section shows the MLs results, includingtheanomaliesidentificationthatlaterwasincludedinthefourth XAIs interface, while the second section shows the user study results.

4.1 MLs results

Inthissectionperformancesofclassifiersandanomaliesdetectorsare reported.

Theresultsareshownbothforthe”Stacklights”andthe”MTconnect”datacollectio nsforwhatconcernstheclassificationmodels.
Ontheotherhand,asonlythe”MTconnect”datawereusedforanomalies detection, results are illustrated for data collections of different sizes.

4.1.1 Classifiers results Stacklights data

AfterbuildingtheclassifiersbasedoninformationgatheredfromtheStack lightsindicators,thedifferentmodelswereevaluatedusing**Accuracy**, **Precision**,**Recall**and**F1-score**.Thesemetricswerecalculatedbothonthe training and on the testing set.In addition, the four measures were calculated onthesetgeneratingthroughCrossValidation(seeSection3.2.4).Atthe end,theclassifiers’performanceswerereportedintoatableinsuchaway that it was possible to compare them (see Figure 4.1).As it is possible to see from the figure, the results on the testing set were lower than 50% for all the models.Hence, as this was not acceptable, it was decided to not proceed with Stacklights Machine 54 data.

	Train Accuracy	Train Recall	Train Precision	Train F1	Validation Accuracy	Validation Recall	Validation Precision	Validation F1	Test Accuracy	Test Recall	Test Precision	Test F1
Logistic Regression Tuned	0.857143	0.888889	0.800000	0.842105	0.81	0.9	0.766667	0.786667	0.142857	0.2	0.333333	0.250000
Logistic Regression	0.857143	0.888889	0.800000	0.842105	0.71	0.8	0.666667	0.686667	0.142857	0.2	0.333333	0.250000
KNN Classifier Tuned	1.000000	1.000000	1.000000	1.000000	0.81	0.6	0.800000	0.666667	0.142857	0.0	0.000000	0.000000
KNN Classifier	0.809524	0.555556	1.000000	0.714286	0.66	0.4	0.500000	0.433333	0.285714	0.2	0.500000	0.285714
Decision Tree Classifier Tuned	0.571429	0.000000	0.000000	0.000000	0.57	0.0	0.000000	0.000000	0.285714	0.0	0.000000	0.000000
Decision Tree Classifier	1.000000	1.000000	1.000000	1.000000	0.67	0.3	0.300000	0.300000	0.428571	0.6	0.600000	0.600000
Neural Network Classifier Tuned	0.904762	0.888889	0.888889	0.888889	0.76	0.8	0.566667	0.653333	0.142857	0.2	0.333333	0.250000
Neural Network Classifier	1.000000	1.000000	1.000000	1.000000	0.71	0.7	0.566667	0.620000	0.285714	0.2	0.500000	0.285714
Random Forest Classifier Tuned	0.666667	0.444444	0.666667	0.533333	0.57	0.0	0.000000	0.000000	0.142857	0.0	0.000000	0.000000
Random Forest Classifier	1.000000	1.000000	1.000000	1.000000	0.72	0.6	0.600000	0.566667	0.285714	0.4	0.500000	0.444444

Figure 4.1:Classifier Results Stacklights Machine 50 data

4.1.2 Classifiers results MTconnect data

TheevaluationresultsfromthecreatedwithMTconnectdatawerecalculate andvisualizedusingasimilarmethodologytotheoneaforementioned. However,asinthiscasetheperformancesobtainedonMachine50were optimal, also the evaluation was performed also on models built with Machine 54 data.From figures 4.2 and 4.3 it is possible to see that all the values were above70%(0.7),butthemodelwiththeoverallbestperformanceswasthe Random Forest.

	Train Accuracy	Train Recall	Train Precision	Train F1	Validation Accuracy	Validation Recall	Validation Precision	Validation F1	Test Accuracy	Test Recall	Test Precision	Test F1
Logistic Regression Tuned	0.823529	0.661765	0.978261	0.789474	0.822355	0.661789	0.974885	0.787344	0.838235	0.683824	0.989362	0.808696
Logistic Regression	0.822304	0.656863	0.981685	0.787078	0.821128	0.656911	0.978158	0.785132	0.838235	0.683824	0.989362	0.808696
KNN Classifier Tuned	1.000000	1.000000	1.000000	1.000000	0.802641	0.948509	0.734886	0.828032	0.805147	0.948529	0.737143	0.829582
KNN Classifier	0.830882	0.943627	0.770000	0.848018	0.740162	0.877356	0.688636	0.771407	0.772059	0.904412	0.715116	0.798701
Decision Tree Classifier Tuned	0.811275	0.634804	0.981061	0.770833	0.811327	0.634869	0.980905	0.769984	0.812500	0.632353	0.988506	0.771300
Decision Tree Classifier	1.000000	1.000000	1.000000	1.000000	0.931356	0.955857	0.911230	0.932985	0.933824	0.926471	0.940299	0.933333
Neural Network Classifier Tuned	0.568627	0.884804	0.542042	0.672253	0.658170	0.641734	0.668669	0.645202	0.569853	0.904412	0.541850	0.677686
Neural Network Classifier	0.566176	0.990196	0.535809	0.695353	0.626208	0.617164	0.638696	0.610300	0.573529	0.985294	0.540323	0.697917
Random Forest Classifier Tuned	0.816176	0.639706	0.988636	0.776786	0.815008	0.637338	0.988242	0.774080	0.812500	0.632353	0.988506	0.771300
Random Forest Classifier	1.000000	1.000000	1.000000	1.000000	0.964447	0.980367	0.950297	0.965008	0.944853	0.941176	0.948148	0.944649

Figure 4.2:Classifier Results MTconnect Machine 50 data

	Train Accuracy	Train Recall	Train Precision	Train F1	Validation Accuracy	Validation Recall	Validation Precision	Validation F1	Test Accuracy	Test Recall	Test Precision	Test F1
Logistic Regression Tuned	0.890259	0.863524	0.910995	0.886624	0.889002	0.880741	0.898694	0.887930	0.852399	0.862319	0.850000	0.856115
Logistic Regression	0.872996	0.786600	0.949102	0.860244	0.865606	0.786451	0.935200	0.852934	0.852399	0.775362	0.922414	0.842520
KNN Classifier Tuned	1.000000	1.000000	1.000000	1.000000	0.889048	0.990093	0.823857	0.899095	0.878229	1.000000	0.807018	0.893204
KNN Classifier	0.916153	0.985112	0.864924	0.921114	0.870582	0.972747	0.808197	0.882461	0.870849	1.000000	0.797688	0.887460
Decision Tree Classifier Tuned	0.893958	0.796526	0.987692	0.881868	0.889033	0.791481	0.981873	0.876082	0.892989	0.797101	0.990991	0.883534
Decision Tree Classifier	1.000000	1.000000	1.000000	1.000000	0.960562	0.977716	0.946485	0.961176	0.955720	0.971014	0.943662	0.957143
Neural Network Classifier Tuned	0.847102	0.990074	0.768786	0.865510	0.843452	0.955401	0.780791	0.858757	0.800738	0.978261	0.725806	0.833333
Neural Network Classifier	0.847102	0.937965	0.792453	0.859091	0.763167	0.667099	0.840907	0.717622	0.837638	0.949275	0.779762	0.856209
Random Forest Classifier Tuned	0.893958	0.796526	0.987692	0.881868	0.890260	0.796420	0.979216	0.877828	0.892989	0.797101	0.990991	0.883534
Random Forest Classifier	1.000000	1.000000	1.000000	1.000000	0.975339	0.977593	0.972742	0.975138	0.970480	0.978261	0.964286	0.971223

Figure 4.3:Classifier Results MTconnect Machine 54 data

As Accuracy, Precision, Recall and F1-score do not provide a quantifica-tion about the number of stoppages correct predictions or running states correct predictions, confusion matrix were generated for Machine 50 and Machine 54 based model (see Figures 4.4 and 4.5).

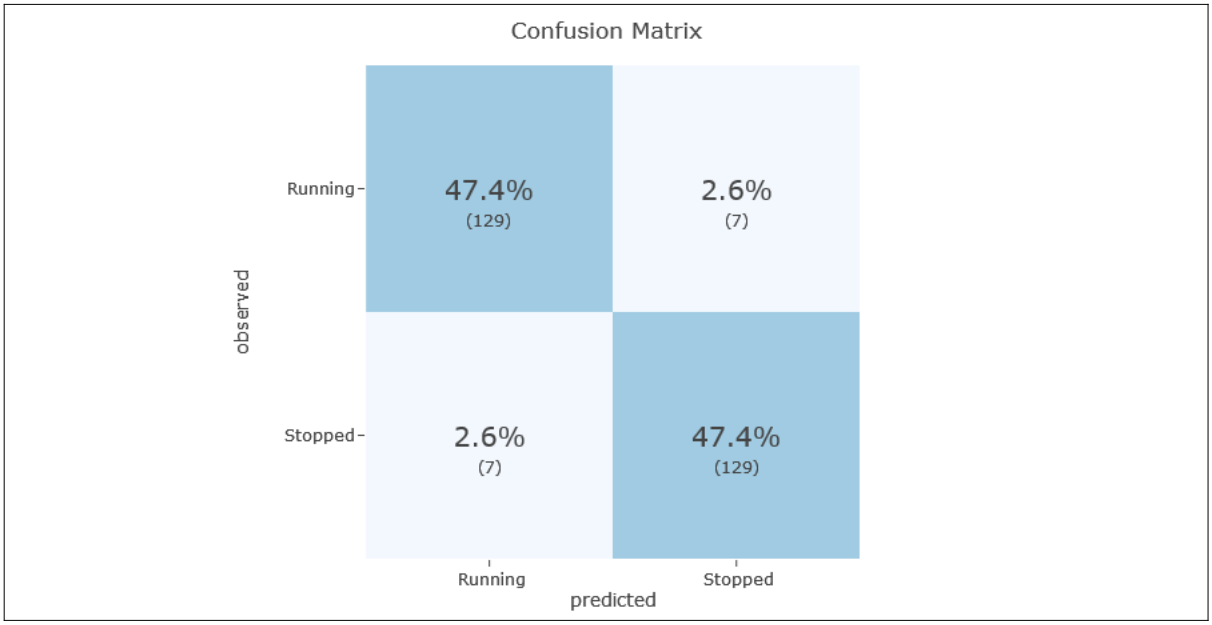

Figure 4.4:Confusion matrix Machine 50 data

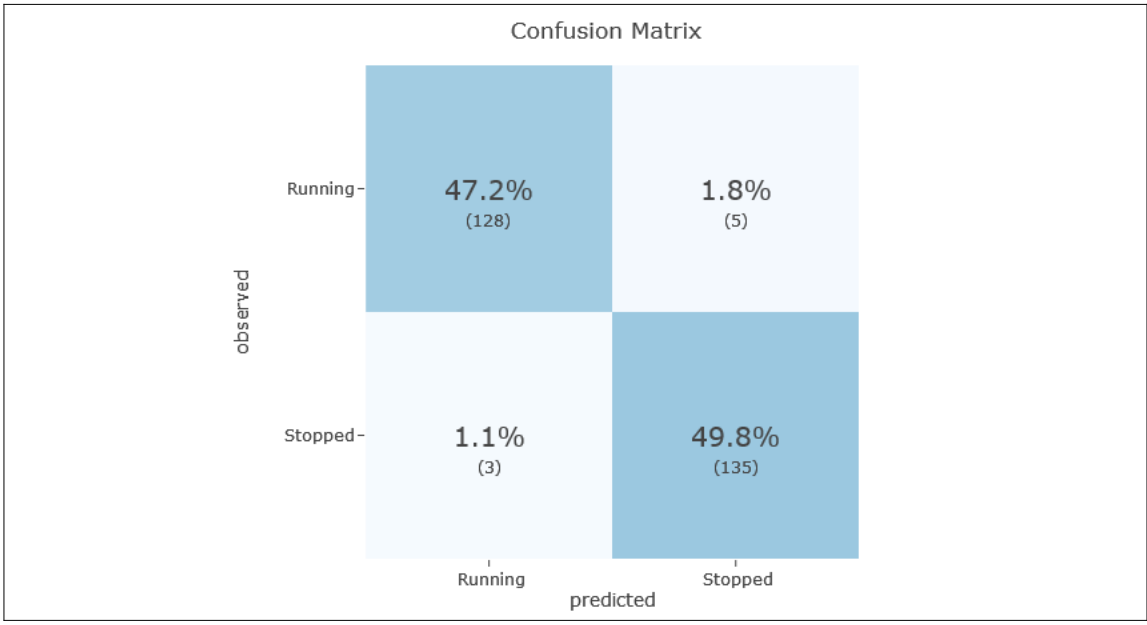

Figure 4.5:Confusion matrix Machine 54 data

Inordertoobtainfurtherinsightsabouttheperformancesofthetwo RandomForestbasedmodels,oneforMachine50andoneforMachine54, ROC-AUC and PR-AUC scores graph were produced.This allowed to capture at glance the goodness of the models in producing the correct predictions.As it is shown from pictures 4.6 and 4.7, the roc-auc and pr-auc scores are of 0.99 and 1.00, respectively for machine 50 and 54.

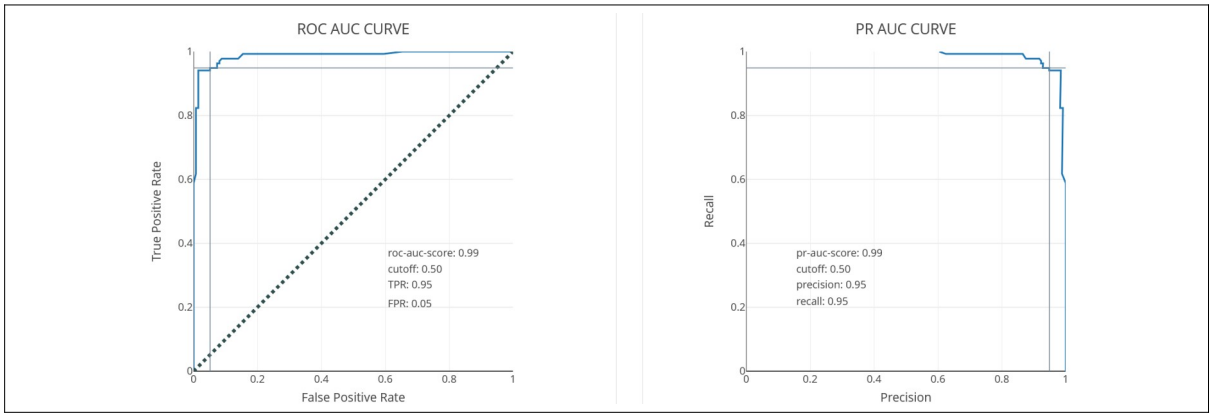

Figure 4.6:ROC-AUC and PR-AUC graphs MTconnect Machine 50 data

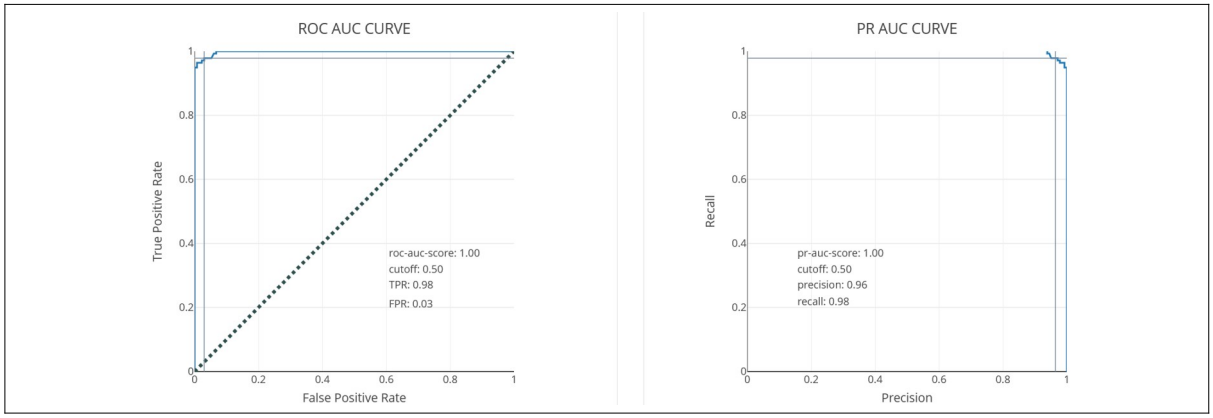

Figure 4.7:ROC-AUC and PR-AUC graphs MTconnect Machine 54 data

Aslastresult,theimpactofeachfeatureonthefinalprediction wascalculatedusingSHAPLEYvalues(seeSection2.4.9). Thefeatures importance was displayed using a bar chart graph (see Figures 4.8 and 4.9)and itwaspossibletoaffirmthatthecurrentstateofthemachinewasthemost influential on the models' outcomes.

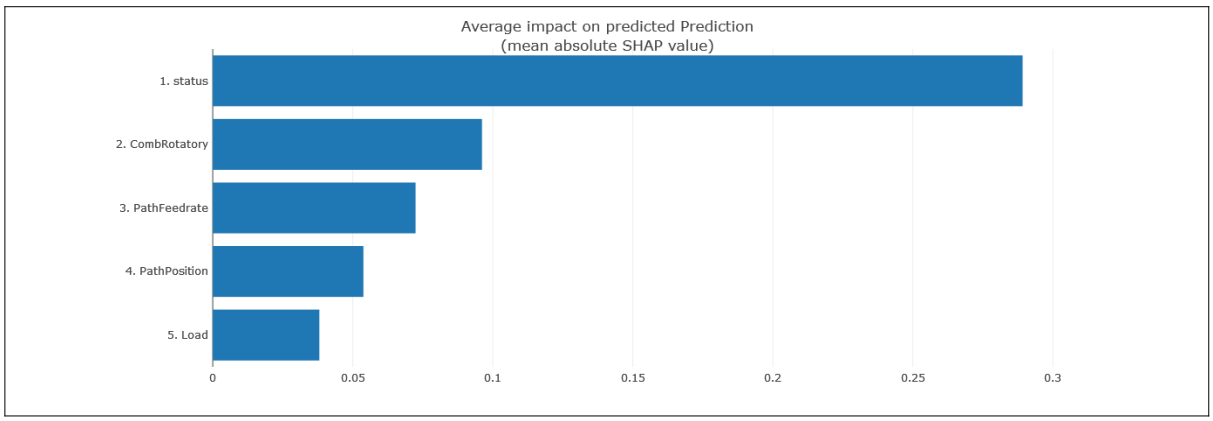

Figure 4.8:Features importance MTconnect Machine 50 data

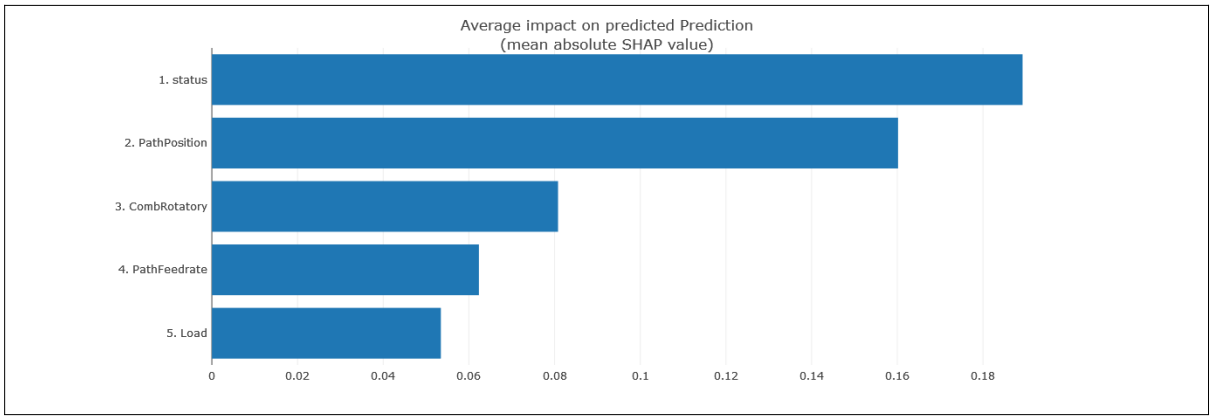

Figure 4.9:Features importance MTconnect Machine 54 data

4.1.3 Anomalies detection results MTconnect data

BeforeusingtheanomaliesdetectionasXAIsinterfacetherewastheneed toevaluatetheanomaliesdetectionmodelbasedontheIsolationForest algorithm,inordertocheckwhetherthemodelwasusableevenoutsidethe testing environment.The evaluation metrics were calculated for each one of the parameters involved in the abnormal values identification and then reported withinatable.Thisevaluationwasperformedonfourdatacollections:two from Machine 50 and two from Machine 54.In both the cases, one set of data was collected in one week of March and the other during the whole month (see Figures 4.10, 4.11, 4.12 and 4.13).

	Accuracy	Precision	Recall	F1-score
Anomalies on combination	0.989263	0.512367	0.425220	0.464744
Anomalies on Load	0.990517	0.573503	0.470939	0.517185
Anomalies on Rotatory Load	0.993169	0.703214	0.581250	0.636441
Anomalies on Rotatory Active Velocity	0.989440	0.750000	0.513514	0.609626
Anomalies on Path Feed rate	0.989585	0.532468	0.420205	0.469722
Anomalies on Path Position	0.989327	0.651282	0.453032	0.534362

Figure 4.10:Performances of anomalies Identification Machine 50 one week of March

	Accuracy	Precision	Recall	F1-score
Anomalies on combination	0.867060	0.053514	0.814975	0.100433
Anomalies on Load	0.900785	0.094505	0.864225	0.170380
Anomalies on Rotatory Load	0.946183	0.118990	0.877591	0.209566
Anomalies on Rotatory Active Velocity	0.853961	0.086403	0.938107	0.158233
Anomalies on Path Feed rate	0.907274	0.073186	0.789336	0.133953
Anomalies on Path Position	0.757831	0.035839	0.944629	0.069058

Figure 4.11:Performances of anomalies Identification Machine 50 month of March

	Accuracy	Precision	Recall	F1-score
Anomalies on combination	0.957986	0.149701	0.478469	0.228050
Anomalies on Load	0.980948	0.304425	0.437659	0.359081
Anomalies on Rotatory Load	0.981972	0.358925	0.430876	0.391623
Anomalies on Rotatory Active Velocity	0.974090	0.035316	0.056716	0.043528
Anomalies on Path Feed rate	0.960561	0.144649	0.410926	0.213976
Anomalies on Path Position	0.980141	0.321821	0.496368	0.390476

Figure 4.12:Performances of anomalies Identification Machine 54 one week of March

	Accuracy	Precision	Recall	F1-score
Anomalies on combination	0.589753	0.018769	0.906765	0.036777
Anomalies on Load	0.856533	0.062291	0.904252	0.116553
Anomalies on Rotatory Load	0.918172	0.071970	0.738061	0.131152
Anomalies on Rotatory Active Velocity	0.852570	0.026616	0.529316	0.050684
Anomalies on Path Feed rate	0.915141	0.077024	0.805692	0.140606
Anomalies on Path Position	0.898642	0.066171	0.657660	0.120244

Figure 4.13:Performances of anomalies Identification Machine 54 month of March

Asitcanbeseenfromthefigureabove,thereisalwaysoneparameter outperformingtheothers.Hence,thevisualevaluationwasperformedonly onanomaliesidentifiedonthemostreliableproductionparameter.Intwo outoffourcasesthiswasrepresentedbythe”RotatoryLoad”,whileforthe modelbasedonMachine54longtermdatathemostreliableattributewas the”PathFeedrate”andforthemodelbasedonMachine54shorttermdata themostreliableattributewasthe”PathPosition”.Thevisualassessment wasperformedmanuallybyinvestigatingiftherewasanhighpercentageof anomalies before a stoppage.Percentages were displayed in a vertical bar chart graph together with the breakdowns, where the x-axes represented the date and the y-axes the values (see Figures 4.14, 4.15, 4.16 and 4.17).

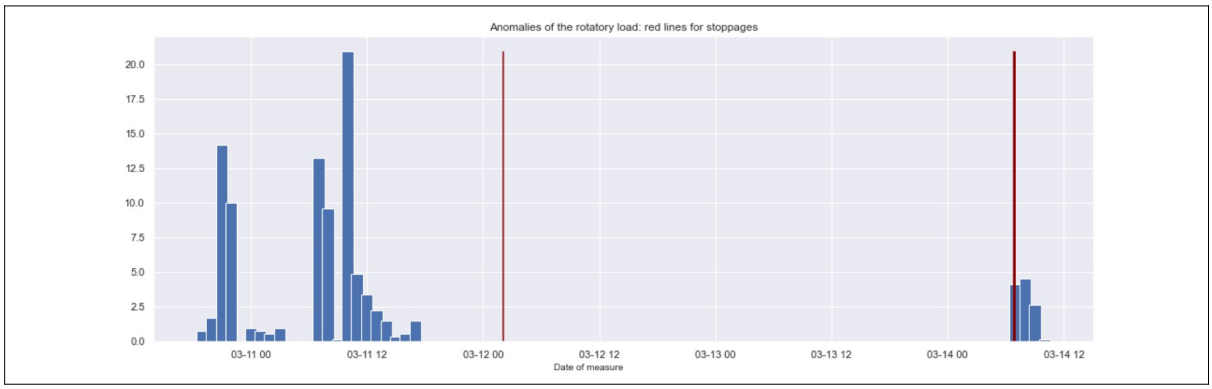

Figure 4.14:Anomalies on Rotatory Load Machine 50 one week of March

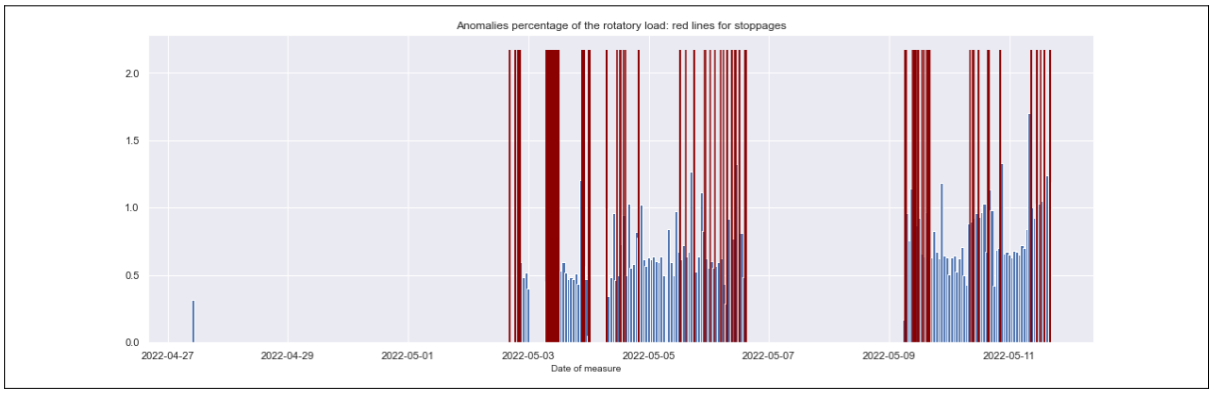

Figure 4.15:Anomalies on Rotatory Load Machine 50 month of March

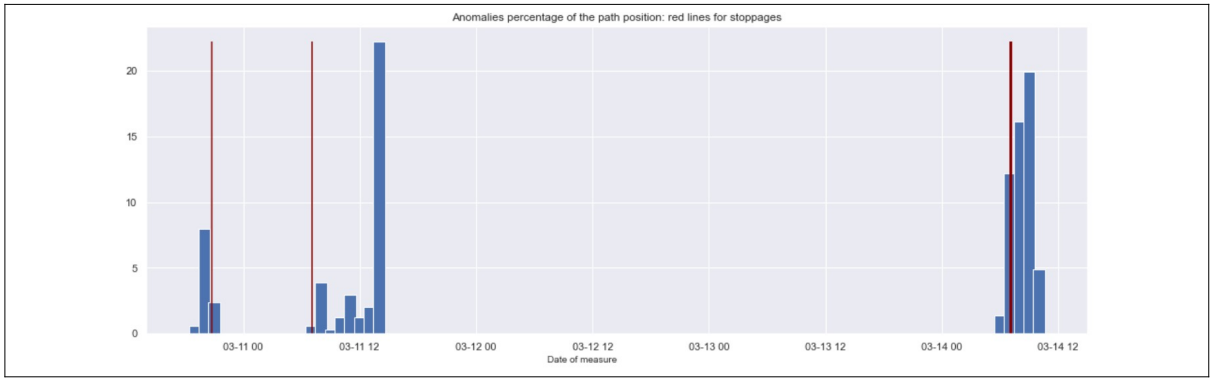

Figure 4.16:Anomalies on Path Position Machine 54 one week of March

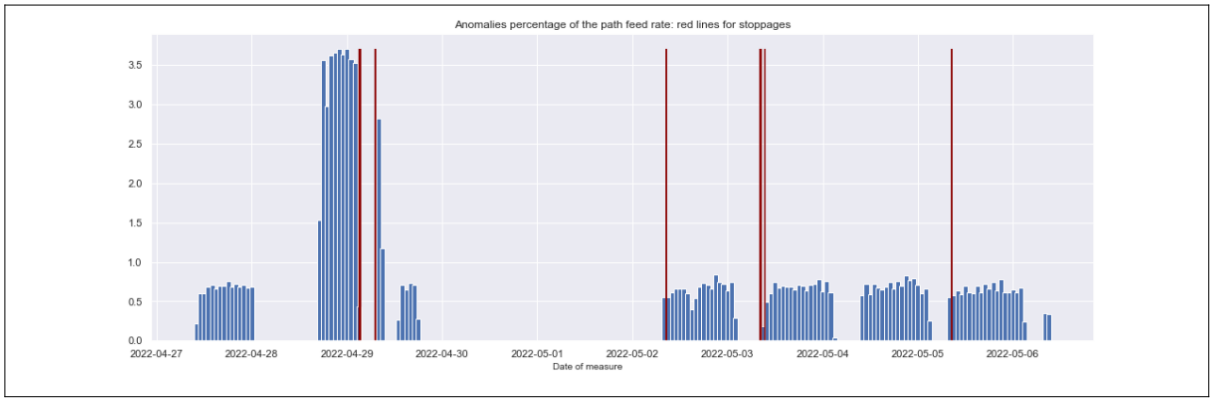

Figure 4.17:Anomalies on Path Feed rate Machine 54 month of March

Fromthefiguresabove,in4.17itcanbeseenapeakofanomaliesof 3.6% right before a stoppage.This is replicated in 4.16,where a percentage ofanomaliesgreaterthan10%isprecedingabreakdown.Usingdatafrom Machine50,itwaspossibletoidentifyanomaliesonthespecifictool(see Appendix D).

4.2 XAIs results

4.2.1 XAIs interfaces final layout

The final layout of the four XAIs involved in the user study was finalized at the end of the pilot study.All the interfaces integrated a summary of the prediction atthebeginningthroughacompositionoftextandapiechartrepresenting probabilitiesofstoppageandrunningstate.AlltheXAIswereshowingthe

same classifier prediction and the layout is shown in figures 4.18,4.19,4.20 and 4.21.In this explanation (figure 4.18) it was possible to see the accuracy

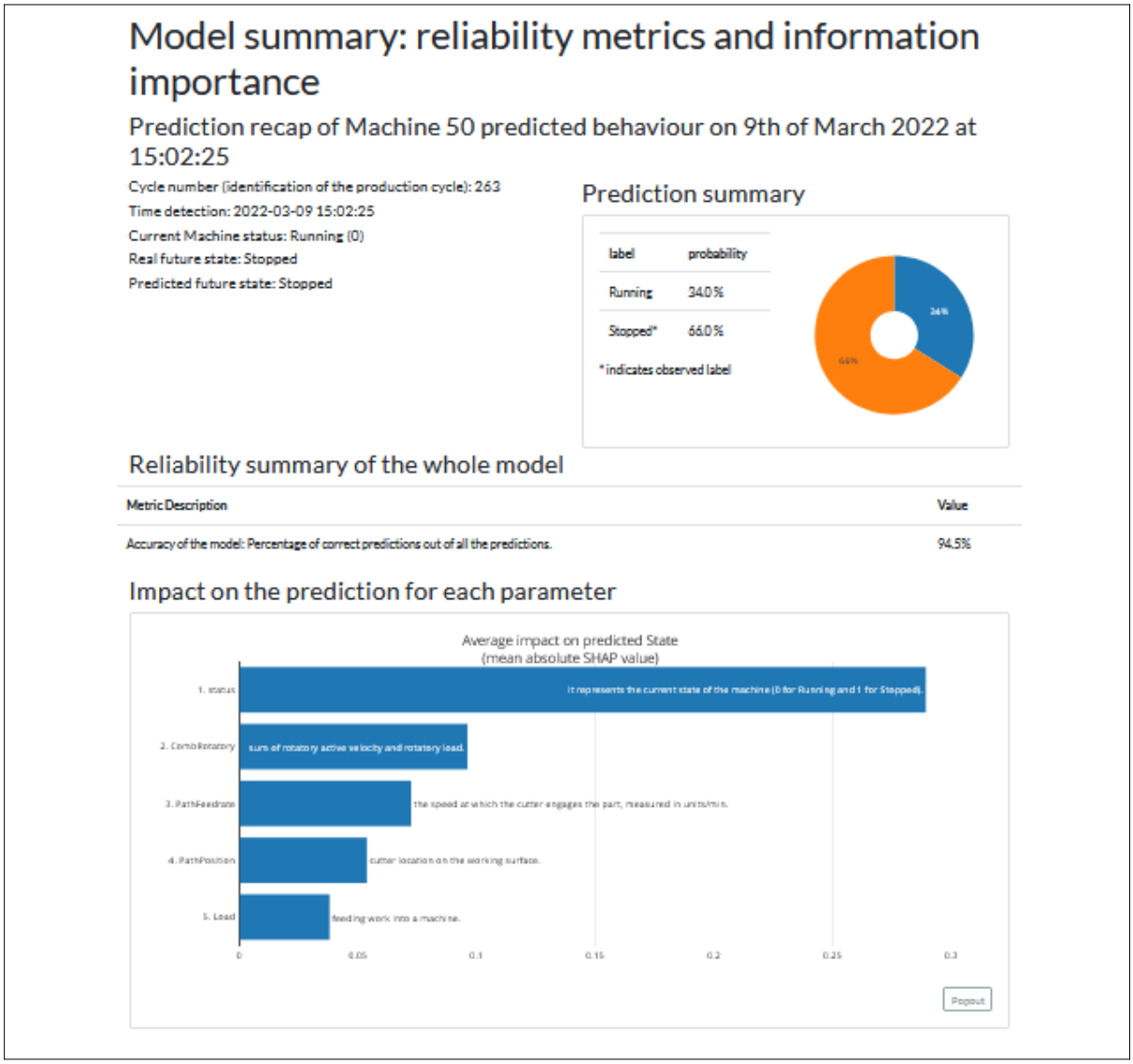

Figure 4.18:Anomalies on features' impact

of the model, with its definition annexed, and the features' impact on the model outcome.In this way, users were able to understand which factors the classifier depended on.For each feature displayed by the bar chart, the description was aligned to the name placed on the y-axes.

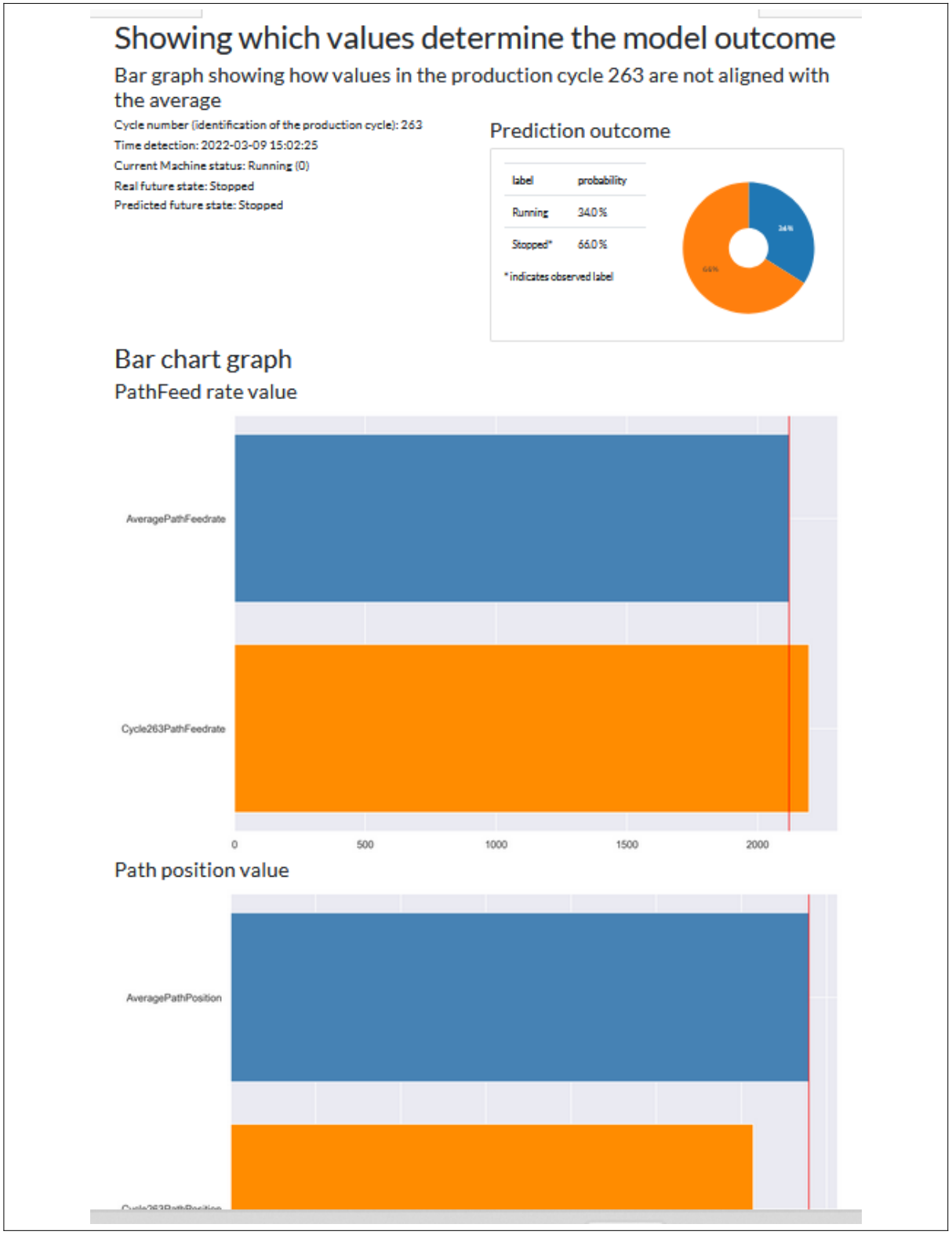

Figure 4.19:Explainable interface with features' values

Inthisinterface(figure4.19)itwaspossibletoseewhichparameters determinedthemachinetostopinthenextproductioncycle.Theirvalues weredisplayedinabarcharttogetherwiththenormalfunctioningaverage values.Fromthegraphsitwaspossibletoseehowtheparameters(inthis casePathFeedrateandPathPosition)werenotalignedtotherestofthe data.Inthisinterface(figure4.20)itwaspossibletoseehowtherandom

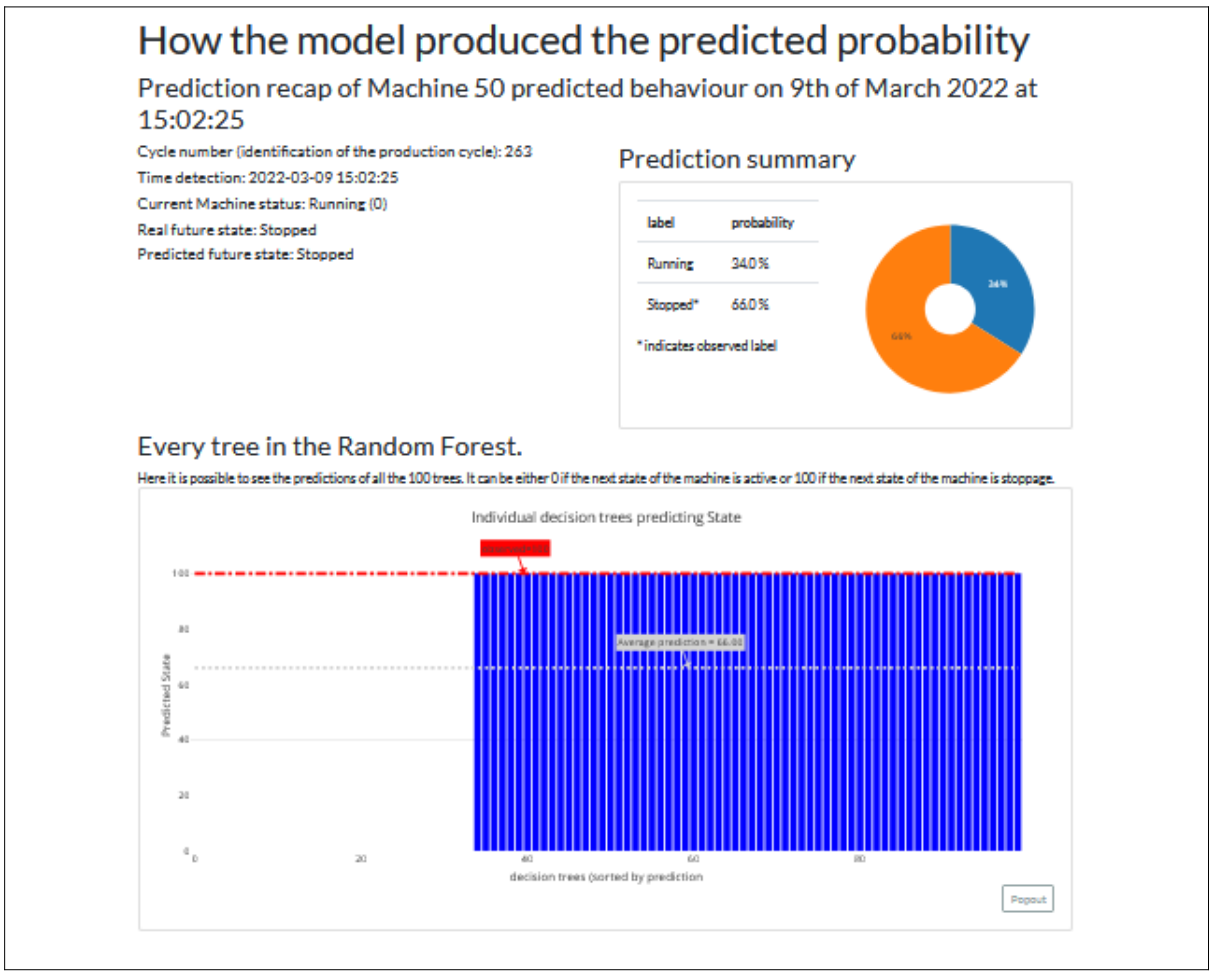

Figure 4.20:Explainable interface with model decision process

forestproducedthedisplayedprediction.AstheRandomForestclassifier is composed by 100 less complex models (Decision Tree),their output were displayedinshapeofverticalbars.Theexplanationshowedhowthemost common result corresponded to the final prediction.In this explanation (figure 4.21) it was possible to see the actual breakdown and which tool caused it.In addition, the two parameters on which the tool caused most of the anomalies

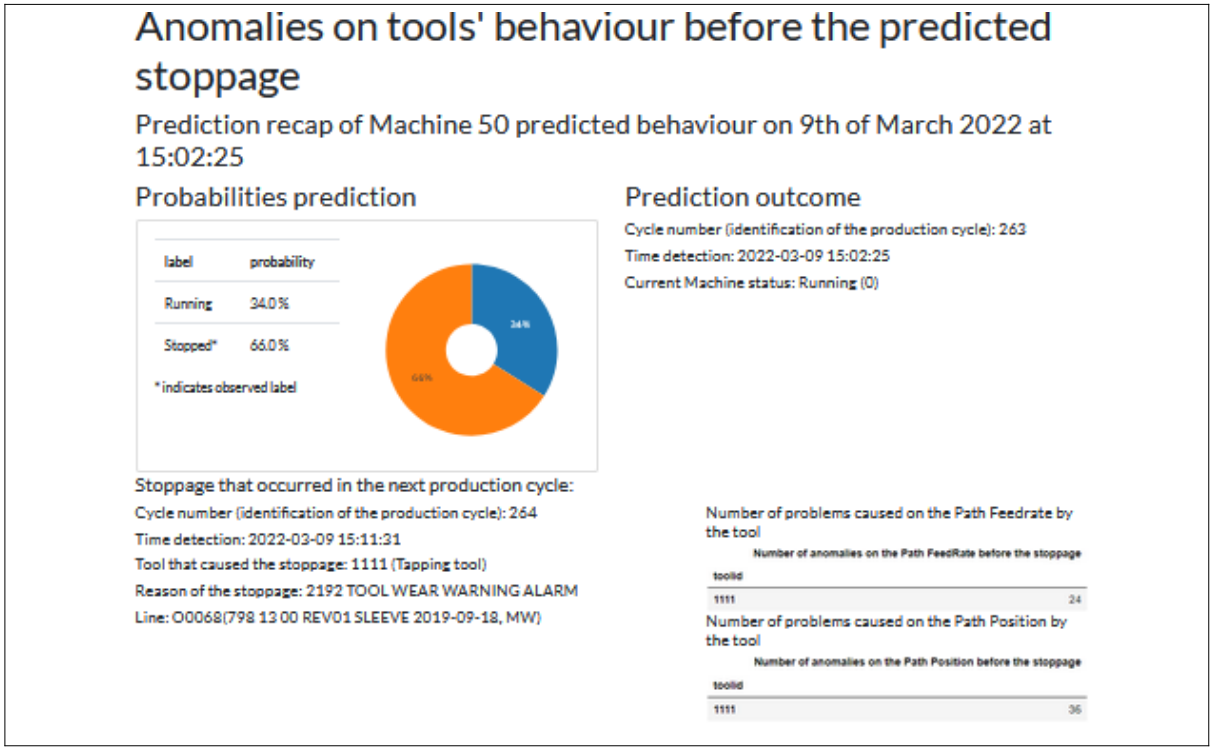

Figure 4.21:Explainable interface with tool anomalies

were displayed, with the relative number of abnormal values.For this interface the anomaly identification model (see Section 3.3.2) was used.Even though the evaluation results were not optimal (see Section 4.1.3), it was included to understandifuserstrustmorethemodelifconcretedataaboutthemachine were shown.

4.2.2 Nearest Neighbour Results

The Nearest Neighbour was implemented through questionnaires with closed probesandinterviewswithopenendedquestions. Hence,inorderto understand the clearest interface for the users, it was necessary to analyze the surveys and interviews results.

4.2.2.1 Questionnaires results

In this section answers to questions related to understandability, satisfaction, details level, completeness, actionability, accurancy and results trustability are reported.Responseswereorganizedinpiechartdisplayingtheparticipants

percentage per each option.

Understandability

Forthefirst,thesecondinterface75%oftheparticipantsansweredthat they were understandable.For what concerns the third and the forth interfaces instead, 83.3% of the participants indicated that the former was understandable and 91.7% said it about the latter.

Satisfaction

Forthefirstinterface75%ofsubjectsindicatedthatitwassatisfactory, 58.3%agreedonthesatisfactionaspectforthesecondinterfaceand66.7% responded positively for the third interface.An high percentage of participants, 83.3%, stated that the fourth interface was satisfactory.

Details level

Forthefirstandthesecondinterface66.7%ofparticipantsrespondedit wasdetailed,while50%answeredpositivelyforthethirdinterface.Onthe other hand, 83.3% of subjects indicated that the fourth interface was detailed.

Completeness

For the first interface only 33% of subjects answered that the explanation wascomplete,whilethe50%respondednegatively.Forthesecondandthe fourthinterfaces50%and58.3%oftheparticipantsrepliedthattheywere complete respectively, while for the third interface 58.3% answered that it was complete.

Actionability

Fortheactionability50%ofthesubjectsanswered”yes”forthefirst interface, 58.3% responded positively for the second interface, while only 25% said that the third interface was actionable.On the contrary, 75% affirmed that the fourth interface was actionable.

Accuracy

For the accuracy and reliability 75% of the participants indicated that the firstinterfacewasreliable,33.3%respondedpositivelyforthesecondand fourth interfaces and 41.7% answered ”yes” for the third explanation.

Trustability

For the first interface 58.3% of the users answered that the first interface wastrustworthy,41.7%respondedpositivelyforthesecondandtheforth interface and 50% considered trustworthy the third explanation.

4.2.2.2 Interview results:thematic analysis

The interviews analysis brought to the identification of seven different themes that are described below and are shown in Figure 4.22.

Theme	Code	Qualitative data
Pairing anomalies with features values	Mix	Mix between "model decision path while use" and "anomalies"
	Parameters tweak	I would like to know which tool caused the anomalies and which parameters I have to tweak
	Compare	Having the possibility to compare information, such as the two bar graphs
	Pairing	In the second it would be nice to have also the number of anomalies like the fourth one.
	Pairing	See information displayed in second one within the fourth interface
Reasons necessity	Why anomalies	Why the anomalies were detected
	Where anomalies	Where in the program the machine stopped
	Why anomalies	Explains why the problem happened
	When anomalies	See when the anomalies occurred
Interfaces combination	Combination	See a combination of all the interfaces
	More information	The more information we have the better is
	One screen	See all the interfaces in one screen
Predictions listing	More predictions	Check more than one prediction at time and compare them
	Good predictions	See predictions where the machine behaviour didn't go wrong
	Past behaviour	See how the machine behaved before the stoppage
Lack of background	No ML knowldge	Without ML knowledge is difficult to grasp what the model decision path is about
	Unable to connect	Not able to connect the graph displayed with the model prediction
	Unable to say	Not able to say how the model derived the final conclusion
Abstractness	Real life	Try the interfaces in real life
	Real life	I would like to use it in real life
Lack of clarity	See thresholds	See threholds in the features values interface and if the breakdown occurs because the bar overcomes the red
	Value scale	Value scale is different and there is no threshold
	Meaning	Meaning of values in the x-axes
	Cannot understand	Cannot understand that what's matter is the difference from the threshold

Figure 4.22:Thematic analysis results

Pairing anomalies with features values One thing that quickly emerged fromtheinterviewsinspectionisthatpeoplewouldhavelikedtosee information displayed in the second and in the fourth interfaces paired together. For instance, one participant stated that it would have been convenient to see which tool caused problems and which parameters he had to modify in order to re-establish the machine running state.

Motivations necessity A large number of participants affirmed that they would have liked to see additional information in the fourth interface, so that they could motivate why the anomalies were detected on that tool.They stated that they would have appreciated to know also when anomalies were detected.

Interfaces combination The analysis showed that numerous participants asserted that a combination of the information displayed in the four interfaces wouldhaveprovidedabetterexplanationthanjustthesingleones. In particular,participant5(P5)affirmedthatalltheinterfacesshouldbe integratedinonescreen,sothatusershadageneraloverviewonthemodel functioning.

Predictions listing This theme was referred to the fact that some subjects would have liked to see a history of the past predictions in order to understand how the model works.For example, in one interview it was stated that all the interfaces should have shown predictions of machine running state as well.

Lack of background From the responses analysis, it emerged that a large number of people, especially the ones belonging to the no MLs experts group, was not able to understand the third interface visualizing the model decision

process.Thiswasmainlyduetothelackofthenecessarybackgroundfor understanding certian MLs concept.

Abstractness This theme was identified as two participants affirmed that themodelneededtobeusedinreallifetobetrusted. Theystatedthat even though the four explanations could contribute in increasing confidence, itneedstobebuiltbefore,byadoptingthemodelinrealmanufacturing environment.

Lack of clarity The majority of participants stated that the second interface wasinducingconfidencetowardstheresults,butitwaslackinginformation. Theywereassertingthatthetwobarchartshaddifferentvaluescalesandit is not clear if the difference from the threshold had an important influence on the final prediction.

Chapter 5

Discussion

ThischapterisdividedinthreesectionswheretheMLsmodelresults,the XAIs interfaces final layout and the user studies responses are discussed.

5.1 ML results discussion

The classifier building for predicting the machine's state (running or stoppage) inthenextproductioncycle,producedsuccessfulresults.However,thereis theneedtounderlinedthat,inordertokeeptheperformancegoodness,it wasnecessarytouseMTconnectdata.Indeed,whenusingtheStackLights datacollection,theclassifiersevaluationmetricswerealwaysassuming valueslowerthan50%onthetestsets,whichrepresentedthepoormodels reliability[23].This may be due to the lack of information, as the model was basing the prediction only on the previous machine's states and their temporal length.On the other hand, by using the MTconnect data set, high performances values (overall greater than 70%) were obtained either if the model was trained on data of machine 50 or machine 54.In particular, the performances of the RandomForestRFsbasedmodelwerefarexceedingtheothers:itpresented Accuracy,Precision,RecallandF1-scoreallabove90%onthetrainingset, on the Cross Validation set and on the test set.Considering the high results, the model evaluation was further investigated. Firsttheconfusionmatrixdisplayedhighpercentagesofcorrectstoppage predictionsTPs(47.4%Machine50and47.2%Machine54)andcorrect running state prediction TNs (47.4% Machine 50 and 49.8% Machine 54).On the contrary, percentages of wrong stoppage forecasts FPs (2.6% for Machine 50and1.8%forMachine54)andwrongrunningstateforecastsFNs(2.6% for Machine 50 and 1.1% for Machine 54) were low.This justified the reason

behindtheelevatedevaluationmetricsvalues,astheywerebasedonthe numbers shown in the confusion matrix (see Section 2.4).
Second, the ROC-AUC and PR-AUC graphs showed that the RFs classifier had almost a perfect behavior with an ideal balance of the positive and the negative classes.Indeed,theareaunderthecurveiscloseto1forboththegraphs which means that that the model achieves a true positive rate of 100% before producing any false positives.These results are aligned with what was found during the literature review, where it was stated that RFs is the best algorithm fortherealizationofbinaryclassifiers[42,41].However,itimportantto underlinethattheseperformancescouldbereachedonlywithdataperfectly balanced which did not reflect the reality.
Third, the features importance grahp displayed that with both the machines, 50 and 54, the parameter impacting the forecast the most was the current machine state.Hence,whetherthemachineisstopped,themodelislikelytopredict abreakdownalsointhenextproductioncycle.Themajordrawbackofthis behaviouristhatrunningstatesfollowingstoppagesmightbeindicatedas failures, leading to incorrect forecasts.
ForwhatconcerntheTSsanomaliesidentificationmodelsbasedonthe IsolationForestalgorithm,itcanbesaidthatthebestAccuracy,Recall, PrecisionandF1-score(99%,70%,58%and63%respectively)valuesare obtainedwiththeRotatoryLoadstoredintheoneweekdataofMachine 504.10.ThisoutcomewasjustifiedbytheoptimalcyclicalityoftheTime Series representing this production parameter.However, when the model was applied on a data set representing a wider temporal window (whole month of March),theevaluationmetricsPrecisionandF1-scoredrasticallydecreased (11% and 20% respectively).This was due to the strong unbalance between anomalous and normal data.Moreover, as in one month different parts types areproduced,theproductioncyclesdurationwasnothomogeneouscausing averagevaluestobelabeledasoutliers.Asimilarsituationwasidentified with machine 54.In this case the best detection was achieved on the Rotatory Load only for the short period data set, and on the Path Feed rate for the one month data collection (Accuracy 91%, Precision 7%, Recall 80% and F1-score 12%).Furthermore, it was possible to visually detect cluster of anomalies right before a stoppage for machine 54.On the other hand, with short term machine 50 data a cluster of anomalies was observable more than 12 hours before the actual stoppage whilst with long term data, clusters were not identifiable due to the large number of breakdowns (represented as red lines in figure 4.15).

5.2 XAI resuts discussion

ThefinallayoutofthefourXAIsinterfaceswasbasedonamixtureofthe insightsgatheredthroughthepilotstudyandtheliteraturereview. The firstimportantaspecttodiscussistheinformationsimplicity.Indeed,no mathematical data or formula were displayed on any of the four explanations. Thereasonbehindthischoicewasmadeinordertoreducetheusagetime: whileconsultinganXAIsinterface,usersdonothavetoinvesttimefor understandingcomplexstatisticalMLsconcepts.Second,itisimportantto underline the high visualizations involvement.This approach was based on the literature review [13], where it was stated that visualizations increase humans'levels of trust.Finally, a one screen design was involved in the implementation ofallthefourinterfaces.Thismethodwaschoseninordertoallowend-userstoaccessatfirstglancealltheinformationneededforunderstanding the predictions.Finally, each of the four XAIs facilities displayed a

5.3 User study resuts discussion

Fromboththeinterviewsandthequestionaries,itwasapparentthatthe prefered XAIs interface was the fourth one.TODO: Why However,itisimportanttomentionthatallthe participantswithaMLs backgroundindicatedthatthethirdinterface,theoneshowingthemodel decisionpathgraph,wastheclearestone. Thisoutcomewasduetothe pointthat,asMLsexpertsdidnothavemanufacturingknowledge,they consideredeasytounderstandmathematicalprocessesbehindtheforecast insteadofmachinesdata.Ontheotherhand,featureexpertswhomisseda strong AIs background, were trusting the model's results after seeing concrete informationcomingfromthemanufacturingenvironment.Thishypothesis was confirmed the theme "Lack of background" found during the interview: most of the managers and operators were not able to understand MLs concepts showninthethirdinterface.Byvisualizinganomaliesinstead,evenifthey were not used by the model in creating the forecast, users were able to match model's outcome to real situation [13].Moreover, in this way users were able tomatchtheirpreviushypothesisaboutwhatcausedthemachinestoppage with the model results [13].Furthermore, from the interview thematic analysis the themes of "Interface combination" or "Interface pairing" were identified. This results also matched with what found in the literature review about end users involving different XAIs facilities in order to build confidence towards

the model results [32].In the end, it is important to mention that, contrarily to what observed in the literature review, only seeing labels attribution (running or stopped) was not enough for inspiring a sense of trust, but they needed to be paired with features values.

5.4 Limitations

It is important to take into considerations that might have affected the results. First,asstatedinsection6.1,theMLsclassifierwasrealizedwithaperfect balanced data collection, which does not truly reflected the reality.In addition, itisimportanttomentiontheuserstudieswereconductedwithastricttime constraint,sosomeanswersmighthavebeenprovidedwithsuperficiality. To what concern the anomalies identification, there is the need to emphasize that production cycles with different shapes were not distinguished during the training and the testing phase.Hence, normal values in one production cycles might have been recognized such anomalous within another one with different shape.Another limitation was represented from the low distribution of serious breakdownswithinthedatacollections:eventhoughfigure4.15displayed alargeamountoffailures,theycanbeconsideredmild. Furthermore,it wasnoticedthatwithinthedatagatheredthroughMTconnect,theregapsof multiple seconds which might have been the cause of the poor performances obtainedinanomaliesidentification.Finally,asitemergedduringtheuser evaluation,peoplewilltrustmodel’sresultsaftertryingitmultipletimesin the real manufacturing environment.

Chapter 6

Conclusions and Future work

6.1 Conclusions

Thestudydescribedinthisreportshowedthepossibilityofbuildings MLs modelstoassistmanagersandoperatorsinamanufacturingenvironment andwhichXAIsinterfacemightbethemostsuitableforinducingthemto actualusetheprediction.Forthefirstquestion,itcanbeaffirmedthatthe reliabilityoftheRFsclassifierinpredictingfuturemachines'behaviouris acceptable.However,thereistheneedtounderlinethatalthoughrealdata were used during the process, this were artificially manipulated as required by AIsstandards.Forthesecondquestion,anunexpectedresultwasobtained. Indeed,notonlyitwaspossibletoexplainthemodelresultsbydisplaying abnormalvaluesprecedingabreakdown,butthiswastheclearestXAIs techniquesbasedontheuserstudyoutcomes. However,theMLsmodel foranomaliesidentificationwasnotreliableonlongtermdata.Finally,for thethirdquestion,itwasshownthatusersneededconcreteandsituational informationinordertousetheprediction.Inparticular,resultsshowedthat mostsuitableXAIstechniquewhichcanleadenduserstotrustandhave confidence in the model results is the one showing the presence of anomalous or normal data preceding the prediction.Indeed they would have liked to have a combination of the second and the fourth XAIs interfaces so that they would have known how to intervene to solve the breakdown.

6.2 Future work

Asfutureworkitmightbeinterestingtofurtherinvestigatetheanomalies identification,bybuildingdifferentmodelsbasedondifferentproduction

cycle’sshapes.Moreover,inordertoreducetimecostsfordatacleaning procedures,theMTconnectdatamightbegatheredinawaytonotpresent timegapsinthefinaldataset. Inparticular,insomeproductioncycles, information were missing for few seconds and this was compromising the time seriesshape.Asolutiontothismightberepresentedbygatheringdatain microsecondsintervals.Inaddition,itmightbevaluabletoconductastudy including Holter-Winter algorithm in order to identify anomalies, considering thatthismethodologieswasnotincludedinthestudyforitslowreliability. From users perspective, it might be worth to conduce a study where the XAIs interfacesvisualizeanhistoryofthepastmodelpredictionsintegratedwith other information.In addition, it might be interesting to conduce A/B testing to compare productivity of managers and operators with and without the aid of XAIs.

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Appendix A

Correlation matrices

A.1 Correlation matrix Stack

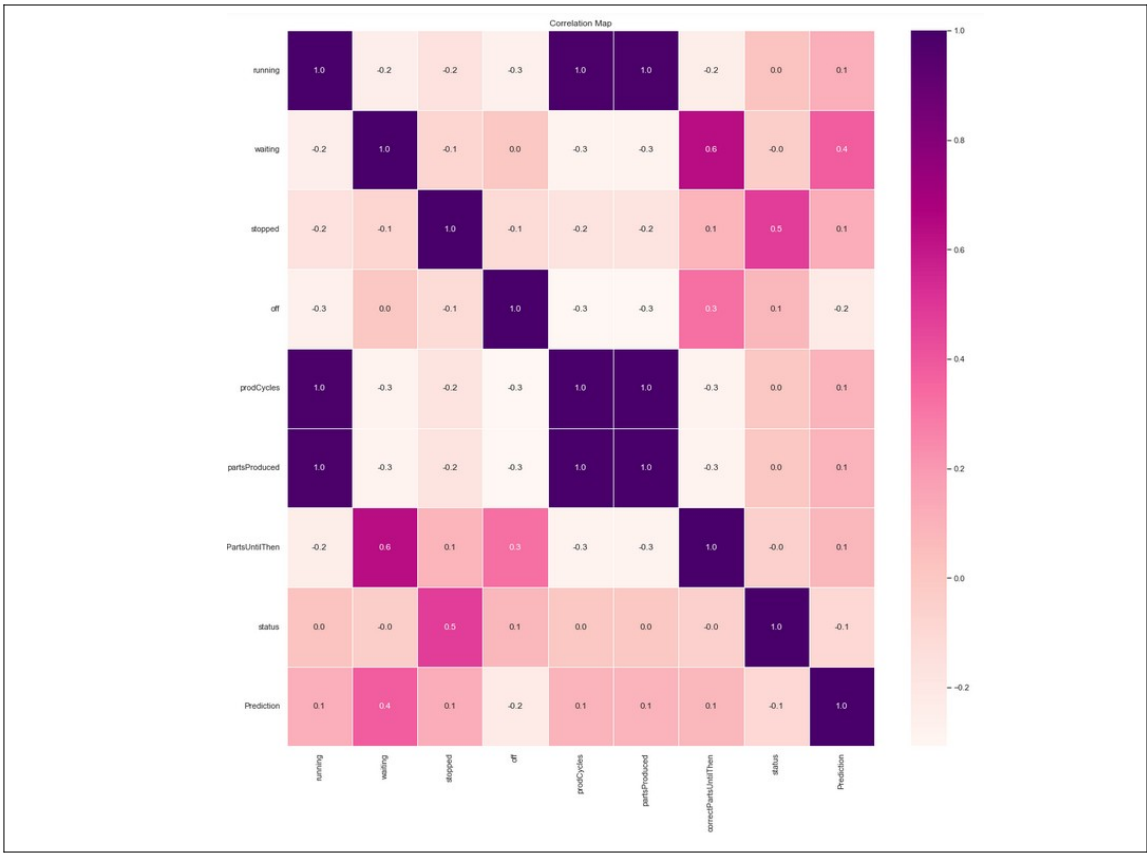

Figure A.1:Correlation matrix Stack lights data machine 50

A.2 Correlation matrix

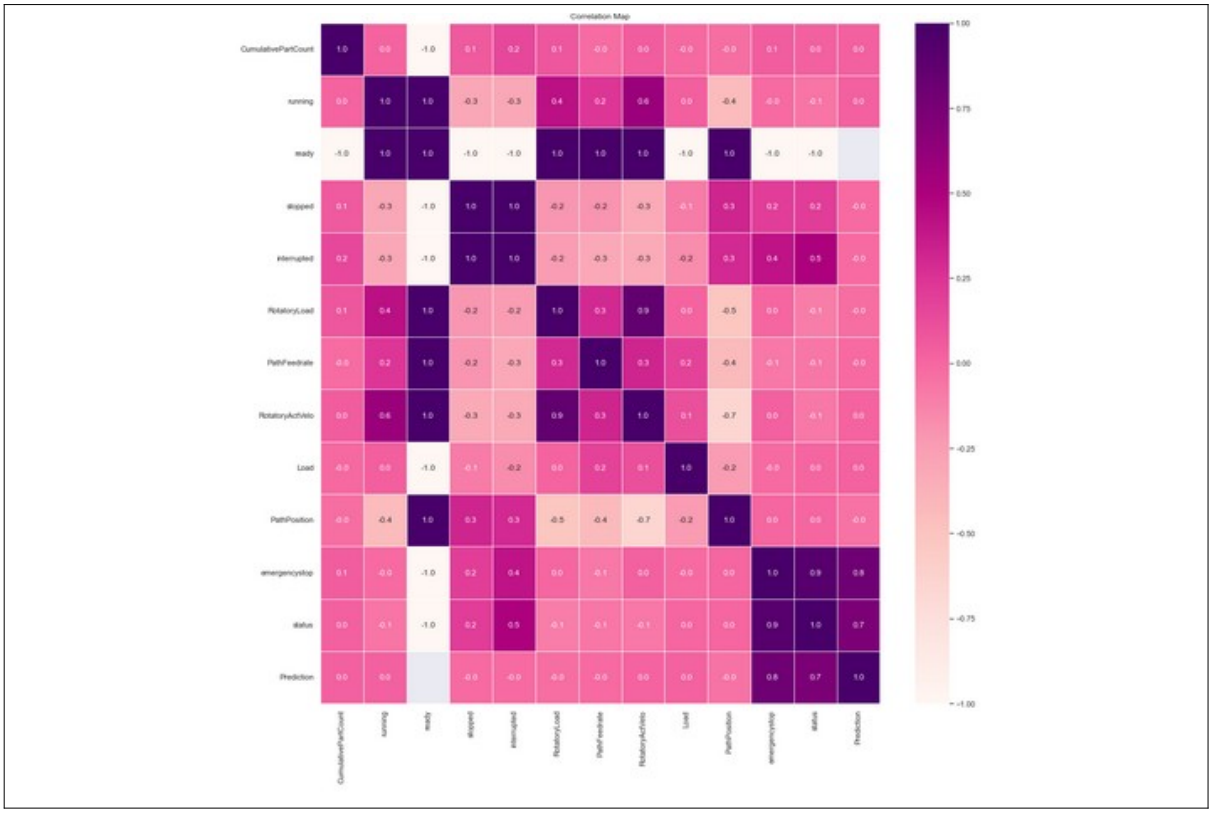

Figure A.2:Correlation matrix MTconnect data machine 50

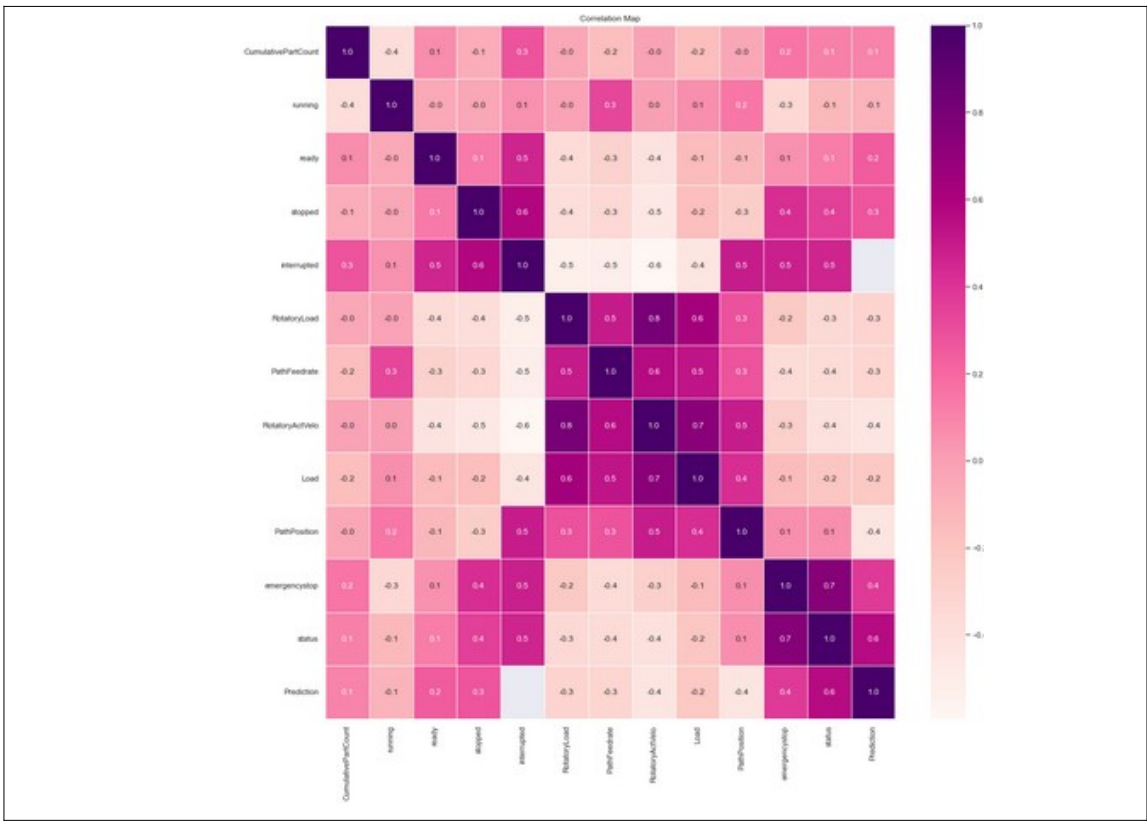

Figure A.3:Correlation matrix MTconnect data machine 54

Appendix B

Timeseriesrepresentingpro-duction cycles

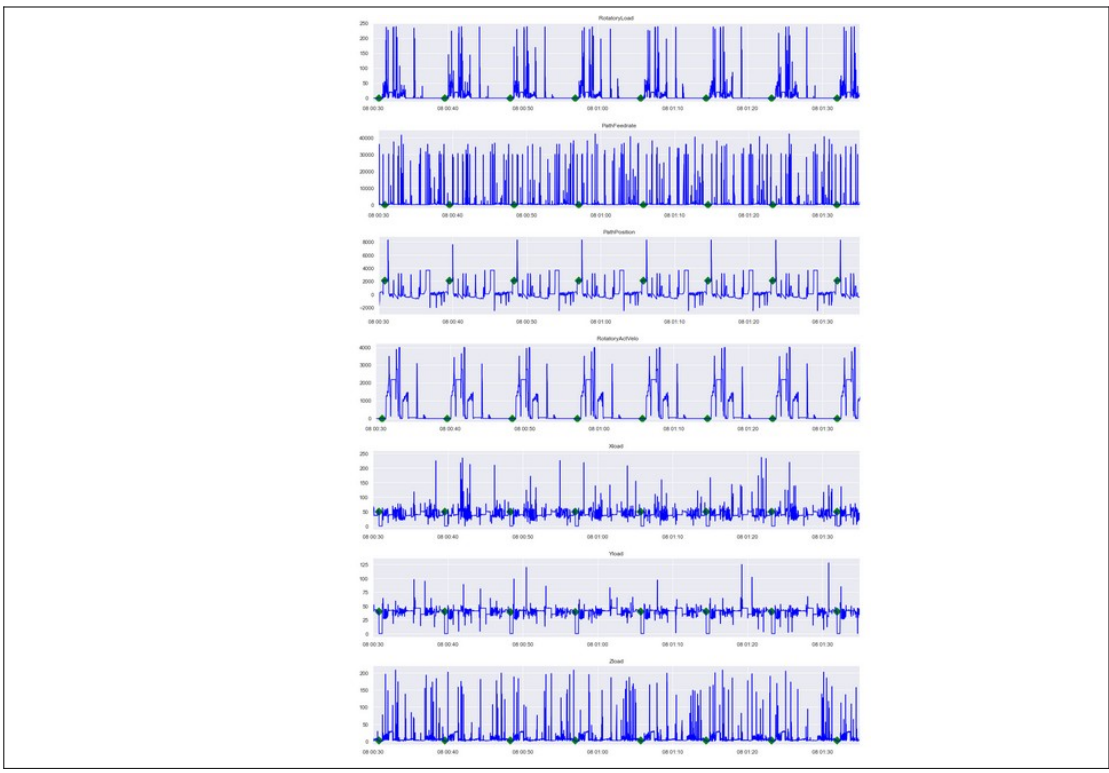

Figure B.1:Time series illustration with production cycles flag represented as green dots machine 50

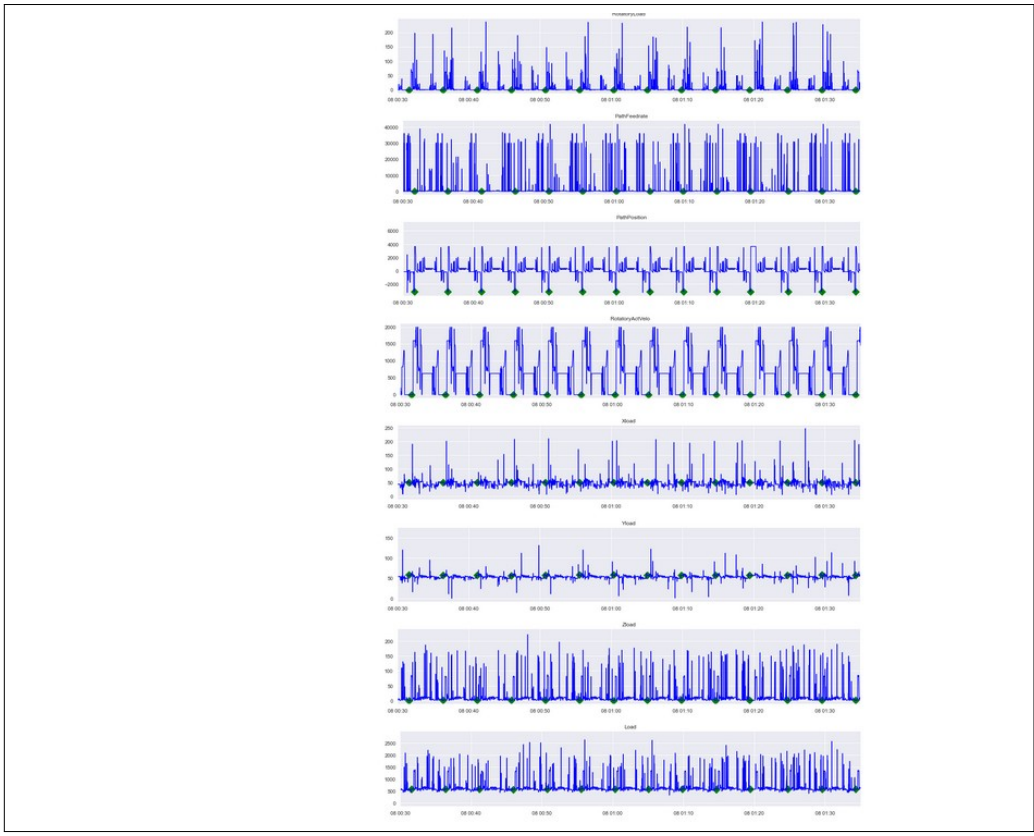

Figure B.2:Time series illustration with production cycles flag represented as green dots machine 54

Appendix C

Python methods:classifier and anomalies identification

Listing C.1:Class Balancing

```
def classBalancing(df):
    y=df['Prediction']
    df=df.drop(columns='Prediction')
    X=df
    sm=SMOTE(sampling_strategy='auto', k_neighbors=2,
    random_state=2)
    X_res, y_res=sm.fit_resample(X, y)
    X_train_res_full, X_test_res_full, y_train_res, y_test_res
    =train_test_split(X_res, y_res, random_state=2)
    X_train_res=X_train_res_full[['PathFeedrate', 'Load',
    'PathPosition', 'status', 'CombRotatory']]
    X_test_res=X_test_res_full[['PathFeedrate', 'Load',
    'PathPosition', 'status', 'CombRotatory']]
    X_train, X_test, y_train, y_test= train_test_split(X, y,
    random_state=2)
    lr =LogisticRegression solver='liblinear')
    kn=KNeighborsClassifier()
    dt=DecisionTreeClassifier(random_state=2)
    nn=MLPClassifier(random_state=2)
    rf=RandomForestClassifier(random_state=2)

    models=[lr, kn, dt, nn, rf]
    models_names=['Logistic Regression', 'KNNClassifier',
```

```
' Decision Tree Classifier', " Neural Network Clas
"RandomForest Classifier"]

train_accuracy=[ ]
cv_accuracy =[]
test_accuracy=[]

data =[[ X_train, X_test, y_train, y_test], [ X
X_test_res, y_train_res, y_test_res] ]

for dataset in data:

train_accuracy_
i=[]
cv_accuracy_i =[]
test_accuracy_i=[]

for model in range ( len (
model_=clone ( models [ model ] )
model_ .fit(dataset[ 0 ], #−
−Tr
ain
```

```
d a t a s e t[ 2 ] )


                    y _ p r e d _ t r a i n=model_ .p r e d i c t(d a t a s e t[ 0 ] )
                    t r a i n _ a c c u r a c y _ i. append ( accuracy_score (d a t a s e t[ 2 ] y _
                    p r e d _ t r a i n) )

                    #−−V a l i d a t i o n s c o r e s
                    c v _ r e s u l t s=c r o s s _ v a l i d a t e( model_, d a t a s e t[ 0 ], d a
                    t a s e t[ 2 ],        s c o r i n g =[" accuracy "]      ,
                    r e t u r n _ t r a i n _ s c o r e =True )
                    cv_accuracy_i. append ( abs ( np . mean (c v _ r e s u l t s [ ' t e s t _ a
                    c c u r a c y' ] ) ) )

                    #−−Test     s c o r e s
                    y _ p r e d _ t e s t=model_ .p r e d i c t(d a t a s e t[ 1 ] )
                    # p r i n t( y _ p r e d _ t e s t )
                    t e s t _ a c c u r a c y _ i. append ( accuracy_score (d a t a s e t[ 3 ], y _ p
                    r e d _ t e s t) )

          t r a i n _ a c c u r a c y. append (t r a i n _ a c c u r a c y _ i)
```

```
        cv_accuracy. append ( cv_accuracy_i )
        t e s t _ a c c u r a c y. append (t e s t _ a c c u r a c y _ i)

o r i g i n a l _ d f=pd . DataFrame (l i s t( zip (t r a i n _ a c c u r a c y[ 0 ],
cv_accuracy [ 0 ], t e s t _ a c c u r a c y[ 0 ] ) ),        columns =[" Train
Accuracy " ,     " V a l i d a t i o n Accuracy " ," Test        Accuracy " ] )
o r i g i n a l _ d f.index=models_names

balanced_df=pd . DataFrame (l i s t( zip (t r a i n _ a c c u r a c y[ 1 ],
cv_accuracy [ 1 ], t e s t _ a c c u r a c y[ 1 ] ) ),        columns =[" Train
Accuracy " ,     " V a l i d a t i o n Accuracy " ," Test        Accuracy " ] )
balanced_df.index=models_names

r e t u r nX,    o r i g i n a l _ d f,  balanced_df,        X_train,       X_test,
y _ t r a i n,   y _ t e s t,   y_pred_train, y_pred_test,                X_train_res,
X_test_res, y _ t r a i n _ r e s, y _ t e s t _ r e s,          X _ t r a i n _ r e s _ f u l l,
X _ t e s t _ r e s _ f u l l
```

Listing C.2:Cross-validation for best parameters

```
random_state =2

models_names =[ ' L o g i s t i c  Regression',  'KNNC l a s s i f i e r',
' Decision  Tree  C l a s s i f i e r',  " Neural  Network C l a s s i f i e r" ,
"RandomF o r e s t  C l a s s i f i e r"]


c l f s =[ L o g i s t i c R e g r e s s i o n ( s o l v e r = ' l i b l i n e a r',
random_state = random_state ),  K N e i g h b o r s C l a s s i f i e r( ),
D e c i s i o n T r e e C l a s s i f i e r( random_state = random_state ),MLP
C l a s s i f i e r( random_state = random_state ),
R a n d o m F o r e s t C l a s s i f i e r ( random_state =2)]

params ={0:  { ' penalty' : ['  ' l2' ],  'C ' :  np .arange ( 0 . 0  0.2,
0 . 0 1 ) },
1:  { ' n_neighbors' : np .arange (5 ,  10 ,  1) ,
' metric' : ['euclidean' , ' manhattan' ],  ' weights' :
[ ' uniform',  ' d i s t a n c e' ] },
2:  { ' c r i t e r i o n' : ['gini',' entropy' ],
' max_depth' : np .arange ( 2, 4, 1 ),
np .arange (10 ,20 ,1)} ,
```

’ min_samples_leaf’ :

3: { ’ h i d d e n _ l a y e r _ s i z e s’ : [(2 2,), (22 , 2 2)],
’ solver’ : [’ sgd’, ’adam ’], ’ a c t i v a t i o n’ : [’l o g i s t i c’,
’ tanh’, ’ relu’] },
4: { ’ c r i t e r i o n’ : [’gini’,’ entropy’],
’ max_depth’ : np .arange (1, 4, 1),
np .arange (3 ,10 ,1)}}

” min_samples_leaf ” :

```
def Grid_Search_cv ( model_index,X, y ) :

    g r i d=GridSearchCV (c l f s[ model_index ], params [ model_index
    verbose=0 ,cv=5 ,          n_jobs=−1 ,          r e t u r n _ t r a i n _ s c o r e =
    True )

    g r i d _ r e s u l t s =g r i d.f i t(X,          y )

    r e t u r n  g r i d _ r e s u l t s.b e s t _ e s t i m a t o r _.get_params( )
```

```
def BestParameters ( X_train, y _ t r a i n) :
    b e s t _ p a r a m s _ l r=Grid_Search_cv (0 , X_train, y _ t r a i
    n)
    best_params_kn=Grid_Search_cv (1 , X_train, y _ t r a i
    n)
    best_params_dt=Grid_Search_cv (2 , X_train, y _ t r a i n)
    best_params_nn=Grid_Search_cv (3 , X_train, y _ t r a i
    n)
    best_params_rf=Grid_Search_cv (4 , X_train, y _ t r a i n)

    best_params =[ best_params_lr, best_params_kn,
    best_params_dt, best_params_nn, best_params_rf ]

    f o r i in range ( len ( models_names) ) :
        p r i n t(”  ”)
        p r i n t(’ \ 0 3 3 [ 1m’+’ Best parameter f o r ’+
        s t r( models_names [ i] ) + ’\033[0m’ )
            p r i n t( best_params [ i] )
    r e t u r n best_params
```

Listing C.3:Model evaluation

```
def modelEvaluation X_test, y _ t r a i n, y _ t e s t,
best_params) :
    t r a i n _ a c c u r a c y=[]
    t r a i n _ r e c a l l=[]
    t r a i n _ p r e c i s i o n=[]
    t r a i n _ f 1=[]

    cv_accuracy =[]
    c v _ p r e c i s i o n=[]
    c v _ r e c a l l=[]
    cv_f1 =[]

    t e s t _ a c c u r a c y=[]
    t e s t _ p r e c i s i o n=[]
    t e s t _ r e c a l l=[]
    t e s t _ f 1=[]

        f o r ix in range ( len

        o r i g i n a l _ m o d e l = clone (c l f s[ ix] )

        optimized_model= clone (c l f s[ ix] ) .set_params (**best_ params [ ix] )

            models =[ optimized_model, f o r
```

o r i g i n a l _ m o d e l]

model=clone (model)

model .f i t(X_train, y _ t r a i n)

#−−Train s c o r e s
y _ p r e d _ t r a i n=model .p r e d i c t(X_train)
t r a i n _ a c c u r a c y. append (accuracy_score (y _ t r a i n, y _ p r e d _ t r a i n))
t r a i n _ r e c a l l. append (r e c a l l _ s c o r e(y _ t r a i n, y _ p r e d _ t r a i n))
t r a i n _ p r e c i s i o n. append (p r e c i s i o n _ s c o r e(y _ t r a i n,

```
y _ p r e d _ t r a i n) )
t r a i n _ f 1. append ( f 1_ s c o re ( y _ t r a i n,    y _ p r e d _ t r a i
n) )

#−−V a l i d a t i o n                    s c o r e s
c v _ r e s u l t s=c r o s s _ v a l i d a t e( model,      X_train,
y _ t r a i n, s c o r i n g =[" accuracy " ,       " r e c a l l" ,
" p r e c i s i o n" ,
" f1 " ] ) cv_accuracy. append ( abs ( np . mean (c v _ r e s u l t s[ 't e s t _ a
c c uracy' ] ) ) )
c v _ r e c a l l. append ( abs ( np . mean (c v _ r e s u l t s[ 't e s t _ r e c a l
l' ] ) ) )
c v _ p r e c i s i o n. append ( abs ( np . mean (c v _ r e s u l t s[ 't e s t _ p r
e c i s i o n' ] ) ) )
cv_f1 . append ( abs ( np . mean (c v _ r e s u l t s[ ' t e s t _ f 1' ] ) ) )

#−−Test s c o r e s
y _ p r e d _ t e s t=model .p r e d i c t( X_test )
t e s t _ a c c u r a c y. append ( accuracy_score ( y _ t e s t, y _ p r
e d _ t e s t) )
t e s t _ r e c a l l. append (r e c a l l _ s c o r e( y _ t e s t,
y _ p r e d _ t e s t) )
t e s t _ p r e c i s i o n. append (p r e c i s i o n _ s c o r e( y _ t e s
t, y _ p r e d _ t e s t) )
t e s t _ f 1. append ( f 1 _ s c or e ( y _ t e s t, y _ p r e d _ t e s t) )
# p r i n t( X_test )
# p r i n t( len ( y _ p r e d _ t e s t) )

index_names =[ ' L o g i s t i c    Regression    Tuned ',    ' L o g i s t i c
Regression',        'KNNC l a s s i f i e r Tuned ', 'KNNC l a s s i f i e r',
    ' Decision   Tree C l a s s i f i e r       Tuned ',    ' Decision    Tree
    C l a s s i f i e r',' Neural    Network C l a s s i f i e r    Tuned ', ' Neural
    Network    C l a s s i f i e r',
    'RandomF o r e s t C l a s s i f i e r
    Tuned ',      'RandomF o r e s t              C l a s s i f i e r'
    ]

h t _ r e s u l t s =pd . DataFrame (l i s t( zip ( t r a i n _ a c c u r a c y,
t r a i n _ r e c a l l,  t r a i n _ p r e c i s i o n, t r a i n _ f 1,   cv_accuracy,
c v _ r e c a l l,  cv_precision,        cv_f1,     t e s t _ a c c u r a c y, t e s t _ r e c a l l,
```

```
t e s t _ p r e c i s i o n,  t e s t _ f 1) ),    columns =[" Train        Accuracy " ,
" Train     Recall " ,     " Train     P r e c i s i o n " ,                " TrainF1 " ,
" V a l i d a t i o n
Accuracy " ,     " V a l i d a t i o n          Recall " ,  " V a l i d a t i o n
P r e c i s i o n" ," V a l i d a t i o n        F1 " ,
" Test      Accuracy " , " Test   Recall " , " Test  P r e c i s i o n " ,
" TestF1 " ] )
h t _ r e s u l t s.index=index_names

r e t u r n  h t _ r e s u l t s
```

Listing C.4:Anomalies labels

```
def  realAnomalies (d f _ l i              cols,              o r i g i n a l) :
elms=len (d f _ l i s t)
    f o r              c                          in              c o l s:
        o r i g i n a l[ ' realAn'+ c ]=np . nan
    f o r              i                          in              range ( elms ) :
                                    df=                          d f _ l i s t[ i ]
      d e s c r i p t i o n=df .d e s c r i b e( )
       f o r              c                          in              c o l s:
        upThreshold=d e s c r i p t i o n[ c ] .i l o c[ 1 ]+
        (4∗d e s c r i p t i o n[ c ] .i l o c[ 2 ] )
        downThreshold=d e s c r i p t i o n[ c ] .i l o c[ 1 ]−
        (4∗d e s c r i p t i o n[ c ] .i l o c[ 2 ] )
        i f( len ( df ) >0):
                o r i g i n a l[ ' realAn'+ c ] .loc [o r i g i n a l[ ' c y c l e P e r i o
                ']== i ]=np . where ( ( df [ c ] > upThreshold )              |
                ( df [ c ] < downThreshold ), 1, 0 )
```

Listing C.5:Anomalies identification model training

```
def anomaliesIdentif original, cols,
outPerc) :
elms=len (df_list)
models=[ ]
outliers_fraction =outPerc
for i in
range ( elms ) : df=
df_list[ i ]
df=df . drop ( columns =[ ' toolid' , ' basetime' ] )
model=
IsolationForest( contamination = outliers_fraction)
model .fit( df [ cols] .values )
models . append ( model )
return models
```

Listing C.6:Anomalies identification model testing

```
def anomaliesIdentif original, models,
cols) :
elms=len (df_list)
original[ ' anomalyCycle' ]=np . nan
sumAn=0
for i in range ( elms ) :
df= df_list[ i ]
df=df . drop ( columns =[ ' toolid' , ' basetime' ] )
if( len ( df ) >0):
anomcol=pd .Series ( models [ i] .predict( df [ cols] .values),
index=df .index )
original[ ' anomalyCycle' ] .
loc [original[ ' cyclePeriod']== i ]=anomcol
```

Appendix D

Anomalies in tool

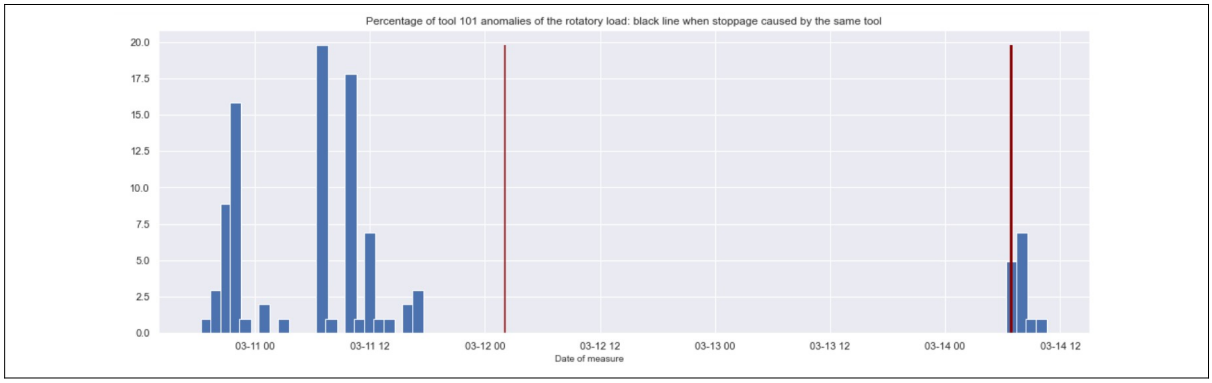

Figure D.1:Anomalies in tool 101 Machine 50

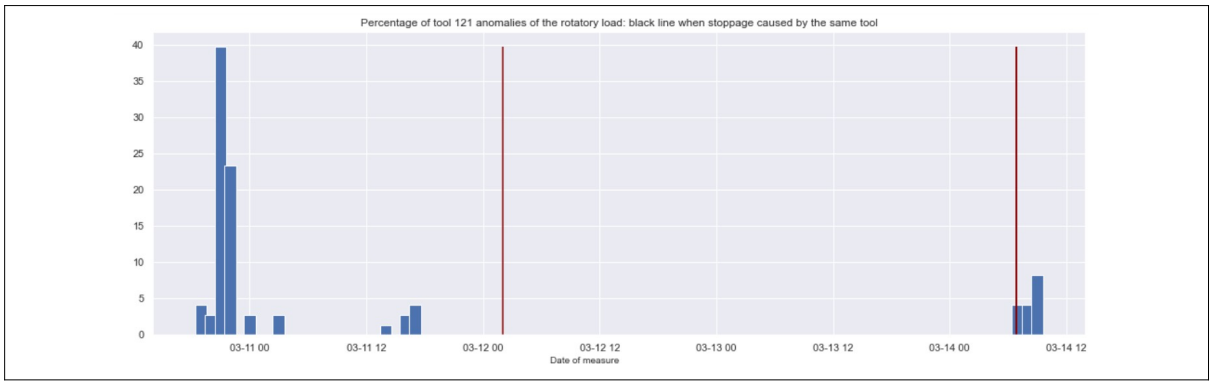

Figure D.2:Anomalies in tool 121 Machine 50

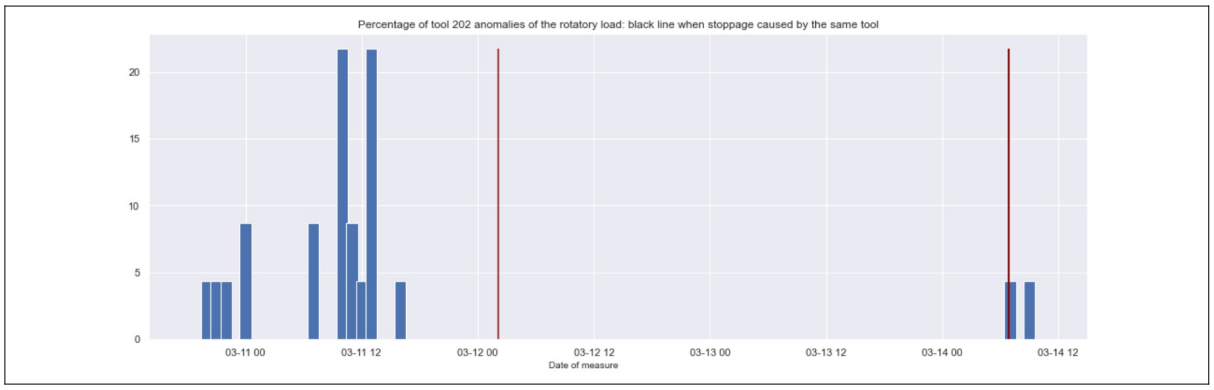

Figure D.3:Anomalies in tool 202 Machine 50

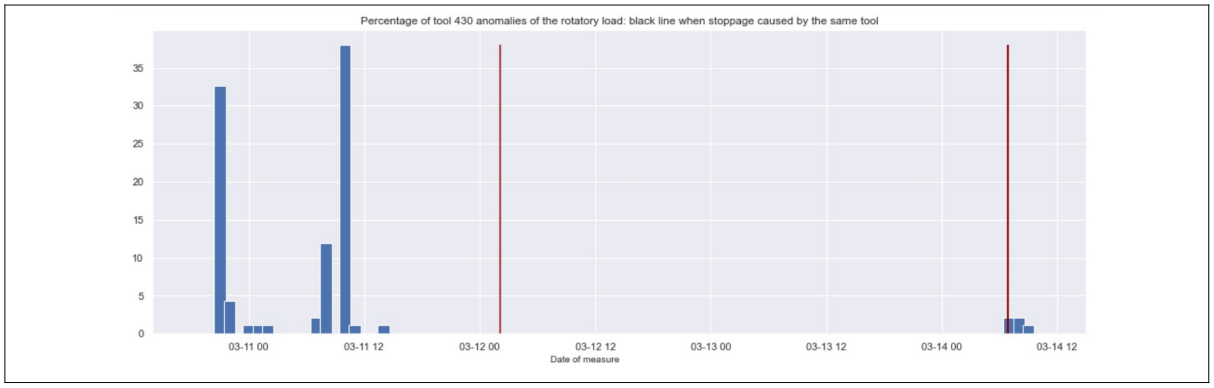

Figure D.4:Anomalies in tool 430 Machine 50

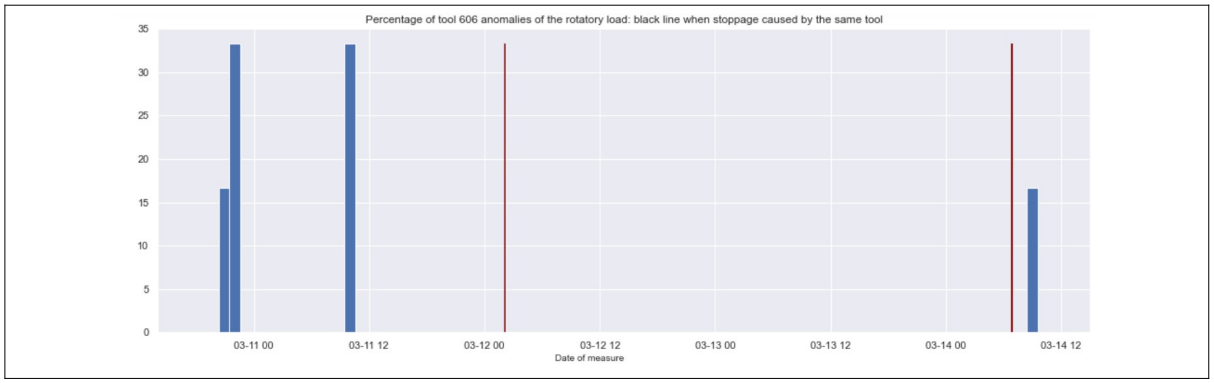

Figure D.5:Anomalies in tool 606 Machine 50

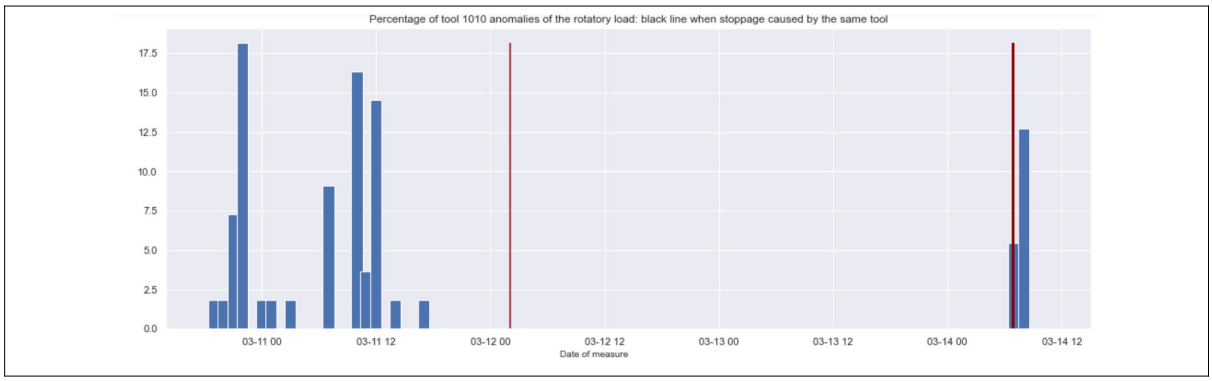

Figure D.6:Anomalies in tool 1010 Machine 50

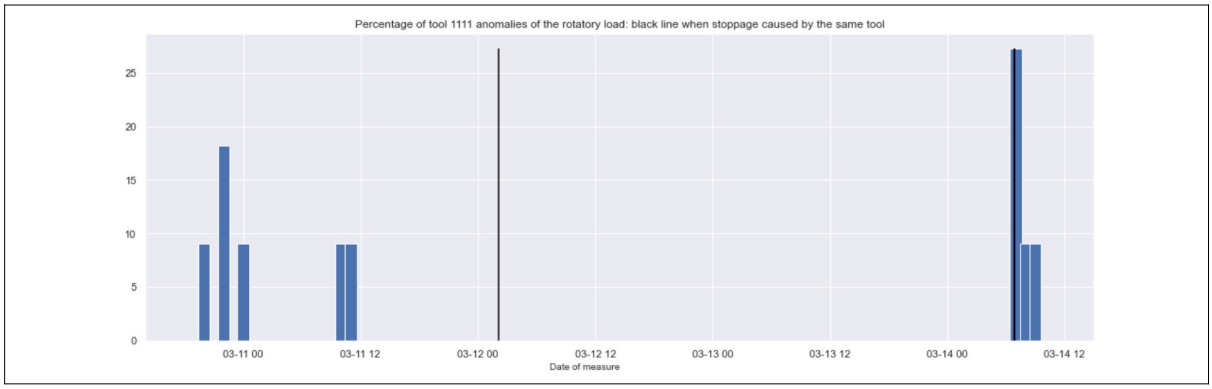

Figure D.7:Anomalies in tool 1111 Machine 50

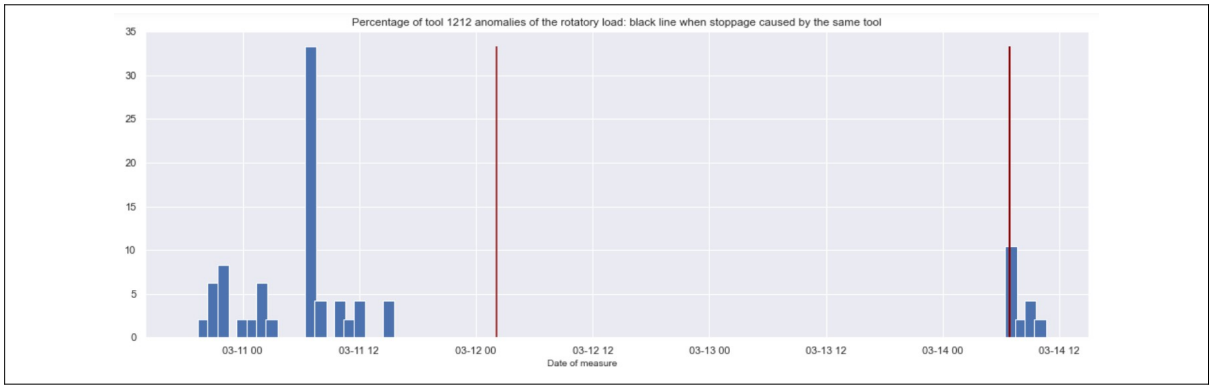

Figure D.8:Anomalies in tool 1212 Machine 50

Appendix E

Understandability surveys

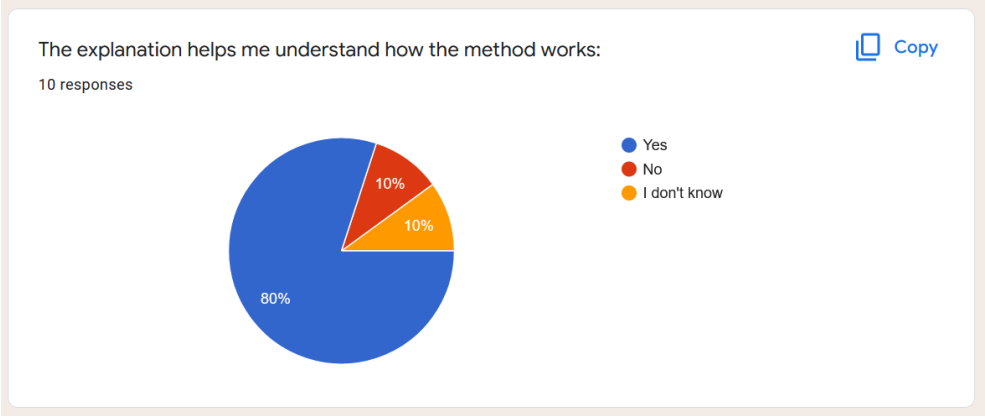

Figure E.1:Understandability performances summary interface

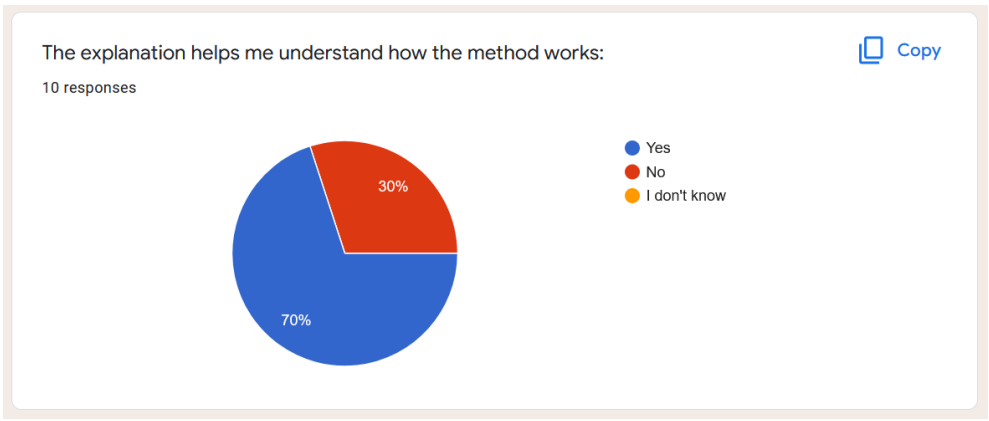

Figure E.2:Understandability feature values interface

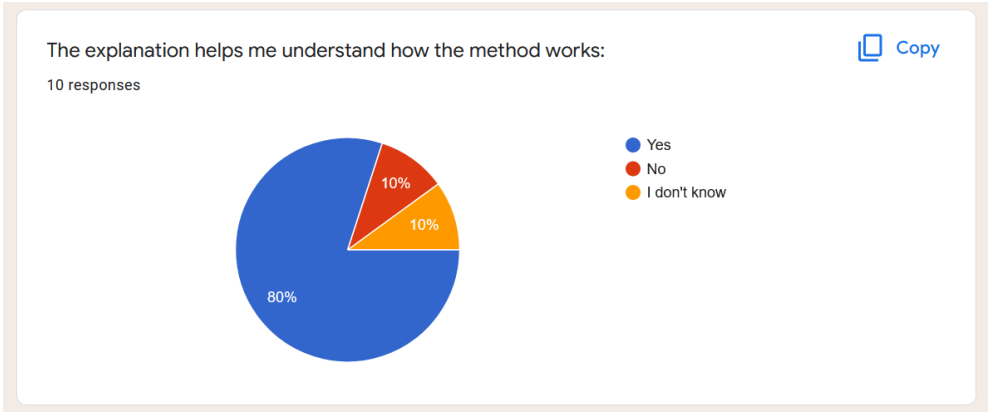

Figure E.3:Understandability model decision interface

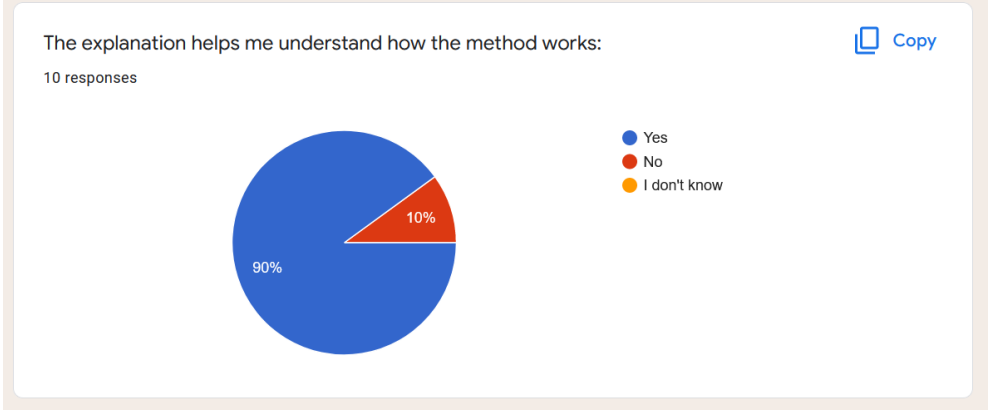

Figure E.4:Understandability tool anomalies interface

Appendix F

Satisfaction surveys

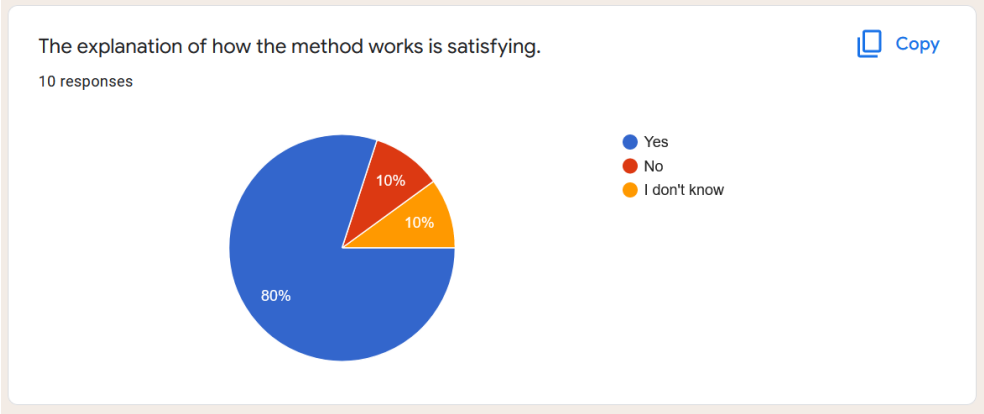

Figure F.1:Satisfaction performances summary interface

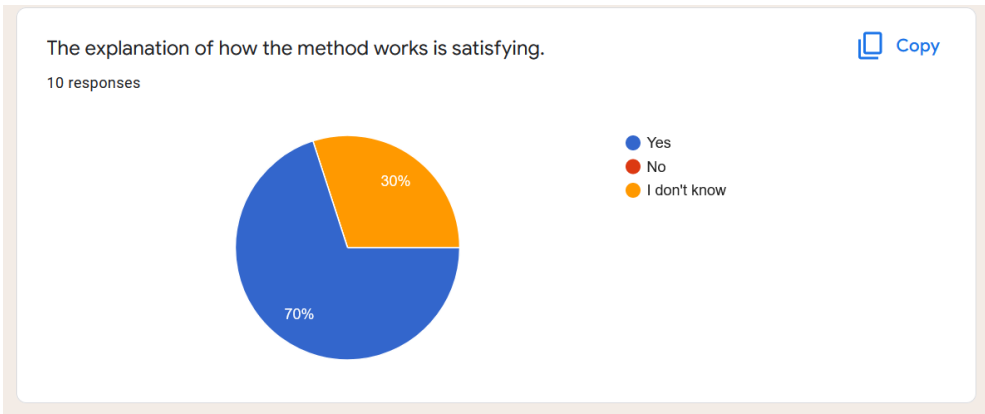

Figure F.2:Satisfaction feature values interface

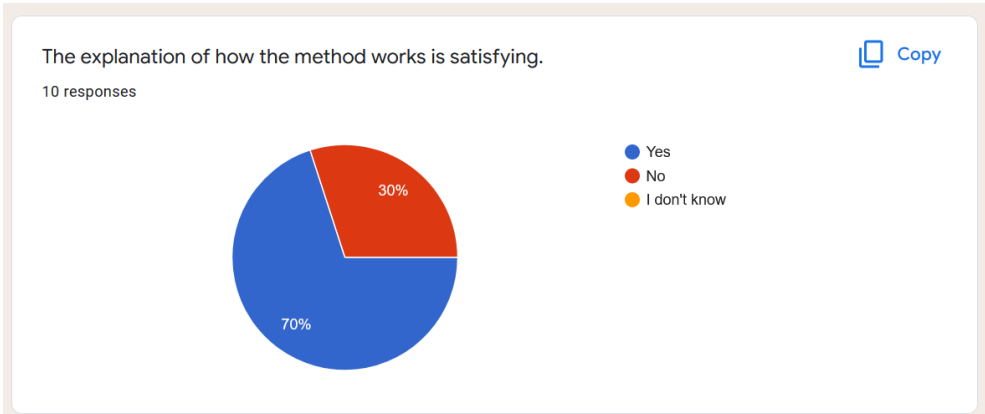

Figure F.3:Satisfaction model decision interface

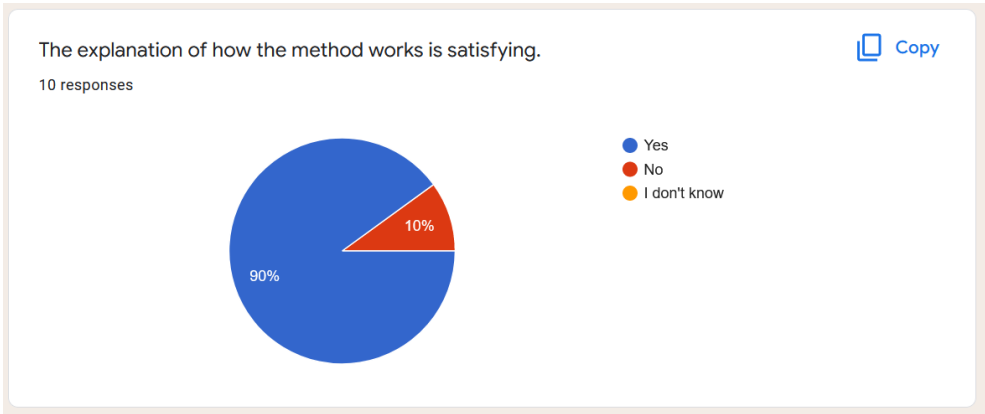

Figure F.4:Satisfaction tool anomalies interface

Appendix G

Detailing levels surveys

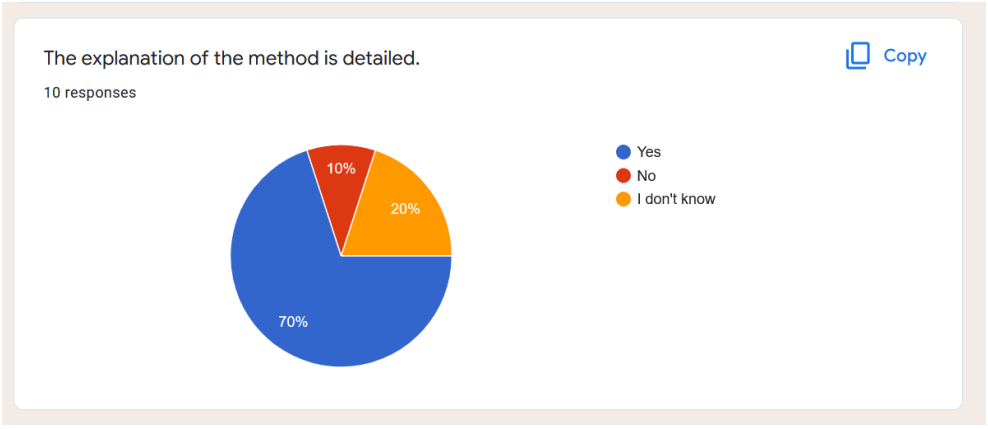

Figure G.1:Detailing performances summary interface

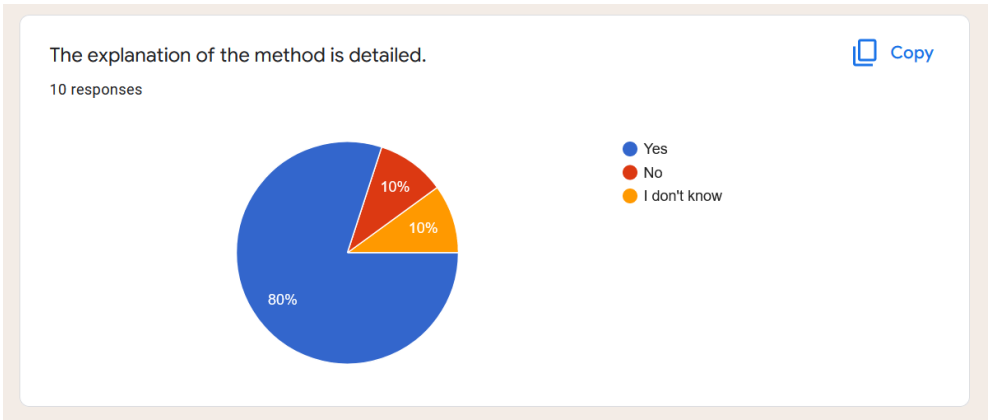

Figure G.2:Detailing feature values interface

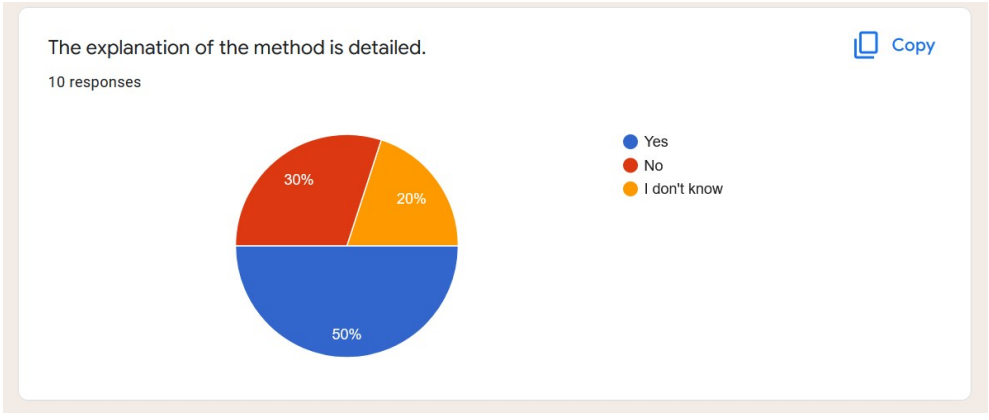

Figure G.3:Detailing model decision interface

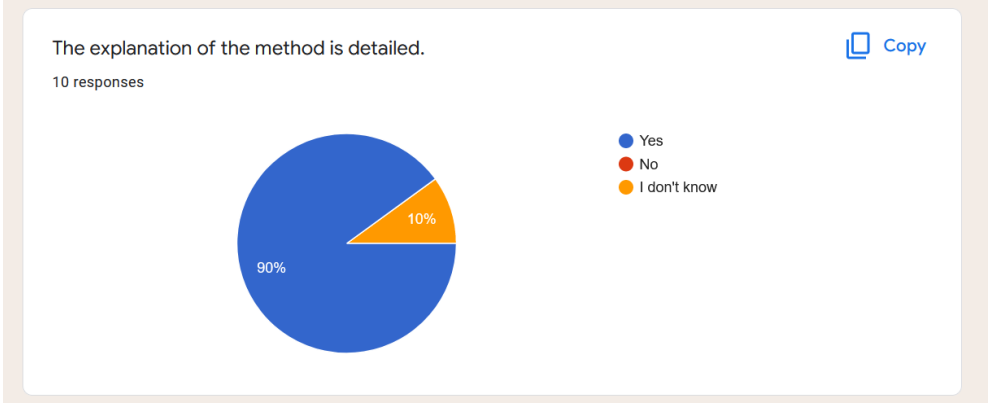

Figure G.4:Detailing tool anomalies interface

Appendix H

Completeness levels surveys

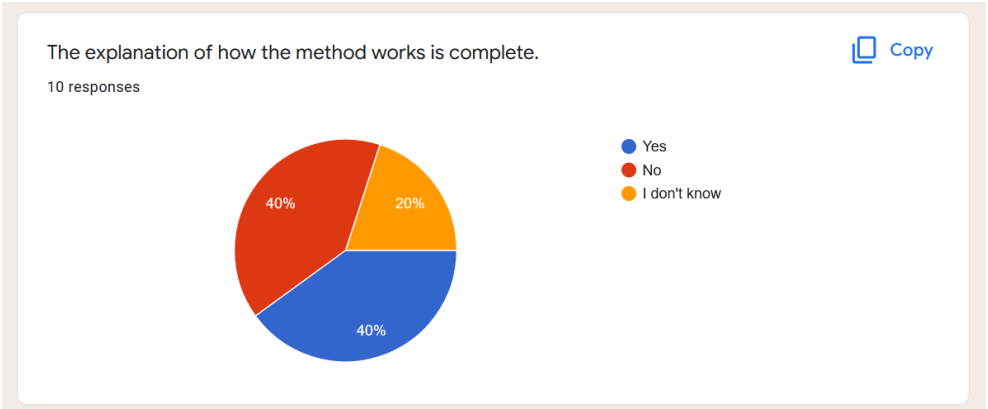

Figure H.1:Completeness performances summary interface

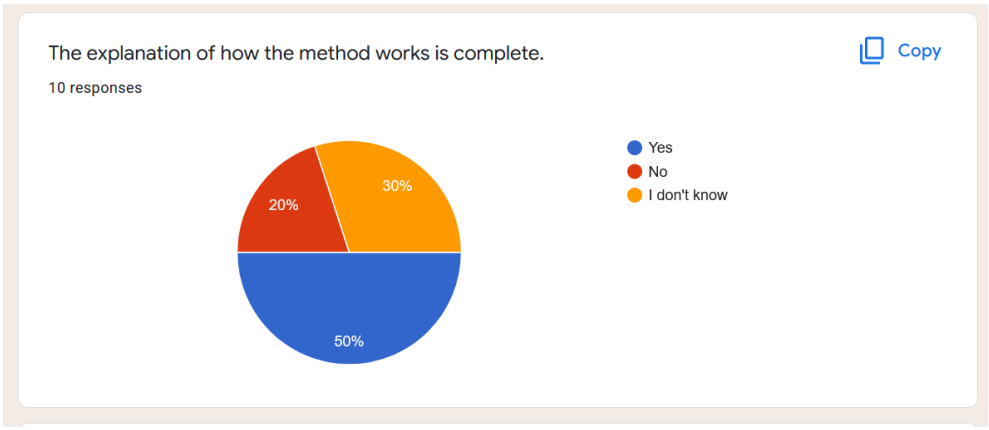

Figure H.2:Completeness feature values interface

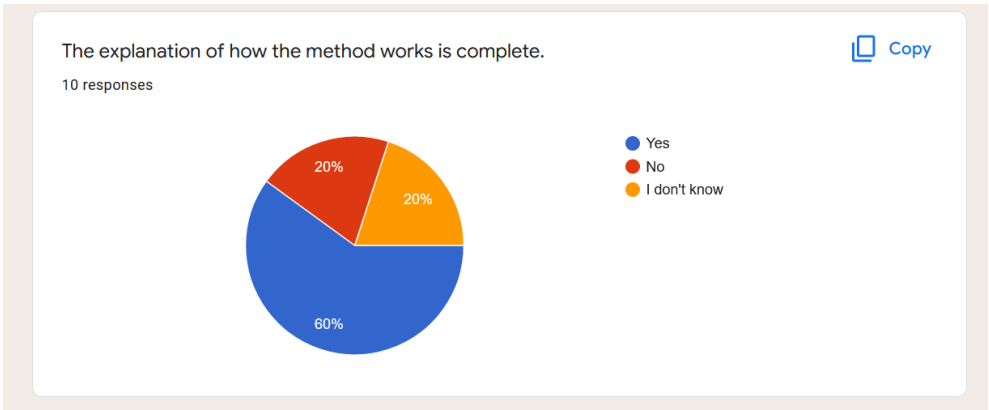

Figure H.3:Completeness model decision interface

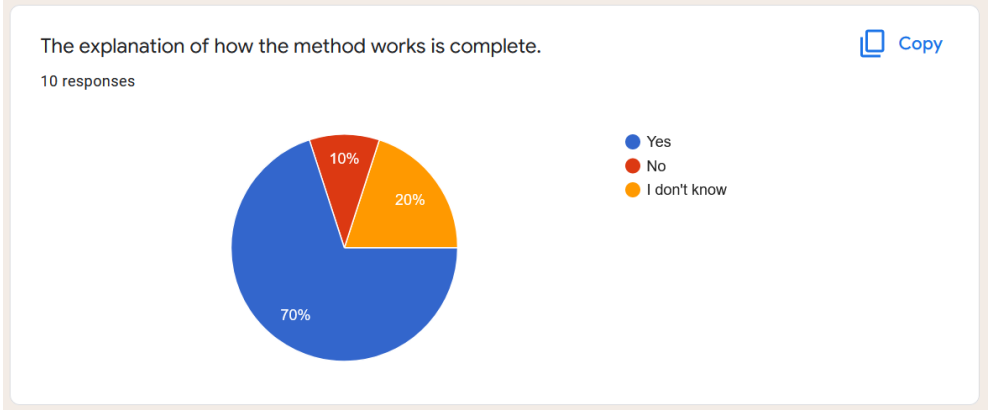

Figure H.4:Completeness tool anomalies interface

Appendix I

Actionability levels surveys

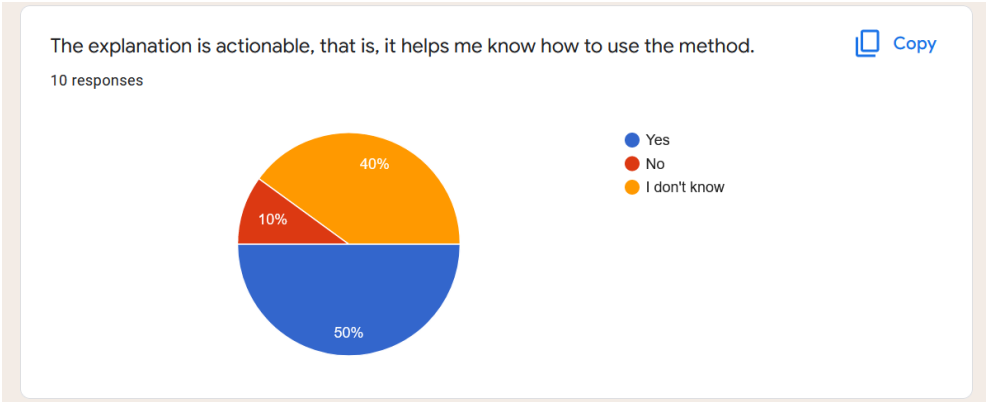

Figure I.1:Actionability performances summary interface

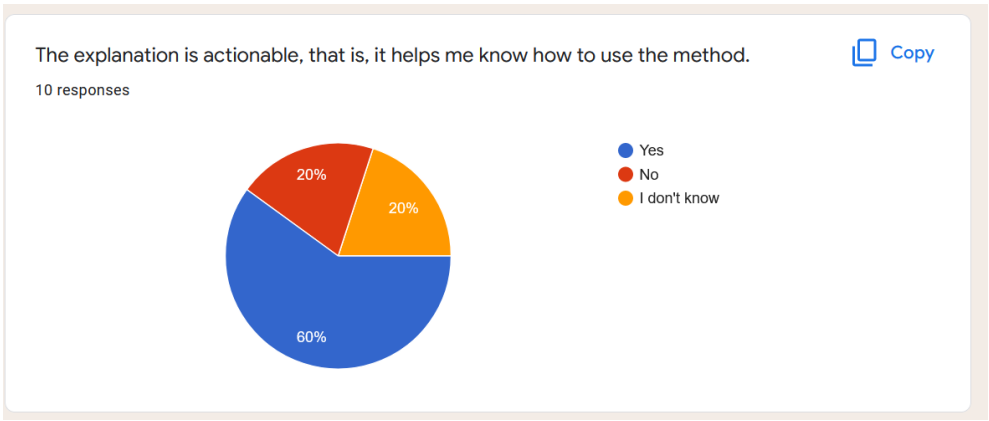

Figure I.2:Actionability feature values interface

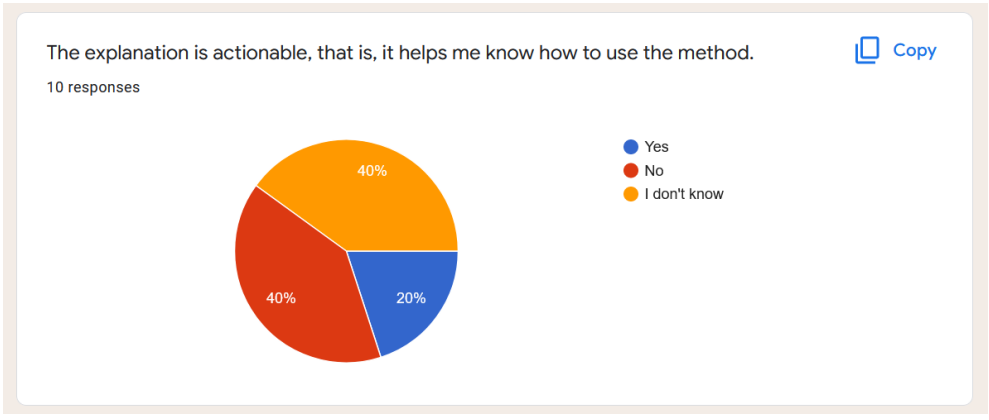

Figure I.3:Actionability model decision interface

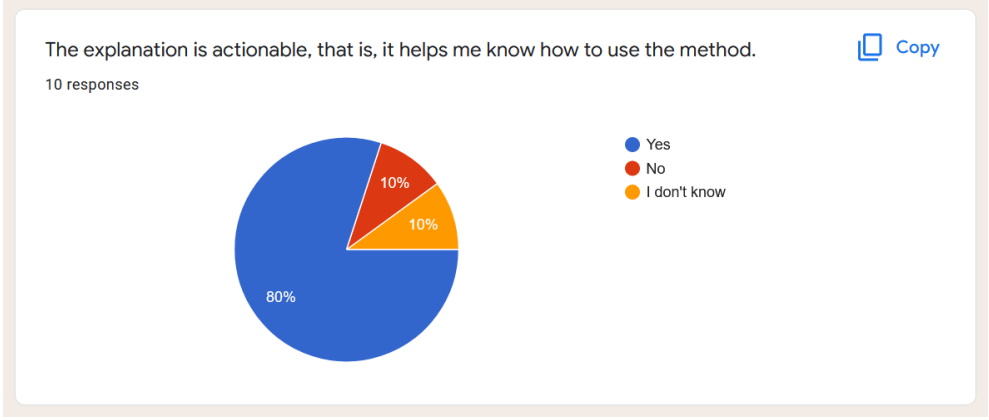

Figure I.4:Actionability tool anomalies interface

Appendix J

Accuracy levels surveys

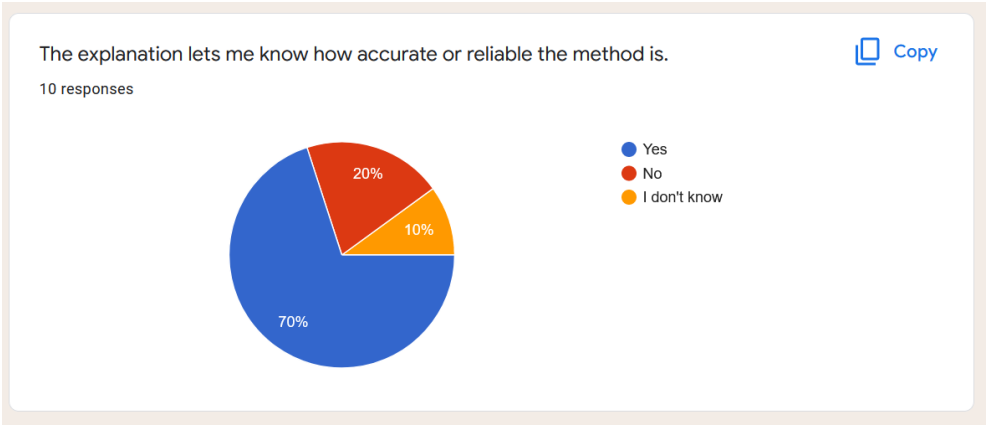

Figure J.1:Accuracy performances summary interface

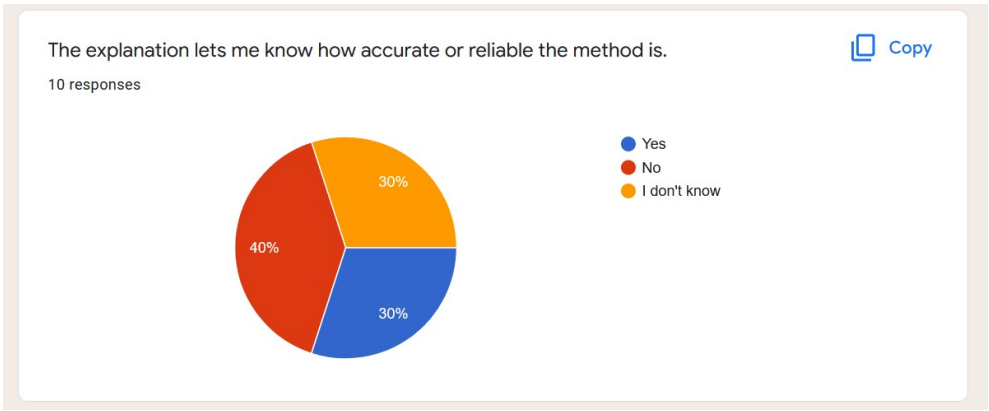

Figure J.2:Accuracy feature values interface

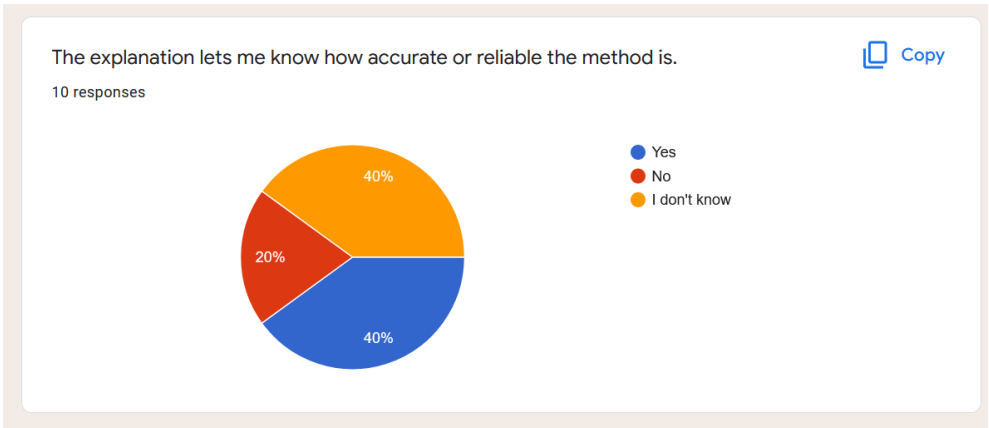

Figure J.3:Accuracy model decision interface

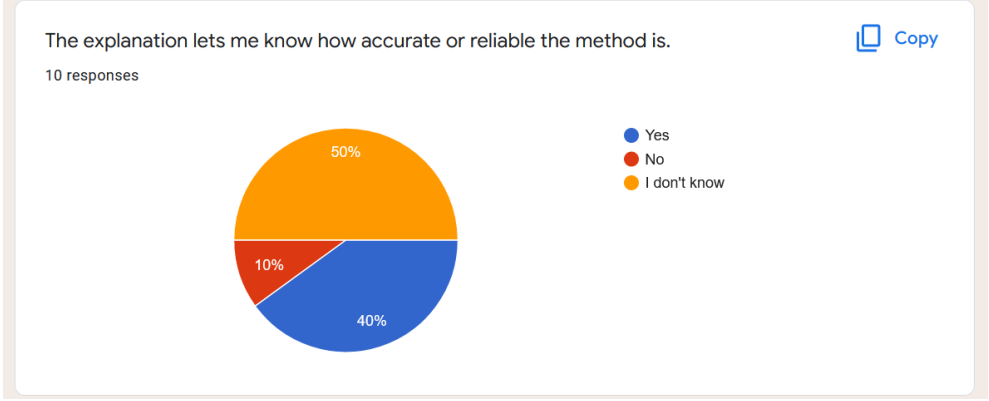

Figure J.4:Accuracy tool anomalies interface

Appendix K

Trustability levels surveys

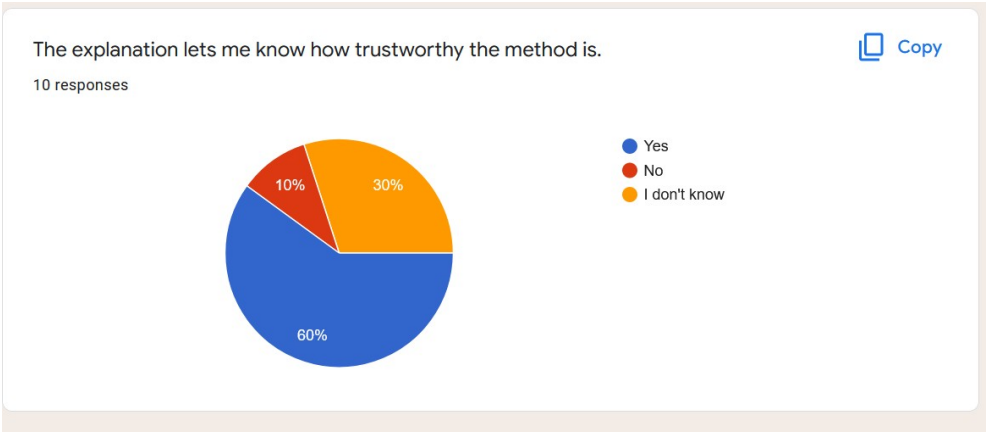

Figure K.1:Trustability performances summary interface

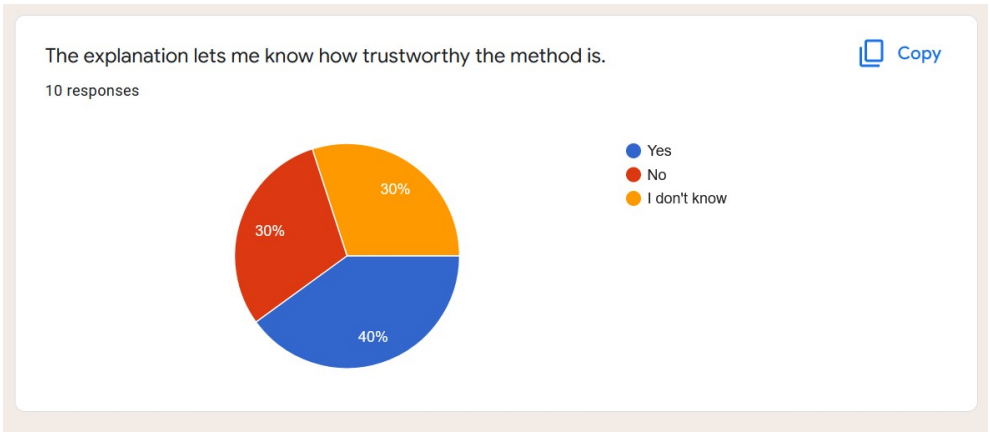

Figure K.2:Trustability feature values interface

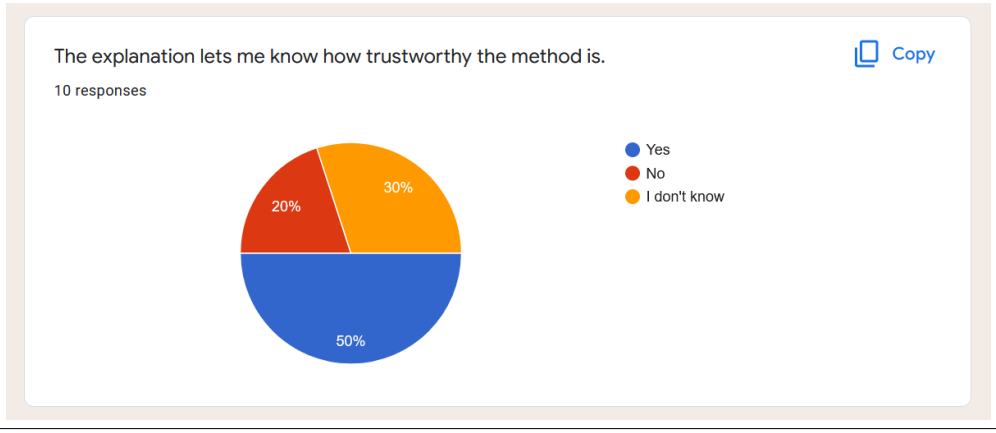

Figure K.3:Trustability model decision interface

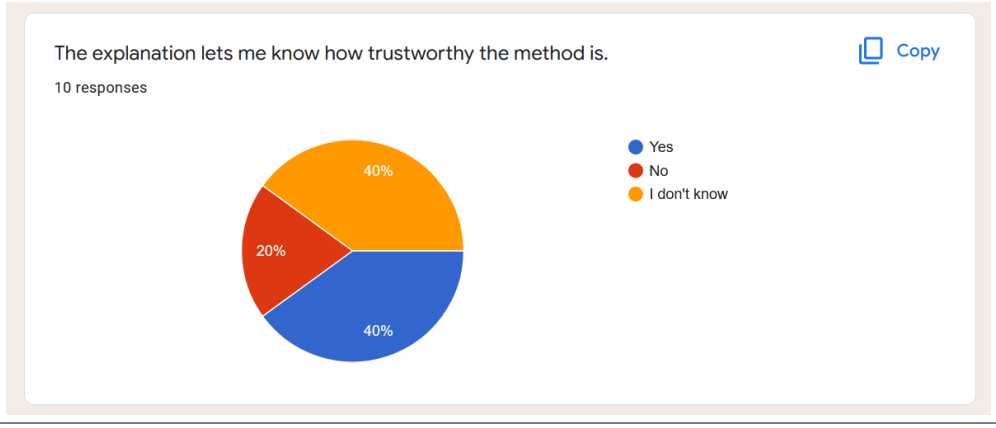

Figure K.4:Trustability tool anomalies interface

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For DIVA

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Monitoringmachineswithinamanufacturingenvironmentisofvitalimportanceforcompaniesthatwant toreachhighlevelsofprofitandproductivity.Moreover,itisfundamentaltomaintainmachinesin areliablestateandsolvetheirfailuresquicklytolimitfinanciallosses.Therefore,information technologyissteadilyincreasinginsidefactoriestomonitormachines’behaviorandassistoperators andmanagersinperformingpredictivemaintenance.However,ArtificialIntelligence\glspl{AI}is stillnotwidelyadoptedwithinthemanufacturingfield.Hence,thepurposeofthisstudyisto integrateMachineLearning\glspl{ML} technologies,whichcansupportoperatorsinmonitoring manufacturingmachines’behavior,withinamanufacturingsetup.Inparticular,aninvestigationon howtorealizeaclassifierabletopredictmachines’statehasbeenconducted.Inaddition, differenttechniquesofExplainableArtificialIntelligence\glspl{XAI} havebeentestedtodiscover themethodologyallowingpeoplewithouta\glspl{ML}backgroundtotrustpredictions.Four \glspl{XAI}facilitieshavebeentested:threeidentifiedthroughliteraturereviewandan experimentalonebasedona\glspl{ML}modelforoutliersidentificationinproductionparameters.To predictfuturemachines’states,classifiersbasedondifferentalgorithmswerebuilt,andafter evaluatingthem,itemergedthattheRandomForest-basedmodelwastheonewiththebestperformance. Inaddition,auserstudywascarriedoutinordertodiscernend-userscognitiveperspectivesabout thefour\glspl{XAI}interfaces.Finalresultsshowedhowan\glspl{XAI}facilitybasedon situationaldata,suchastheonerepresentinganomalousproductionvalues,gainedhighapproval amonguserswithnoneorbasic\glspl{ML}knowledge.

€€€€,
”Keywords[eng]”:€€€€
Explainable Artificial intelligence, Machine Learning, Random Forest, Anomalies Identification, User Understandability, User Trustability, Interface, Evaluation, User study, Isolation Forest, Thematic analysis, Surveys€€€€,
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